GEK-36523A

# INSTRUCTIONS

# ELECTROHYDRAULIC CONTROL

# FOR

# BOILER FEEDPUMP TURBINE

# 3S7513TC300A17

GENERAL ELECTRIC

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*These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company*

## INTRODUCTION

The electronic portion of an electrohydraulic control system for a boiler feedpump turbine drive is described in these instructions.

Any adjustment or maintenance of this equipment should be performed either by qualified turbine personnel or supervised by such personnel.

This equipment is an electronic speed governor whose speed set point can be varied by an automatic boiler controller or by an operator from a remote control unit. Electronic redundancy and mechanical overspeed trip are used to provide a high degree of reliability and safety.

The elementary diagram and alignment diagram for this unit are included in these instructions. Since both the number and arrangement of auxiliary components vary with particular application, it is necessary to refer to the drawings furnished with each equipment to determine the correct connections and auxiliary components furnished in each instance.

## RECEIVING, HANDLING AND STORAGE

### RECEIVING AND HANDLING

Immediately upon receipt, the equipment should be carefully unpacked to avoid damaging the apparatus. Particular care should be exercised to prevent small parts being mislaid or thrown away in the packing material.

As soon as the equipment is unpacked it should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric Sales Office should be notified promptly.

### STORAGE

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If the equipment is to be placed in long term storage, the interior cabinet temperature should be 5°C above ambient.

## DESCRIPTION

### SPEED CONTROL

The speed control loop will be described first. Starting at zone B6 and B7 of the alignment drawing, the coil of the electrohydraulic servo valve receives a low power signal from the electronic unit and controls the flow of oil to a double acting hydraulic cylinder which positions the turbine steam control valve. Turbine speed is sensed by three magnetic sensors mounted near the periphery of a shaft-driven gear. Pulses whose frequency is directly proportional to turbine

## DESCRIPTION (continued)

SPEED CONTROL (cont'd)

speed are generated as each gear tooth passes a sensor. Primary and secondary speed channels in the electronic control produce a DC voltage proportional to the frequency of the sensor outputs. Outputs of two speed channels are fed into high value gate number one, which rejects the lower of the two speed signals and passes on the higher valued speed signal. This redundancy permits one channel to fail without significantly affecting turbine speed. The secondary channel is adjusted for an output 1% less than the primary channel. It is therefore uncontrolling, except in case of failure of the primary channel or if a backup amplifier test switch in the cabinet is actuated which increases the output of the secondary channel to 1% greater than the primary channel. A test meter in the cabinet can be used to determine the controlling channel.

The speed signal from high value gate one is summed with the speed set point to obtain a speed error. (See zone B3 of the alignment diagram.) The speed error is passed through a stability compensation and error amplifier, which includes a regulation adjustment to obtain a valve lift demand signal. This is summed with a valve position signal to obtain a valve position error which is passed through the servo amplifier to the servo valve. Increasing the speed set point increases valve lift demand, valve position error, and valve opening. Speed will increase until the rise matches the set point increases.

The valve position signal is obtained from a Linear Variable Differential Transformer (LVDT) whose core is positioned by the control valve. Three kilohertz excitation for the LVDT is generated in the electronic cabinet. A demodulator in the cabinet produces a DC signal proportional to the AC output of the LVDT, which varies with core position.

A control valve closing bias is applied directly to the servo coil of the control valve in the tripped condition. Before the EHC can reset the feedpump turbine, a valve closing signal from the servo amplifier (2SA) must be present to operate the Close Signal Check (CSC) relay and remove the bias signal that had been previously applied directly to the servo valve coil. Relay circuitry also requires that the electrical connectors must be plugged into the servo valve, the LVDT, and the speed sensors before starting. The LVDT signal is biased to close the control valve if the LVDT connector is removed with the turbine running. The servo valve contains a 5% mechanical bias which will close the control valve in the absence of an electrical signal.

SPEED INDICATION

At zone B9 of the alignment diagram, a third magnetic sensor and circuitry identical to the previously described speed signal are used for speed indication. An analog signal is obtained for speed indication on the EHC remote control panel, local panel, and customer use.

## DESCRIPTION (continued)

SPEED INDICATION (cont'd)

A clamp signal from the 10% speed voltage comparator holds the turbine to less than 10% speed if both the primary and secondary speed channels show less than 10% speed.

At 105% speed, a voltage comparator operates a relay that unlocks the lock out valve in the emergency trip exerciser scheme. At 110% speed another voltage comparator operates the 110SX relay to indicate that 110% speed has been reached. A contact of this relay is provided for the customer.

SPEED REFERENCE OR SET POINT

Identical motor driven LVDT's are used to generate the manual and automatic speed references. The core position is varied by a two-phase stepping motor which moves in discrete steps and locks in position when not running. Raise and Lower pushbuttons on the operator's manual control panel cause the manual reference to run from the LSS stop to the HSS stop in 80 to 90 seconds. Raise Fast and Lower Fast pushbuttons will run the manual reference from stop to stop in approximately 10 seconds. The automatic speed reference circuits contain interface circuitry for compatibility with a specific boiler controller. The auto speed reference is adjusted to run from the LSS to the HSS in 10 seconds. The automatic and manual speed references are fed into a low value gate which rejects the higher reference and passes on the lower reference to control turbine speed. A relay operated from this gate indicates which of the two references is controlling.

An overspeed test can be performed after first running both manual and automatic references to the high speed stop. The Overspeed Test pushbutton is then actuated, causing an additional speed reference to be generated which increases linearly with time at approximately 12% speed per minute. This is added to both manual and automatic references. If turbine speed rises to 120% of rated, a speed signal will override the ramp generator and prevent further speed increase. Upon release of the Test pushbutton, the reference ramps back to normal at the same rate as it increased.

HIGH PRESSURE/LOW PRESSURE STOP VALVE INDICATION

The EHC cabinet contains circuitry for driving HPSV and LPSV position indicators located on the control panel supplied with the EHC cabinet. The signals for these position indicators are obtained from LVDT's mounted on their respective stop valves. This function is for position indication only and is not part of the control circuitry for the feedpump. (Refer to zone C7, C8 of the alignment drawing.)

## DESCRIPTION (continued)

LOW SPEED SWITCH

An electronic Low Speed Switch (LSS) is provided in the EHC cabinet to automatically, if selected, engage the turning gear on decreasing turbine speed less than 2 rpm.

If at turbine speeds greater than 200 rpm, pulses are lost from the LSPU shown at zone A9 of the alignment drawing, the LSS will prevent the LSSX relay from energizing and prevent automatic engagement of the turning gear by the EHC.

The LSS provides an alarm under the following conditions:

1. If the turbine is at standstill, 0 rpm, and then is increased in speed and pulses are not detected from the LSPU, the LSS 2K contacts will energize the LSS alarm X relay when 200 rpm is reached.

2. If the pick-up pulses are detected from the LSPU but lost before 200 rpm is reached, the LSS 2K contacts will energize the LSS alarm X relay when 200 rpm is reached.

3. If pick-up pulses are lost from the LSPU at turbine speeds greater than 200 rpm, the LSS 2K contacts immediately energize the LSS alarm X relay.

4. Once the LSS alarm X relay is energized, the automatic engagement of the turning gear is prevented until pulses from the LSPU are obtained.

POWER SUPPLIES

Two redundant power supplies, referred to as line A and line B, are provided, They are the full wave rectifier, capacitor filtered variety. Each receives its 120 VAC from the customer's power.

The 120 VAC is fed to a transformer that has three secondary windings. The output of each of the rectified, filtered supplies is connected to three identical series regulators that are adjusted for 30 VDC, 24 VDC and -22 VDC.

The outputs of the series regulators on the line A supply are adjusted slightly higher so they are carrying load and the line B supply is the backup power supply.

SERIES REGULATOR

Operation of the series regulators is described as follows. Output of the voltage sensing bridge consisting of 15R, 16R, 17R, 1P and 2ZD is fed to 7Q and 8Q. An increase in output voltage raises the voltage on 8Q, increasing its conduction and lowering conduction of 7Q. Decreased current in 7Q decreases current in 4Q, 5Q, and 6Q, dropping the output voltage until balance is restored. Excessive current output through 10R turns 3Q on, 2Q off, and 1Q on, grounding the base of 7Q. With 7Q off, series regulators 4Q, 5Q, and 6Q are off and the output goes to zero. When 1C discharges, 3Q is turned off and output is restored. The circuit will oscillate between off and on until the cause of the high current is removed.

## MONITORING

Located in the front of the cabinet are two meter panels that contain several alarms, meters, indicating lights and test meter that will allow checking or monitoring of the power supplies and most amplifiers in the cabinet.

POWER SUPPLY ALARM

For example, an alarm will occur if the 30 volt line A power supply should exceed the 95% or 105% set point of the meter relay in the cabinet. There will be three lights that will be illuminated. First, the power lamp on the control panel supplied with the cabinet will change from a steady glow to a flashing indication, the alarm lamp in the cabinet will be illuminated and the illuminated pushbutton above the line A 30 volt meter will indicate either a high or low voltage.

The three alarm lamps can be cleared after the malfunction is corrected by depressing the lighted pushbutton above the line A 30 volt meter. Identical functions are provided for both the line A and line B 30 V, 24 V, and -22 V power supplies and 3 KHz 6 V RMS power supply in the EHC cabinet. If any of these power supplies are not at their proper voltage level when the 120 V 60 Hz is applied, startup of the feedpump will be prevented.

GROUND ALARM (24 VDC)

If a ground were to occur on the positive or negative side of either the line A or line B 24 volt power supply, the power lamp on the control panel would flash, the +24 V GND/-24 V GND lamp in the cabinet would be illuminated, and the alarm lamp in the cabinet would be illuminated. These lamps will be cleared when the ground is removed. Depressing the +24, -24 V lamp in the cabinet is not required to clear the lamp.

HSS/LSS ALARM

HSS and LSS indication is provided both in the cabinet and on the control panel for both the manual and auto speed reference travel limits. These lamps will be cleared when the respective speed reference is driven off the limit.

CABLE CHECK ALARM

The cable check will prevent startup of the feedpump if the cables are not properly connected. The power lamp on the control panel will flash, the cable check lamp in the cabinet will be illuminated, and the alarm lamp in the cabinet will be illuminated until the cable is properly reconnected.

LAMP TEST

A lamp test can be performed on the indicating lamps in the cabinet by depressing the Alarm pushbutton. There aren't any lamps in the Backup Speed Amplifier Test pushbutton.

## MONITORING (continued)

LAMP TEST (cont'd)

Depressing the Power pushbutton on the control panel will test the lamps on the control panel. There aren't any lamps in the Lower Fast, Lower, Raise, Raise Fast pushbuttons.

TEST SWITCH

The amount of current being supplied by each series regulator can be observed by the test switch and meter in the cabinet. When measuring series regulator current, the meter is calibrated to read approximately 5 amps full scale.

The voltage output of amplifiers 1 thru 12 can be observed by using the test switch and meter. When measuring amplifier outputs, the meter is calibrated to read 10 volts full scale. The amplifier numbers and function are shown on the alignment drawing and elementary diagram.

## INSTALLATION AND ADJUSTMENT

The electronic control cabinet should preferably be mounted in a clean, dry, air-conditioned environment. Ambient temperature should not exceed 104°F unless otherwise noted on the outline drawing for a specific unit.

Equipment adjustment and the required test equipment are covered in detail in an alignment diagram and alignment instructions furnished with each unit.

The location of subassemblies within the cabinet is shown on the elementary diagram furnished with each control. Printed circuit board locations are shown on sheets 1 and 2 of the elementary diagram.

## OPERATION

Refer to the instruction book for the turbine.

## MAINTENANCE

No periodic maintenance is required. Outage time due to component failure is best minimized by maintaining a complete set of spare parts. Elementary diagrams of the printed circuit boards are included in this instruction book. The assembly drawings showing component location on the printed circuit boards are furnished with each order except where component identifications are marked on the boards.

## TROUBLESHOOTING

Most troubleshooting problems can be quickly isolated to a particular section of the control by use of the built-in test meter, the meter relays, and alarm lights. These, used in conjunction with the alignment diagram, indicate the operating condition of each section. An oscilloscope is recommended for pinpointing a defective component or components. Each printed circuit board can be removed from the rack for servicing without disconnection due to the use of terminal boards.

## RENEWAL PARTS

When ordering renewal parts, address the nearest Apparatus Sales Office of General Electric Company and supply the following information.

a. Catalog number stamped on the part and a complete description of the part, including its use and location.

b. Complete nameplate data appearing on the assembly of which the part is a component.

c. If possible, data on original order on which equipment was first supplied, including all numerical references.

Since operation of a steam turbine-generator is dependent upon operation of the feedpump control, it is suggested that an adequate stock of spare parts be carried on hand to minimize outage time due to component failure.

## METER RELAY LAMP REPLACEMENT

Type 195 Meter Relays have been made with two versions of lamps; one without leads attached to the lamp (GE Pt. No. 4868K32P001) and the other with leads attached to the lamp and lamp holder assembly (GE Pt. No. 4872K12G700). These replacement parts can be ordered directly from: the nearest General Electric Sales Office or Utility & Process Automation Products Department, General Electric Company, 40 Federal Street, West Lynn, Mass. 01910.

In the event of lamp failure, the applicable lamp or lamp assembly can be replaced when the control unit is detached from the rear of the indicator set point unit. Then, the lamp-access plate (see figure) must be removed (screws are beneath adhesive labels on some models) to expose the lamp holder which can be pulled out.

PT.NO. 4868K32P001 LAMP

When installing the bayonet-type Pt. No. 4868K32P001 (without leads), twist the lamp in the holder assembly until the lamp filament will be in the horizontal position when the lamp holder is replaced in the indicator set point unit.

PT. NO. 4872K12G700 LAMP & HOLDER

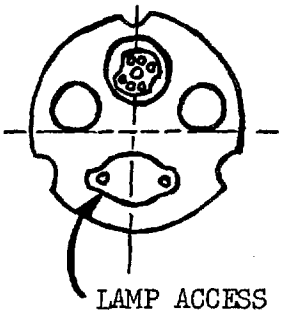
The Pt. No. 4872K12G700 lamp has been fixed in the lamp holder at the factory so the filament position will be in the correct position when the keyed assembly is inserted in the indicator set point unit. Make sure the lamp-lead clips are engaged by the screws when the lamp access cover is replaced.

NOTE: If relay operation is not within $\pm$ 2% of set point index, the lamp holder position must be rotated 180°.

## METER RELAY LAMP REPLACEMENT (continued)

REASSEMBLY

After the new lamp or lamp assembly has been installed, make sure the lamp access plate is replaced (and screws are tight) before control unit is reassembled on rear of indicator set point unit.




# DRAWING LIST

| DESCRIPTION | DRAWING NUMBER |
|---|---|
| Diode Board, 1DB | 44A300243 |
| Reference Low Value Gate, 2LVG | 44A300281 |
| Compensation and Regulation, 1CR | 44A301400 |
| Overspeed Integrator, 1OSI | 44A302692 |
| Signal Low Value Gate, 1LVG | 44A304763 |
| Alignment Instructions | 44A308807 |
| Series Regulator Board #1, 1REC | 44A306392 |
| Renewal Parts | 44A335636 |
| Cable Wiring Diagram, Signal & Low Voltage, 1CO | 44B300423 |
| Control Panel Outline | 44B309405 |
| Low Speed Switch Power Supply, LPS | 44B306144 |
| Cable Wiring Diagram Power, 2CO & 3CO | 44B306148 |
| Series Regulator Board #2, 1SR | 44B306197 |
| Detector Indicator, 1DI | 44C300609 |
| Operational Amplifier, 1OA | 44C300610 |
| Analog Output, 1AO2 | 44C301216 |
| Analog Output, 3AO2 | 44C301622 |
| Frequency to Voltage Converter, 2FVC | 44C301682 |
| Clock and Stepping Motor Drive, CL&SMD | 44C303567 |
| Servo Amplifier, 2SA | 44C304444 |
| 3kHz Oscillator, 7OSC | 44C321478 |
| Signal Converter, 2SC | 44C301648 |
| Voltage Comparator, 1VC | 44C305259 |
| Elementary Drawing, Master | 44C309397 |
| Connection Drawing | 44C309398 |
| Low Speed Switch, LSS | 44C306444 |
| Line A & Line B Power Supply | 44C306526 |
| Cabinet Outline | 44D300043 |
| Alignment Drawing | 44D308148 |
| Control Panel Connection Diagram | 44D309933 |
| Relay Panel Connection Diagram | 44D308137 |
| Summary | 44X308922 |

GENERAL ELECTRIC

44A300243
CONT ON SHEET SH NO.

REV NO. 0

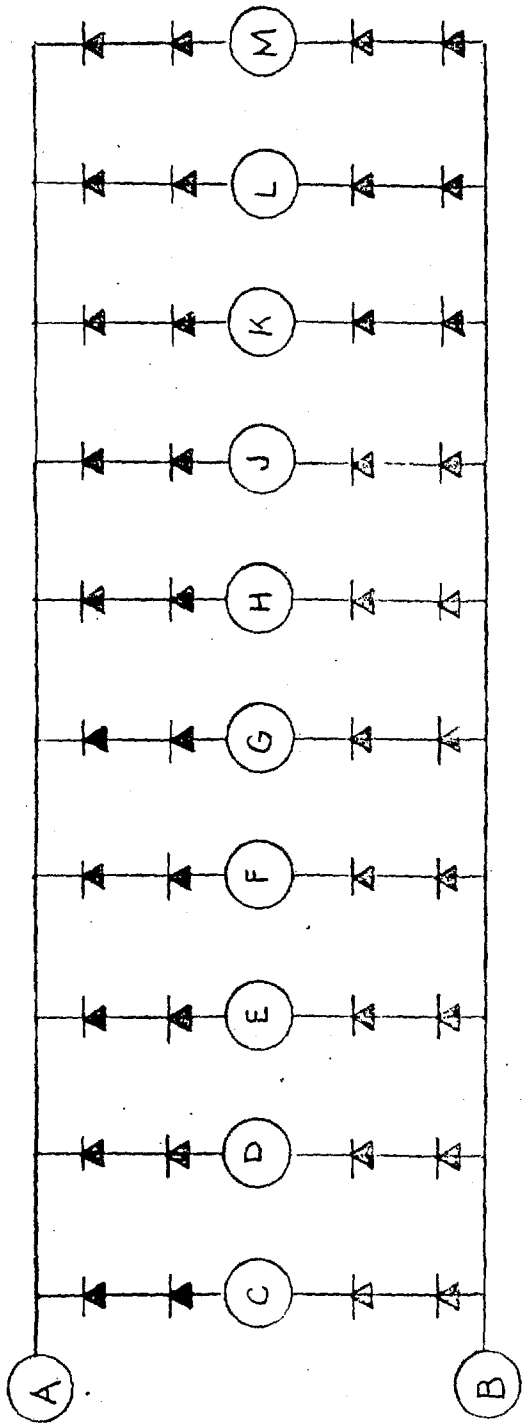
TITLE

# DIODE BOARD IDB

CONT ON SHEET SH NO.

FIRST MADE FOR 44D236370-G01

REVISIO




ALL DIODES 200 MA; 200 V.

11

DFB 12

PRINTS

MADE BY R. Chambers 1/15/68

ISSUED MAR 12 1968

APPROVALS MBC

SPECIALTY CONTROL DIV OR DEPT.

WAYNESBORO, VA. LOCATION

44A300243

CONT ON SHEET SH NO.

CODE IDENT

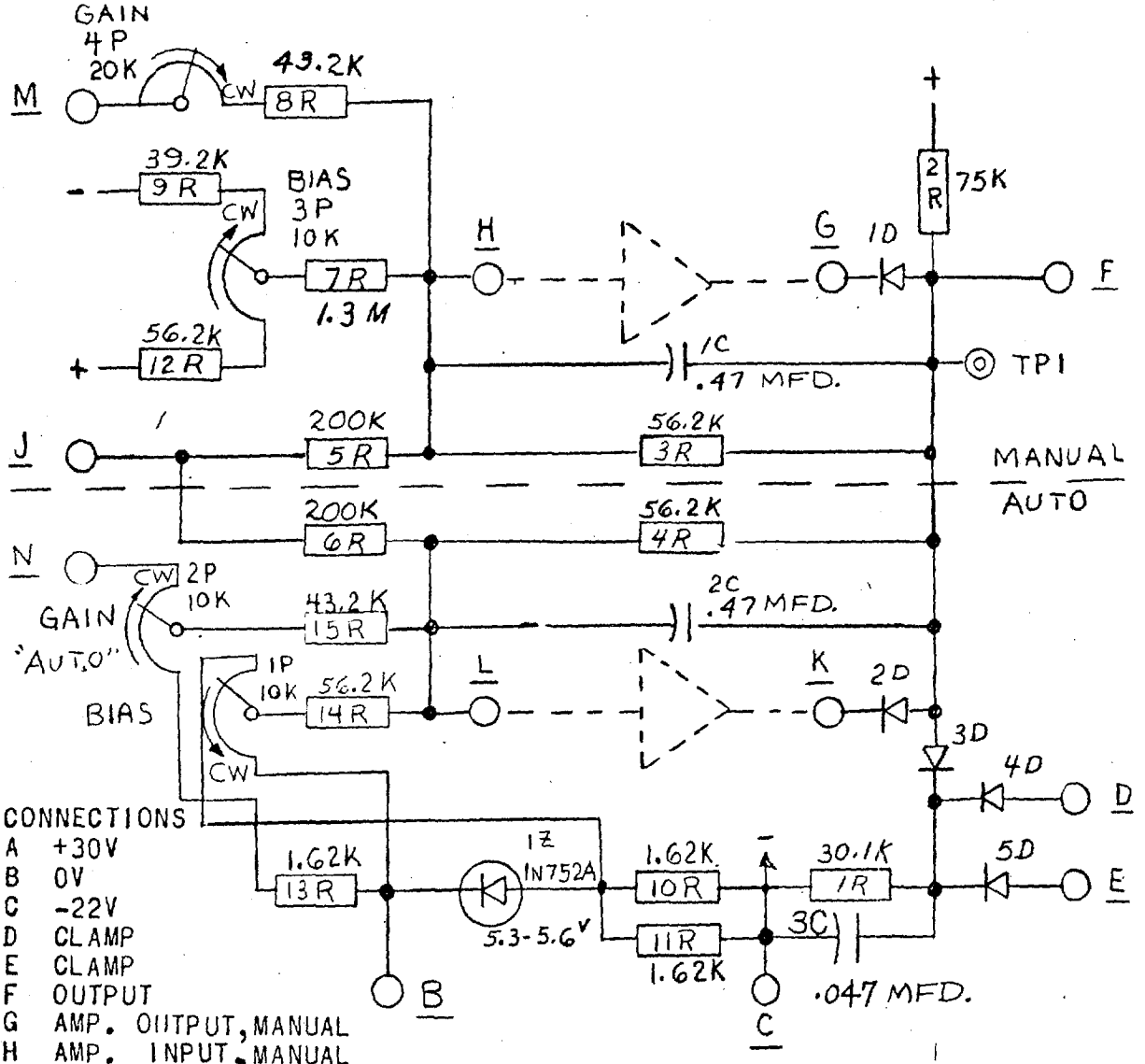
FF-803-WF (2-67)

GENERAL ELECTRIC | 44A300281

REV NO. (1) C. KENNEDY | TITLE | CONT ON SHEET | SH NO.

REFERENCE LOW VALUE GATE (2LVG)

CONT ON SHEET | SH NO. | FIRST MADE FOR 44D236378-G01




CLAMP NOTE: ZERO VOLTS ON D OR E WILL HOLD OUTPUT (F) AT ZERO.

ADJUSTMENT:

1: WITH AUTO INPUT AT HSS AND MANUAL AT LSS, ADJUST BIAS FOR ZERO OUTPUT AT F.

2: WITH BOTH GAINS MAX. (CW) AND INPUTS AT HSS, ADJUST MANUAL GAIN FOR 5 VOLTS OUTPUT.

3: DECREASE AUTO GAIN UNTIL OUTPUT OF THE ASSOCIATED AMPLIFIER SWITCHES NEGATIVE TO POSITIVE.

4: AUTO BIAS IS SET CCW INITIALLY, FINAL SETTING PER APPLICATION REQUIREMENT.

REVISION

| | | |
|---|---|---|
| 1 | CHANGED SR, GR PR VALUES BUKOSKEY | MAY 5 1969 |
| 2 | E.A. Lee | OCT 19 1970 |
| 3 | R. Adams | NOV 17 1971 |

11

12P DFB

FG-W

PRINTS TO

MADE BY K Chambers Mod 11 1968 | APPROVALS R W Simmons | SPECIALTY CONTROL DIV OR DEPT. | 44A300281

ISSUED APR 17 1968 | WAYNESBORO, VA LOCATION | CONT ON SHEET | SH NO.

FF-803-WF (9-66) PRINTED IN U.S.A. | CODE IDENT NO

GENERAL ELECTRIC

44A301400 CONT ON SHEET SH NO.

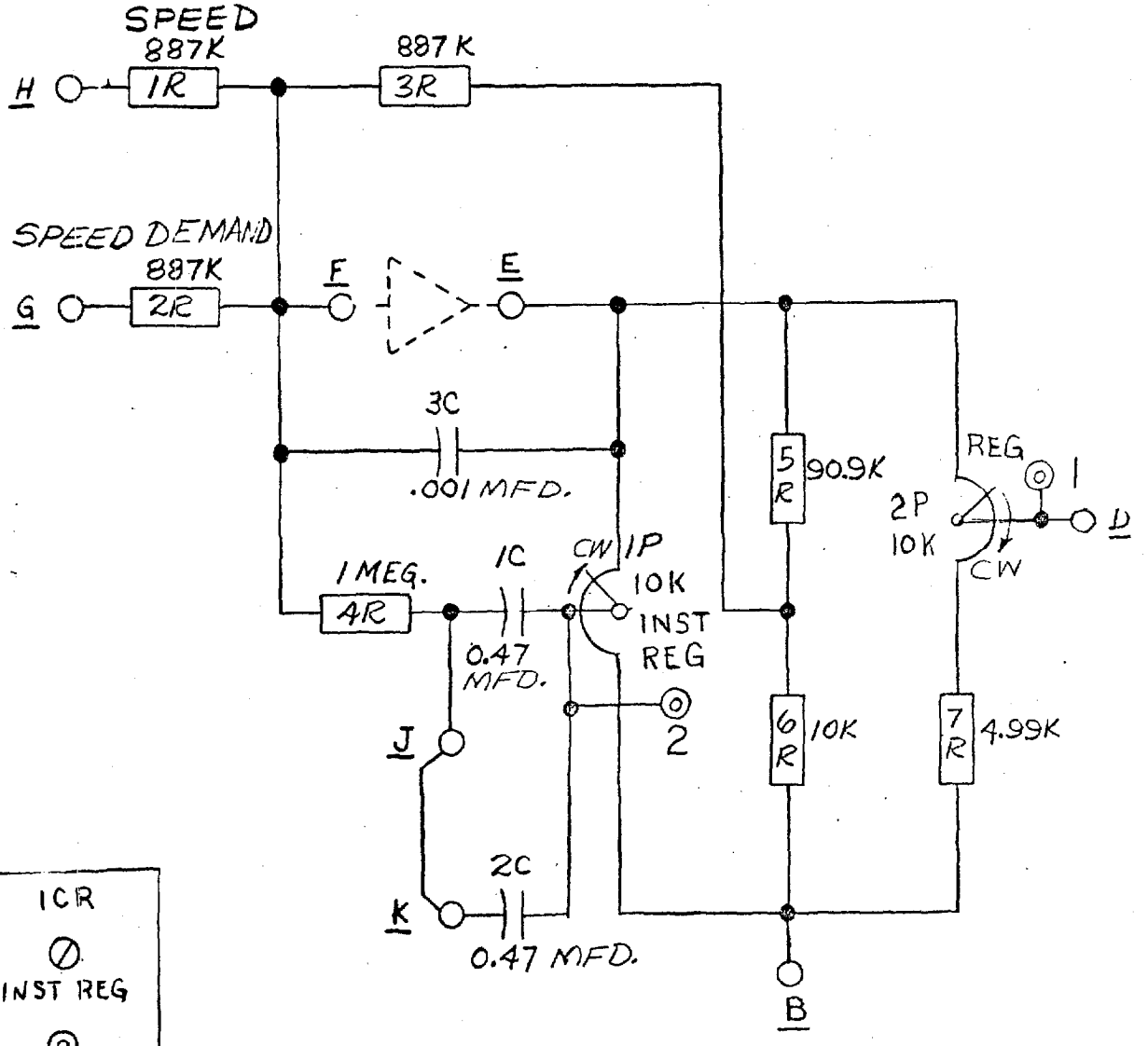
REV NO.

44A301400 CONT ON SHEET SH NO.

TITLE

COMPENSATION & REGULATION (ICR)

FIRST MADE FOR 44D236388-G01




ICR
INST REG
REG

REVISION

| | | |
|---|---|---|
| 1 | R.M. BUKOSKEY APR 1 1969 | |
| 2 | Add Fig. Sept. 21, 1971 | |
| 3 | Nov. 17, 1971 | |
| 4 | Aug. 28, 1972 | |

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PRINTS TO

MADE BY R.M. Bukoskey
ISSUED FEB 18 1969

APPROVALS R.W. Simmons

SPECIALTY CONTROL DIV OR DEPT.
WAYNESBORO, VA. LOCATION

44A301400 CONT ON SHEET SH NO.

CODE IDENT NO

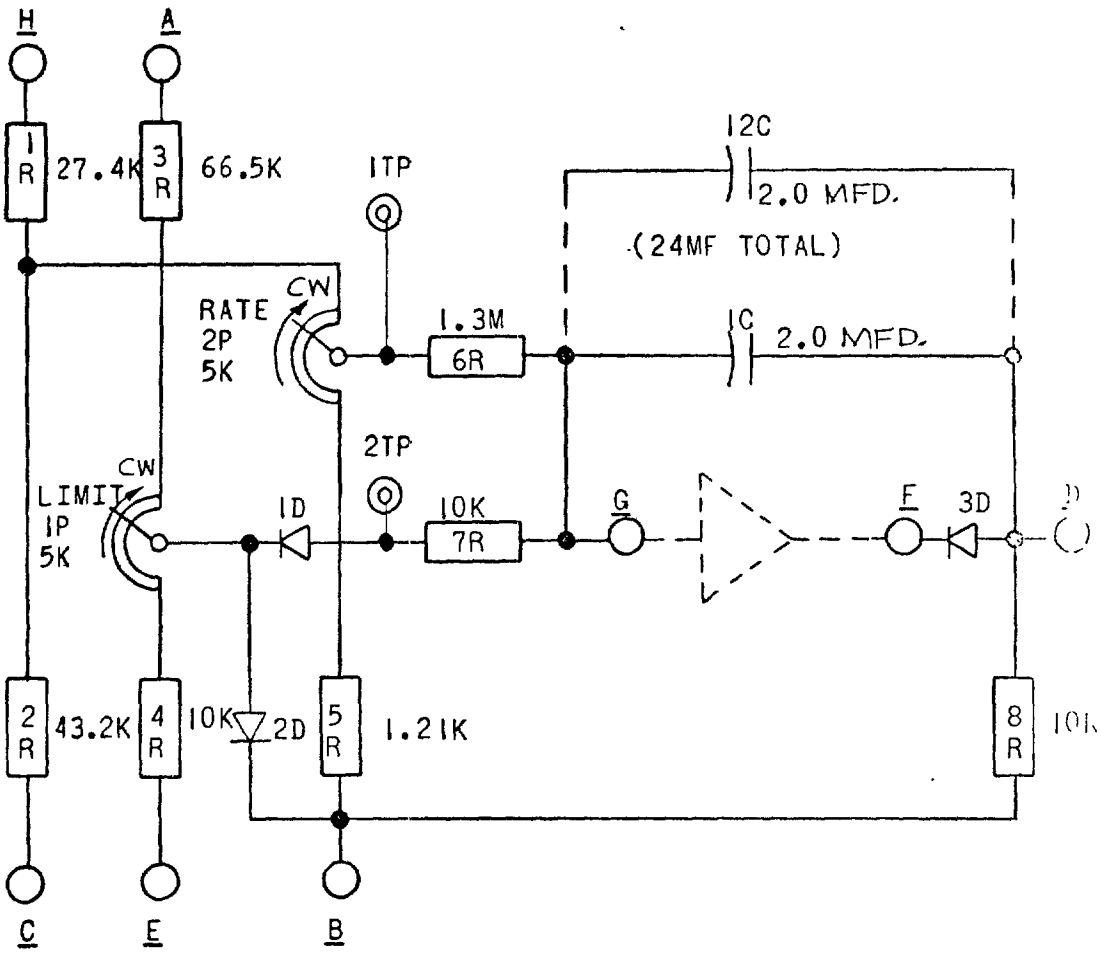
FF-803-WP (9-67)

GENERAL ELECTRIC

44A302692

| REV NO | TITLE | CONT ON SHEET | SH NO |
|---|---|---|---|
| ① C KENNEDY | OVERSPEED INTEGRATOR | | 1ØSI |
| 44A302692 CONT ON SHEET SH NO. | FIRST MADE FOR 44D236373-G02 | | |

REVISION




CONNECTIONS

A + 30V
B 0V
C -22V
D OUTPUT
E SPEED SIGNAL
F AMPLIFIER OUTPUT
G AMPLIFIER INPUT
H POS. SUPPLY THROUGH RELAY
J N.C.

| MADE BY | APPROVALS | | | |
|---|---|---|---|---|
| H. Templeton 3-16-70 | R. Culver | Control Gear Devices DIV OR DEPT | 44A302692 | |
| ISSUED APR 8 1970 | | Waynesboro, Va. LOCATION | CONT ON SHEET | SH NO |

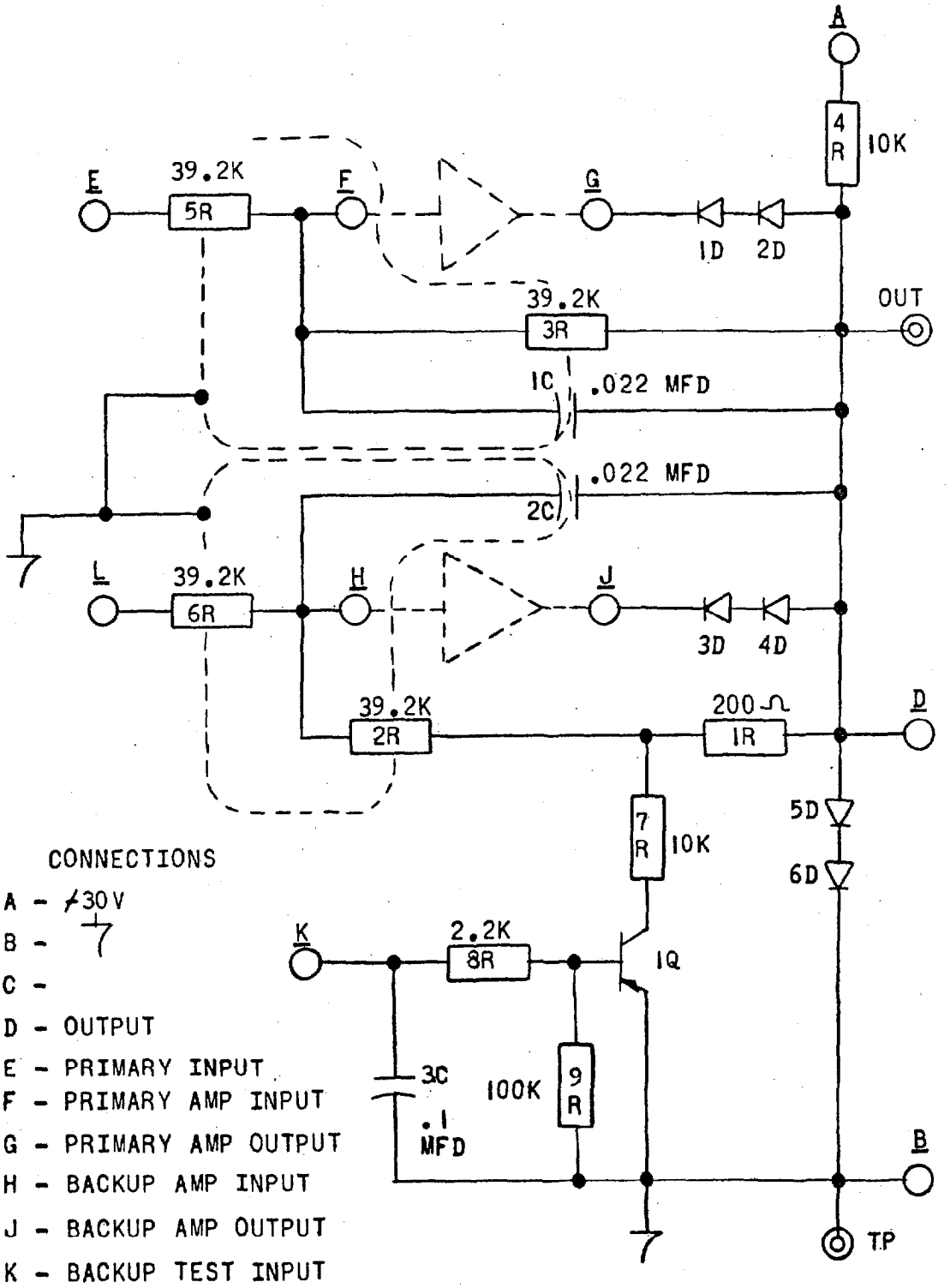
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2-11

GENERAL ELECTRIC

| REV NO. | TITLE | 44A304763 CONT ON SHEET SH NO. |
|---|---|---|
| 44A304763 CONT ON SHEET SH NO. | SIGNAL LOW VALUE GATE | ILVG |
| | FIRST MADE FOR 44D236372-G02 | |

REVISIONS




CONNECTIONS

A - +30V

B - ⏚

C -

D - OUTPUT

E - PRIMARY INPUT

F - PRIMARY AMP INPUT

G - PRIMARY AMP OUTPUT

H - BACKUP AMP INPUT

J - BACKUP AMP OUTPUT

K - BACKUP TEST INPUT

L - BACKUP INPUT

11
12P DFE
FG-W

PRINTS TO

| MADE BY | APPROVALS | | |
|---|---|---|---|
| H. Eagleton 8-13-71 | R. Culver | COMM. & CONT. DEVICES DIV OR DEPT | 44A304763 CONT ON SHEET SH NO. |
| ISSUED Aug. 16, 1971 | D. L. Mack | WAYNESBORO, VA. LOCATION | |

FF-803-WF (2-62)
PRINTED IN U.S.A.

TITLE LOWER COLORADO RIVER AUTHORITY
ALIGNMENT INSTRUCTIONS
ELECTRONIC FEED PUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A17 TURBINE 153090 & 153091

REVISIONS

REFERENCE ALIGNMENT DIAGRAM 44D308148

TEST EQUIPMENT

1. AUDIO OSCILLATOR: HEWLETT PACKARD 201C, 200CD; GENERAL RADIO 1210C, OR EQUIVALENT. OUTPUT MUST BE 10 VOLTS RMS MINIMUM INTO 600 OHMS.
2. AC/DC DIGITAL VOLTMETER: 0.1 MV TO 100 V MAX.
3. DIGITAL FREQUENCY COUNTER: 6 VOLT RMS SENSITIVITY. 30 TO 5000 HZ.
4. TEN TURN POTENTIOMETER; NOT MORE THAN 5000 OHMS RESISTANCE.
5. TWO INPUT OSCILLOSCOPE, CALIBRATED SWEEP: X1 AND X10 PROBES.
6. OPTIONAL: BRUSH RECORDER SPECIAL TESTS ARE DESIRED.
7. VOLT - OHM - MILLIAMMETER

I. PRELIMINARY

1. CHECK INTERCONNECTION OF CONTROL CABINET, OPERATOR'S CONTROL DEVICES, TURBINE DEVICES, ETC.
2. MECHANICALLY ZERO TEST METER IN CABINET AND OPERATOR'S TACHOMETER AND OTHER INSTRUMENTS.

II. LINE "A" AND LINE "B" POWER SUPPLIES LOC. 12 AND 13

1. SET UP CONNECT 120V, 60 HZ TO LINE "A" POWER SUPPLY TERMINALS T24 AND T25. REMOVE POWER FROM LINE "B" TERMINAL T22 & 23.
2. DC SERIES REGULATORS, ADJUST POWER SUPPLIES AND METER CALIBRATIONS PER CHART. IF THE VOLTAGES LISTED CANNOT BE OBTAINED IT MAY BE NECESSARY TO ADJUST THE TAPS ON TRANSFORMER 1T, LOC. 12 & 13.

| LINE "A" SUPPLY | TEST POINTS | ADJUST |
|---|---|---|
| +30V | BUS BARS | 0308 |
| -22V | BUS BARS | 0316 |
| 24V | +F LOC. 0322; -G, LOC. 0324 | 0324 |

711
12P-DFB2
FG-W
12P-FS2

PRINTS TO

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3. METER RELAY ALARMS VARY ALARM SET POINT OF LINE "A" POWER SUPPLY METER RELAYS TO CHECK ALARM OPERATION. SET ALARMS FOR 95% AND 105%.

4. CONNECT POWER TO LINE "B" POWER SUPPLY BY JUMPERING TERMINAL T24 TO T22 AND T25 TO T23, SO THAT POWER IS APPLIED TO BOTH SUPPLIES.

5. ADJUST THE INDIVIDUAL LINE "B" SERIES REGULATOR BOARDS SO THAT THEY ARE CARRYING ONLY 0.1 AMP AND THE LINE "A" SUPPLIES THE REMAINING LOAD. WHILE NOTING THIS CONDITION IS MAINTAINED MAKE FINAL ADJUSTMENTS IF NECESSARY PER CHART.

| SUPPLY | LINE B | LINE A | TESTPOINTS |
|---|---|---|---|
| +30 | 0310 | 0308 | BUS BAR |
| -22 | 0318 | 0316 | BUS BAR |
| +24 | 0326 | 0324 | LINE A, + F, 0322; -G, 0324<br>LINE B, + F, 0328; -G, 0326 |

6. TEST AND SET LINE "B" METER RELAY ALARMS AT 95% AND 105%

III. 3KHZ OSCILLATOR, -0217

1. VERIFY OUTPUT FREQUENCY AT 0217 IS 3000HZ. ± 300 HZ. USING COUNTER AT TERMINALS L&M.

2. ADJUST VOLTAGE LEVEL FOR 6.000 ± .006V. SET METER FOR 100%. CHECK METER RELAY ALARMS AND SET AT 95 AND 105%.

3. OFFSET ADJUSTMENT, USING DIGITAL VOLTMETER ADJUST OFFSET FOR ZERO VOLTS. REMOVE OSCILLATOR BOARD FROM RACK FOR OFFSET ADJUSTMENT.

IV. SSPU SPEED INDICATION CIRCUITS

1. SET UP, REMOVE LEAD FROM T28 AND T29 AND CONNECT AUDIO OSCILLATOR. SET OSCILLATOR TO SSPU RATED HZ.

1
12P
FG
12P

PRIN

2. FREQUENCY TO VOLTAGE CONVERTER

2FVC LOC. 0224. ADJUST COARSE AND FINE POTS TO OBTAIN ⁄5.00V AT TERMINAL J.

3. VOLTAGE COMPARATORS IVC LOC., 0218, 0219, 0220, 0221. SET EACH VOLTAGE COMPARATOR (IVC) PER CHART ON ALIGNMENT DRAWING. RED LIGHT ON IVC SHOULD BE ON WHEN RELAY ENERGIZED. CHECK EACH IVC SETTING BY SLOWLY VARYING FREQUENCY ABOUT THE OPERATING POINT.

4. TACHOMETER 3AØ2 LOC. 0108 WITH OSCILLATOR AT ZERO FREQUENCY ADJUST 2A BIAS FOR ZERO INDICATION ON REMOTE CONTROL PANEL. SET OSCILLATOR TO RATED SSPU FREQUENCY AND ADJUST GAIN AND I OUT AS REQUIRED FOR CORRECT INDICATION OF TURBINE RPM.

4.1 REPEAT ZERO FREQUENCY AND RATED SPEED SETTINGS AND MAKE FINAL ADJUSTMENTS IF REQUIRED.

5. SECOND TACHOMETER 3AØ2 LOC. 0110 WHERE REQUIRED ADJUST 1A & 2A USING PROCEDURE IN STEPS 4 AND 4.1 ABOVE.

6. REMOVE OSCILLATOR FROM T28 AND T29 AND RECONNECT INCOMING LEADS.

V. SSPU SPEED MEASURING CIRCUITS

1. SET-UP JUMPER T4 TO T7 AND T5 TO T8. CONNECT OSCILLATOR AND COUNTER TO T4 AND T5.

2. FREQUENCY TO VOLTAGE CONVERTERS 2FVC, LOC. 0203 AND 0206 HIGH SPEED GATE ILVG, LOC. 0209.

2.1 PRIMARY AT RATED SHAFT SPEED PICKUP FREQUENCY, ADJUST 2FVC LOC. 0203 FOR -5.000 VOLTS AT OUTPUT TESTPOINT OF ILVG, 0209. (OBSERVE WITH TEST METER IN POSITIONS 6 AND 7 THAT AMPLIFIER 7 IS MORE POSITIVE THAN 6; IF NECESSARY ADJUST 2FVC, 0206, TO OBTAIN THIS CONDITION ).

2.2 BACKUP DEPRESS BACKUP SPEED AMPLIFIER TEST SWITCH AND ADJUST 2FVC, 0206, FOR -5.05 VOLTS AT OUTPUT TP, 0209.

3. REMOVE TERMINAL POINT JUMPERS AND OSCILLATOR AND COUNTER LEADS.

VI. CONTROL VALVE RAM SERVO LOOP

1. CHECK LVDT CORE PER LEVER DIAGRAM.

7/11

12P-DFB²

FG-W

12P-FS²

PRINTS TO

2. DETECTOR INDICATOR IDI, LOC. 0216. WITH CONTROL VALVE FULL OPEN ADJUST NULL OF IDI FOR 0.00 ± 0.01 VOLTS AT OUT TP. NOTE VOLTAGE READING ON ALIGNMENT DIAGRAM. WITH VALVE IN CLOSED OVERTRAVEL POSITION RECORD OUT TP VOLTAGE ON ALIGN. DIAG. ADJUST ZERO FOR 0 INDICATION OF CONTROL VALVE POSITION ON OPERATOR'S CONTROL CONSOLE. RETURN VALVE TO FULL OPEN POSITION AND ADJUST CALIBRATE FOR 100% POSITION INDICATION.

3. DETERMINE GAIN OF DETECTOR CIRCUIT IN VOLTS PER INCH BY DIVIDING THE DIFFERENCE IN OUT TP VOLTS AT EXTREME VALVE POSITIONS BY THE TOTAL VALVE TRAVEL IN INCHES, AND RECORD THIS FIGURE ON THE ALIGNMENT DIAGRAM.

3.1 ANALOG CONTROL VALVE POSITION SIGNAL

1AØ2 LOC. 0106. WITH VALVE CLOSED ADJUST 1A BIAS FOR ZERO POSITION ON OPERATORS METER.

3.2 WITH VALVE OPEN ADJUST 1A GAIN AND I OUT FOR FULL OPEN INDICATION.

3.3 REPEAT 3.1 AND 3.3 AS NECESSARY.

4. MAKE ELECTRICAL CONNECTIONS TO SERVO VALVE.

5. SERVO AMPLIFIER, 2SA, LOC. 0214.

5.1 PS10 TRIP CHECK FOR APPROX. -0.7 VOLTS AT "I" TP UNDER TRIP CONDITION.

5.2 CLOSE SIGNAL CHECK WITH TURBINE TRIPPED, DISCONNECT DEMAND TERMINAL E AND LIFT TERMINAL D OF 2SA. CONNECT TEST POT WITH NEGATIVE OUTPUT TO TERMINAL D AND SET POT FOR ABOUT -3 VOLTS AT LIFT TP. THE OUTPUT OF AMPLIFIER 9 ON TEST METER POSITION 9 SHOULD BE ABOUT -10 VOLTS. TURN BIAS ADJUSTMENT CCW UNTIL AMP. 9 GOES TO ≠10 VOLTS. THERE SHOULD BE LITTLE OR NO CONTROL OF 9 AT INTERMEDIATE VOLTAGES. LEAVE 9 AT ≠10 VOLTS AND RESET TURBINE. SLOWLY REDUCE TOWARDS 0 THE TEST POT SETTING AND NOTE THAT 9 SWINGS FROM ≠10 TO A MINIMUM VALUE OF APPROX. -6.2 VOLTS. NOW APPLY MORE NEGATIVE VOLTAGE FROM THE TEST POT AND NOTE THAT 9 HAS A MAXIMUM VALUE OF APPROX. ≠6.2 VOLTS.

5.3 GAIN REFER TO ALIGNMENT DRAWING TO OBTAIN LIFT TEST POINT VOLTAGE REQUIRED TO PRODUCE 1 VOLT AT I TEST POINT, WHEN GAIN IS PROPERLY ADJUSTED.

5.31 VARY THE TEST POT SETTING UNTIL I TEST POINT IS 0.000 VOLTS AND NOTE LIFT TEST POINT VOLTAGE. CHANGE THIS VOLTAGE WITH THE TEST POT BY PLUS LIFT TEST POINT VOLTAGE FROM STEP 5.3, ADJUST GAIN UNTIL I TEST POINT READS MINUS 1.000 VOLTS.

5.32 RECHECK ZERO POINT BY RETURNING TEST POT SO THAT ZERO VOLTS IS ONCE AGAIN AT I TEST POINT AND TEST POT VOLTAGE IS THE SAME AS NOTED IN 5.31.

5.33 ADJUST TEST POT FOR MINUS LIFT TEST POINT VOLTAGE AND NOTE THAT I TEST POINT IS PLUS 1.000 VOLT.

5.4 BIAS DISCONNECT TEST POT AND RECONNECT THE LEAD REMOVED FROM D. WITH THE VALVE RAM AT ITS INTERNAL STOP AND ZERO VOLTS A DEMAND TEST POINT SET BIAS FOR -0.35 ± .01V AT I TEST POINT. RECONNECT TERMINAL E.

6. AFTER REMOVING THE ELECTRICAL CONNECTOR, MOUNT THE SERVO VALVE AND VERIFY THAT THE MECHANICAL BIAS IS IN THE VALVE CLOSING DIRECTION.

## VII. REGULATION AND COMPENSATION ICR, LOC. 0212

1. REGULATION CONNECT THE OUTPUT OF THE TEST POT TO SUPPLY A NEGATIVE VOLTAGE TO TERMINAL E. ADJUST REGULATION TO OBTAIN VALVE TRAVEL FROM NO LOAD LIFT TO FULL LOAD LIFT FOR 3.00 VOLTS CHANGE IN THE OUTPUT OF THE TEST POT.

2. COMPENSATION VERIFY, THAT JUMPER J-K IS NOT CONNECTED.

3. INSTANTANEOUS REGULATION SET TEST POT FOR -1 VOLT OUTPUT AND ADJUST INST. REG. FOR VOLTAGE EQUAL TO 0.0 VOLTS AT INST REG. T.P.

711

12P-DFB²

FG-W

12P-FS²

VIII. MANUAL SPEED REFERENCE (STAND-STILL CHECKS)

1. THIS REFERENCE IS OBTAINED FROM THE UPPER OF THE TWO MOTOR DRIVEN LVDT'S. WHEN POWER IS APPLIED IT SHOULD RUN TO LSS AND LIGHT THE LSS INDICATING LIGHT.

2. SLOW REFERENCE SPEED, CL & SMD, LOC. 0118. SHUT OFF HIGH PRESSURE OIL SUPPLY OR DISCONNECT SERVO VALVE BEFORE RAISING MANUAL REFERENCE. ACTUATE RAISE PUSHBUTTON AND TIME TRAVEL TO HSS LIGHT INDICATION. TIME SHOULD BE 80-90 SECONDS.

3. FAST REFERENCE SPEED, CL & SMD, LOC. 0118. ACTUATE LOWER FAST PUSHBUTTON AND TIME MOTOR TRAVEL FROM HSS LIGHT TO LSS LIGHT. IT SHOULD BE 8 TO 10 SECONDS.

4. BIAS AND GAIN, MANUAL REFERENCE, 2LVG, LOC. 0116. DISCONNECT K OF 2LVG. RAISE MANUAL REFERENCE OFF LSS AND BACK INTO IT AT SLOW RATE WITH LOWER PUSHBUTTON. ADJUST MANUAL BIAS FOR 0.000 VOLTS AT OUTPUT TP, 0116. RAISE TO HSS AT SLOW RATE AND ADJUST MANUAL GAIN FOR OUTPUT VOLTAGE AT TP 0116 AS SPECIFIED BY ALIGNMENT DIAGRAM.

711
12P-DFB 2
FG-W
12P-FS 2

PRINTS TO

TITLE LOWER COLORADO RIVER AUTHORITY
ALIGNMENT INSTURCTIONS
ELECTRONIC FEED PUMP TURBINE CONTROL

CONT ON SHEET SH NO.

FIRST MADE FOR 3S7513TC300A17 TURBINE 153090 & 153091

REVISIONS

1X. AUTO SPEED REFERENCE (STAND-STILL CHECK)

1. AUTO SPEED REFERENCE TRAVEL ADJUST LOC. 0108,0125
REMOVE INPUT LEADS FROM FEEDWATER CONTROLLER.
JUMPER T33 TO T34 AND AUTO SPEED REFERENCE TO LSS. ADJUST 1A BIAS OF 3A02 LOC. 0108 FOR ZERO AT T35 & T36.
WITH AUTO SPEED REFERENCE STILL AT LSS, CHECK TRAVEL TIME TO HSS BY JUMPERING T32 TO T34. TRAVEL SHOULD BE 40 SECONDS AND CAN BE ADJUSTED BY USING FREQUENCY POTENTIOMETER ON 3SC LOC. 0125.
WITH AUTO SPEED REFERENCE AT HSS, ADJUST 1A GAIN OF 3A02 LOC. 0108 TO 1-MA AT T35-T36.

2. AUTO SPEED REFERENCE STOPS LSS 2LVG LOC. 0116

DISCONNECT G AT 0113 AND WITH AUTO SPEED REFERENCE AT LSS ADJUST SC BIAS, 0113 FOR OUTPUT TEST POINT VOLTS 0116 PER ALIGNMENT DIAGRAM.

HSS 2LVG, 0116. WITH AUTO SPEED REFERENCE AT HSS, ADJUST SC GAIN 0113 FOR OUTPUT TEST POINT VOLTS 0116 PER ALIGNMENT DIAGRAM. RE-CONNECT G.

3. LOCAL/REMOTE CONTROL LIGHT, 1VC, LOC. 0117. ADJUST REFERENCE, 0117 FOR 6.5 VOLTS AT L TP. CHECK TAHT LOCAL CONTROL INDICATION IS OBTAINED WHEN AUTO REFERENCE IS AT LSS AND MANUAL REFERENCE IS LOWERED SO THAT 2LVG OUTPUT IS LESS THAN APPROXIMATELY 1 VOLT.

4. AUTO SPEED REFERENCE INDICATION, 1DI LOC. 0128.
ADJUST ZERO AT LSS AND CAL AT HSS TO OBTAIN 0 AND 100 INDICATIONS ON OPERATORS POSITION INDICATOR.

X. OVERSPEED TEST INTEGRATOR, 10SI, LOC. 0112

1. RATE CONNECT VOLTMETER TO OUTPUT TEST POINT 2LVG, 0116. WITH MANUAL AND AUTO REFERENCE AT HSS JUMPER T54 TO T58. VOLTMETER READING SHOULD INCREASE 0.6 ± 0.1 VOLTS PER MINUTE FOR A (12% ± 2%) SPEED INCREASE PER MINUTE. ADJUST RATE, 0112 AS REQUIRED.

2. LIMIT CONNECT OSCILLATOR AND COUNTER TO T4 AND T5. SET FREQUENCY FOR 120% SSPU FREQUENCY. OBSERVE OUTPUT OF AMPLIFIER 5 WITH TEST METER. JUMPER T54 TO T58 AND ADJUST LIMIT, 0112 FOR POINT AT WHICH AMPLIFIER SWITCHES FROM POSITIVE SATURATION TO ZERO VOLTS. CHECK BY REDUCING OSCILLATOR FREQUENCY AND SLOWLY INCREASING IT TO THE SWITCHING POINT FREQUENCY.

TIL
12P-DEB
FB-W
12P-FS

PRINTS TO

XI. LP STOP VALVE POSITION INDICATOR IDI, LOC. 0102. WITH STOP VALVE FULLY CLOSED ADJUST DIFF. TRANS. NULL FOR -3 VOLTS AT OUT TP AND METER ZERO FOR 0 POSITION. ADJUST CAL. FOR 100% WITH VALVE FULL OPEN.

XII. HP STOP VALVE POSITION INDICATOR IDI, LOC. 0104. SAME AS XI.

XIII. LOW SPEED SWITCH POWER SUPPLY, LPS, LOC 0227

1. CHECK FOR 5.6 VDC AT TERMINAL D.

2. CHECK FOR 12VDC AT TERMINAL E.

XIV. LOW SPEED SWITCH LSS LOC. 0225

1. STANDSTILL CHECKS - CHECK THAT TERMINALS F&G ARE +5.6VDC WITH RESPECT TO B. THIS WILL CONFIRM ITDX & 200RPM RELAYS ARE ENERGIZED.
2. RELAY CONTACTS AT TERMINAL D AND E SHOULD BE CLOSED.
3. ITP SHOULD BE 5.6 VDC.
4. REMOVE WIRE FROM F AND JUMPER F TO B. RELAY CONTACTS AT H & K CLOSE. REMOVE JUMPER AND CONNECT WIRE TO F.
5. OPERATE TURNING GEAR TO OPEN CONTACTS AT H & K.

OPERATIONAL CHECKS

1. CHECK FOR 0-5 VOLT PULSE AT TERMINAL N WHEN TURNING GEAR OPERATED.
2. NEGATIVE PULSES SHOULD BE PRESENT AT 2TP WHEN TURNING GEAR OPERATED.
3. ITP SHOULD BE ZERO VOLT WHEN TURNING GEAR OPERATED.
4. RELAY CONTACTS AT D & E SHOULD BE OPEN WHEN TURNING GEAR OPERATED.
5. WITH TURNING GEAR OFF AND TURBINE AT STANDSTILL RELAY CONTACTS AT D & E SHOULD BE CLOSED.

XV. SEE LEVER DIAGRAM FOR RUNNING CHECKS OF SPEED REGULATION.

711
12P-DFB2
FG-W
12P-FS2

PRINTS TO

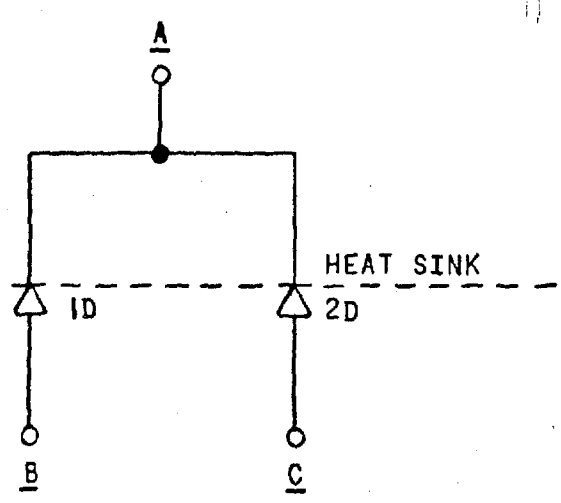
GENERAL ELECTRIC

44A306392

CONT ON SHEET SH NO.

| REV NO. | TITLE |
|---|---|
| 44A306392 | RECTIFIER 1 REC |
| CONT ON SHEET SH NO. | FIRST MADE FOR 44B317827-G01 |

REVISIONS




11

FG-W

12P-DE

PRINTS T

| MADE BY | APPROVALS | | |
|---|---|---|---|
| Eleanor Ellison 4/24/73 | R.G.S.D. Fleet | DRIVE SYSTEMS PROD. DEPT OR DEPT. WAYNESBORO, VA. LOCATION | 44A306392 CONT ON SHEET SH NO. |
| ISSUED D. Moyer April 26, 1973 | R.J. Culver | | |

FF-803-WF (8-71) PRINTED IN U.S.A.

CODE IDENT NO.

3

3S7513TC300A17 EHC BOILER FEEDPMP

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 3S7513KF300A5 | LINE POWER PANEL |
| 1 | 3S7513KF306A9 | METER PANEL |
| 1 | 3S7513KF306A11 | METER PANEL ASM |
| 1 | 3S7513KF307A16 | RELAY PANEL |
| 5 | 44B331761-G01 | PWB VOLT COMPARATR |
| 1 | 44D236389-G02 | PWB SERVO AMPLI |
| 5 | 44D381585-G01 | PWB DETECTOR IND |
| 10 | 44D381589-G01 | PWB OPER AMPLIFIER |
| 1 | 44B371594-G01 | PWB 3KHZ OSC |
| 1 | 44D236379-G01 | PWB ANALOG OUTPUT |
| 2 | 44D236379-G03 | PWB ANALOG OUTPUT |
| 2 | 44D331918-G01 | PWB CL MOTOR DRIVE |
| 3 | 44D236365-G02 | PWB 2FREQ VOLT CNV |
| 1 | 44D236367-G02 | SIGNAL CONVERTER |
| 3 | 44D236370-G01 | PWB DIODE BOARD |
| 6 | 44B317827-G01 | PWB SERIES REG 1 |
| 6 | 44D236371-G01 | PWB SERIES REG 2 |
| 1 | 44D236373-G02 | PWB OVERSPD INTEG |
| 1 | 44D236372-G02 | PWB SIG LO VAL |
| 1 | 44D236388-G01 | PWB COMP+REG ICR |
| 2 | 44D315178-G01 | PWB DIFF TRANS DR |
| 1 | 44D236378-G01 | PWB REF LOW VALUE |
| 1 | 44B337328-G01 | PWB LPF PWR SUPPLY |
| 1 | 44B337327-G01 | PWB LOW SPD SW LSS |
| 3 | 44C319112-G01 | RACK ASM PWB |
| 2 | K9774719L50 | CAP 500UF 50VDC |

3S7513KF300A5 LINE POWER PANEL

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 44B334726-G01 | SUPPLY TRANSF |
| 18 | 44A316724-M19 | CAP 1900UF 75VDC |
| 6 | 44A333453-D36 | RES 360HM 100W |
| 6 | 44B214518-G01 | REACTOR |

3S7513KF306A9 METER PANEL

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 5 | 44A317759-018 | SWITCH |
| 20 | GE387 | LAMP |

1 R.C. JUN 29 1977

3S7513KF306A11 METER PANEL ASM

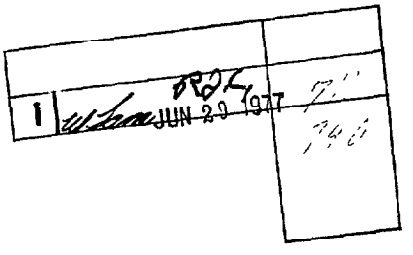
| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 1 | 128A4057H200 | RES .200 MEGO .5W |
| 2 | 44A317759-018 | SWITCH |
| 2 | 44A317759-019 | SWITCH |
| 1 | 44A317759-012 | SWITCH |
| 3 | 44A317759-020 | SWITCH |
| 3 | 128A4057G301 | RES 30.1K OHMS .5W |
| 20 | GE387 | LAMP |

3S7513KF307A16 RELAY PANEL

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 20 | 3S2791G138D4 | MINIATURE RELAY |
| 2 | 3SAA1343A2 | RELAY |
| 1 | 12HMA11B3 | RELAY |
| 1 | 44B312431-005 | TIME DELAY RELAY |
| 1 | 44A230536-001 | SIGNAL FLASHER |
| 34 | 44B232019-004 | RECT 1A 400V |
| 2 | 44A317994-G01 | COMP BD ASM |
| 4 | 3S2791G138D3 | MINIATURE RELAY |
| 2 | K9774723K40 | CAP 40UF 450VDC |
| 2 | 44B314000-G01 | D C REACTOR |
| 2 | K5908303G1 | T B ASM |
| 26 | 933B545P2 | RELAY SOCKET |
| 10 | 44B232011-001 | RECT 200MA 200VDC |

3S7513KL300A15 CONTROL PANEL ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 3 | CR2940U207 | CONTACT BLOCK |
| 2 | CR2940UA202C | PUSH BUTTON |
| 1 | 44A317759-007 | PUSHBUTTON SWITCH |
| 4 | 44A317759-001 | PUSHBUTTON SWITCH |
| 3 | 44A317759-008 | PUSHBUTTON SWITCH |
| 1 | 44C392721-G01 | DIODE BOARD |
| 10 | GE387 | LAMP |
| 3 | CR2940UA200C | OPERATOR |
| 1 | 6105700G43G | IND,LIGHT |



GENERAL ELECTRIC

44B300423 CONT ON SHEET SH NO

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS FRACTIONS ± | DECIMALS ± | ANGLES ± |
|---|---|---|---|---|
| | ✓ | | | |

REV NO 44B300423 CONT ON SHEET SH NO

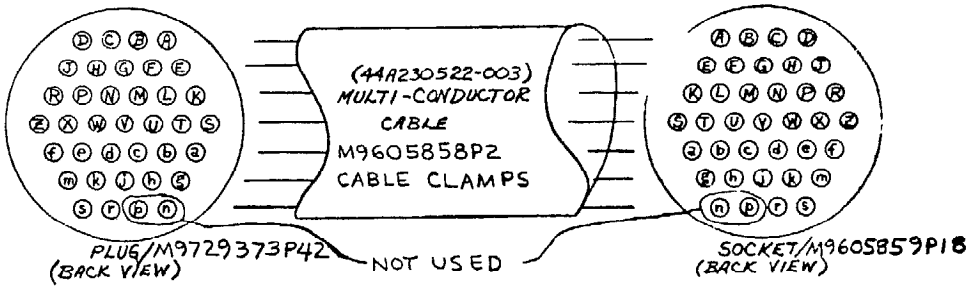
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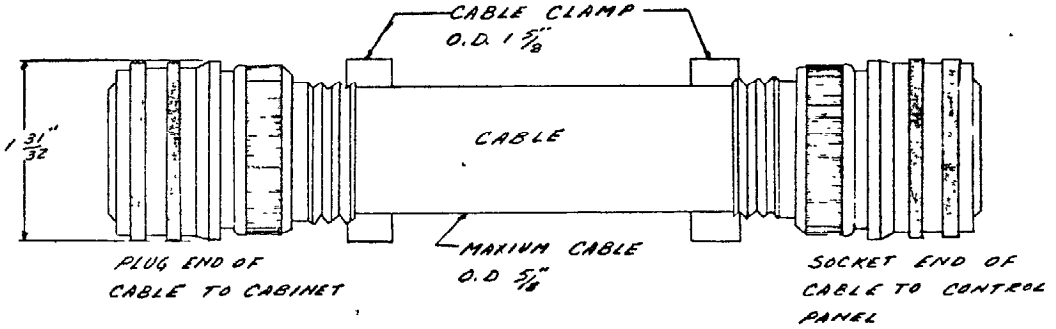
# CABLE WIRING

FIRST MADE FOR 3S7513TC300A2

SIGNAL & LOW VOLTAGE

| | PLUG TERMINAL | WIRE | SOCKET TERMINAL |
|---|---|---|---|
| | V | SHIELD | V |
| SHIELDED CABLE | M | WHITE/BLACK/VIOLET | M |
| | N | BLACK | N |
| | W | BROWN | W |
| SHIELDED CABLE | d | WHITE/BLACK/GRAY | d |
| | c | SHIELD | c |
| | U | RED | U |
| SHIELDED CABLE | T | WHITE/BROWN/RED | T |
| | L | SHIELD | L |
| | F | ORANGE | F |
| SHIELDED CABLE | G | WHITE/BROWN/ORANGE | G |
| | H | SHIELD | H |
| | P | WHITE | P |
| SHIELDED CABLE | X | WHITE/BLACK | X |
| | e | SHIELD | e |
| | k | WHITE/BROWN/YELLOW | k |
| SHIELDED CABLE | j | WHITE/BROWN/GREEN | j |
| | h | SHIELD | h |
| | b | WHITE/BROWN | b |
| SHIELDED CABLE | g | WHITE/RED | g |
| | a | SHIELD | a |
| | S | WHITE/ORANGE | S |
| | K | WHITE/YELLOW | K |
| | E | WHITE/GREEN | E |
| | A | WHITE/BLUE | A |
| | B | WHITE/VIOLET | B |
| | C | WHITE/GRAY | C |
| | D | WHITE/BLACK/BROWN | D |
| | J | WHITE/BLACK/RED | J |
| | R | WHITE/BLACK/ORANGE | R |
| | Z | WHITE/BLACK/YELLOW | Z |
| | f | WHITE/BLACK/GREEN | f |
| | m | WHITE/BLACK/BLUE | m |
| | s | WHITE/BROWN/BLUE | s |
| | r | WHITE/BROWN/VIOLET | r |




NOTES:

1. USE PLASTIC SLEEVING OVER WIRE AND TERMINALS TO INSULATE CONNECTIONS FROM EACH OTHER.

| REVISIONS | | PRINTS TO |
|---|---|---|
| 1 | Wed. AUG 19 1969 | |
| 2 | H. Tommiller | 12P-0-B |
| 3 | ADDED NOTE 3 Feb 13 1973 D K MOYER | FG-W |

MADE BY R M Bukrshey ISSUED MAY 22 1968 U

APPROVALS R M G

SPECIALTY CONTROL DIV OR DEPT WAYNESBORO, VA. LOCATION

44B300423 CONT ON SHEET SH NO

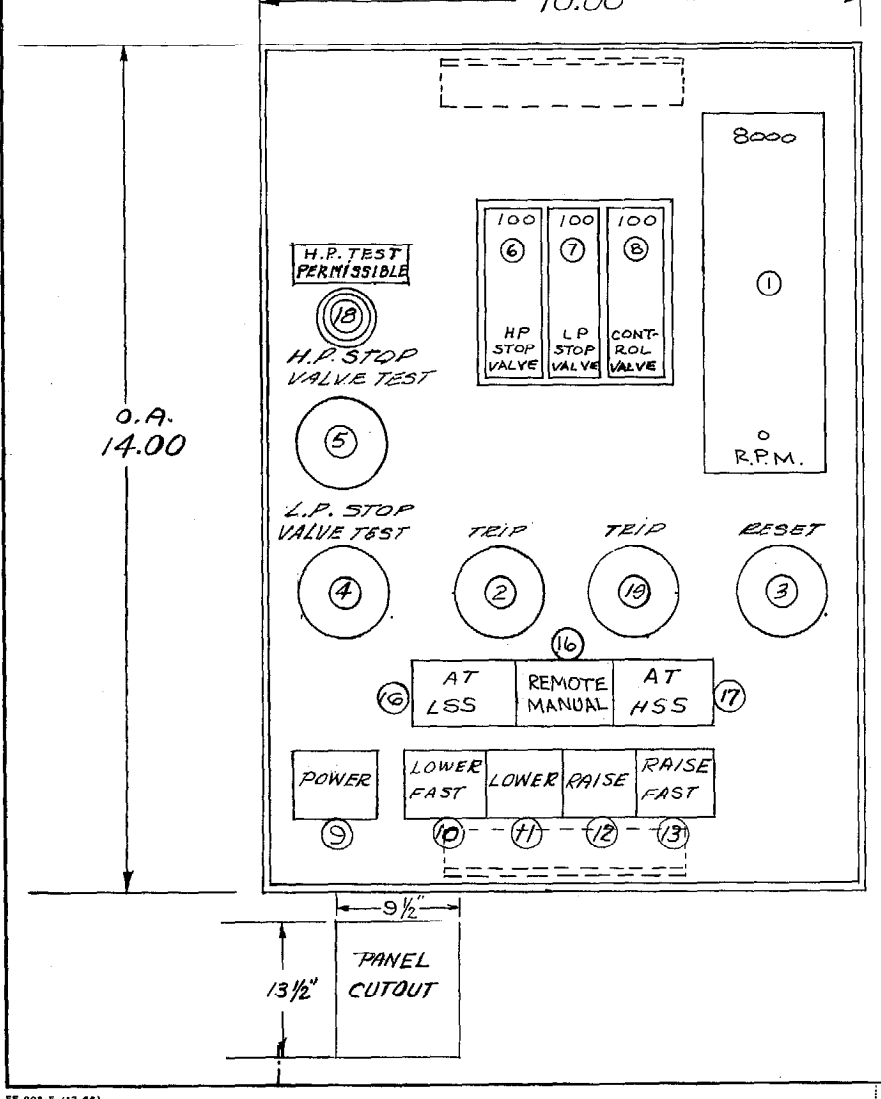
FORM F 118-661 PRINTED IN U.S.A.

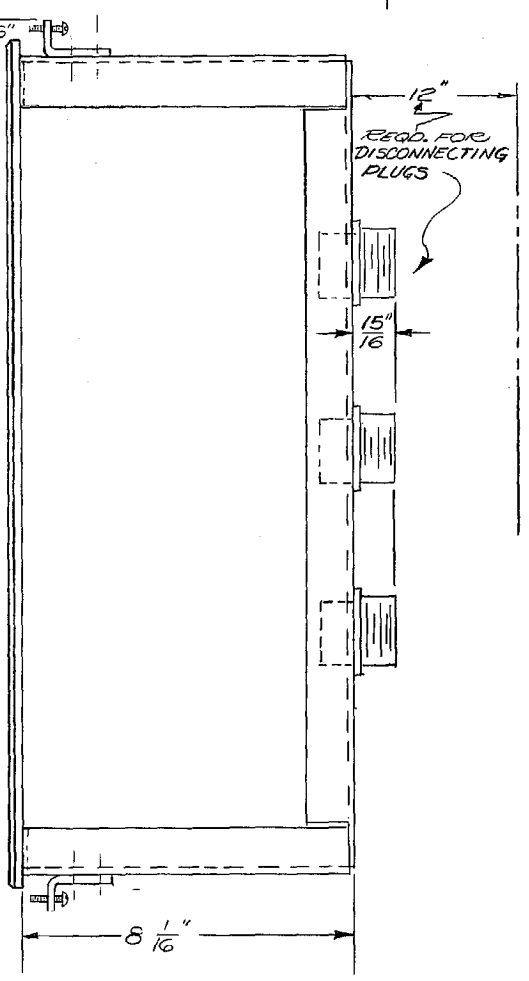
GENERAL ELECTRIC

44B309405
CONT ON SHEET FNL SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±
REV NO 1
CONT ON SHEET , SH NO.

TITLE (OUTLINE)
# FEEDPUMP TURBINE CONTROL PANEL
FIRST MADE FOR 3S7513KL300A15 TURBINE 153090
LOWER COLORADO RIVER AUTHORITY, FAYETTE"




## PARTS LIST

A. 1 INSTRUMENTS, GE TYPE 180.
B. 2,3,19 GE CR2940UA203C PUSHBUTTON WITH GUARD.
C. 4,5 GE CR2940UA202C PUSHBUTTON WITH GUARD.
D. 6,7,8 INSTRUMENTS, GE TYPE 185.
E. 9 ILLUMINATED PUSHBUTTON RED, POWER ON, CONDITIONS NORMAL, YELLOW (FLASHING) POWER ON, CONDITIONS ABNORMAL DEPRESS TO TEST ALL LAMPS.
F. 10-13, PUSHBUTTONS.(WHITE)
G. 16, 17 INDICATING LIGHTS.(WHITE)
I. 9-17, MASTER SPECIALTIES SERIES 12, VERTICAL BARRIER.
J. FINISH GLIDDEN 4618
K. 18 - GREEN LIGHT "TEST PERMISSIBLE"

| REVISIONS | | | PRINTS TO |
|---|---|---|---|
| 1 | T.R. Kuster MAY 8 1975 | | 11 |
| 2 | T.R. Kuster FEB 12 1976 | | 12P-DFB |
| | | | FG-W |
| | | | 12P-FS2 |

MADE BY C. Campbell 4-11-75
ISSUED APR 18 1975
APPROVALS R.G. Bell, R. Watson
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44B309405
CONT ON SHEET FNL SH NO 1

FF-800-F (12-68)
PRINTED IN U.S.A.

GENERAL ELECTRIC

44B306144

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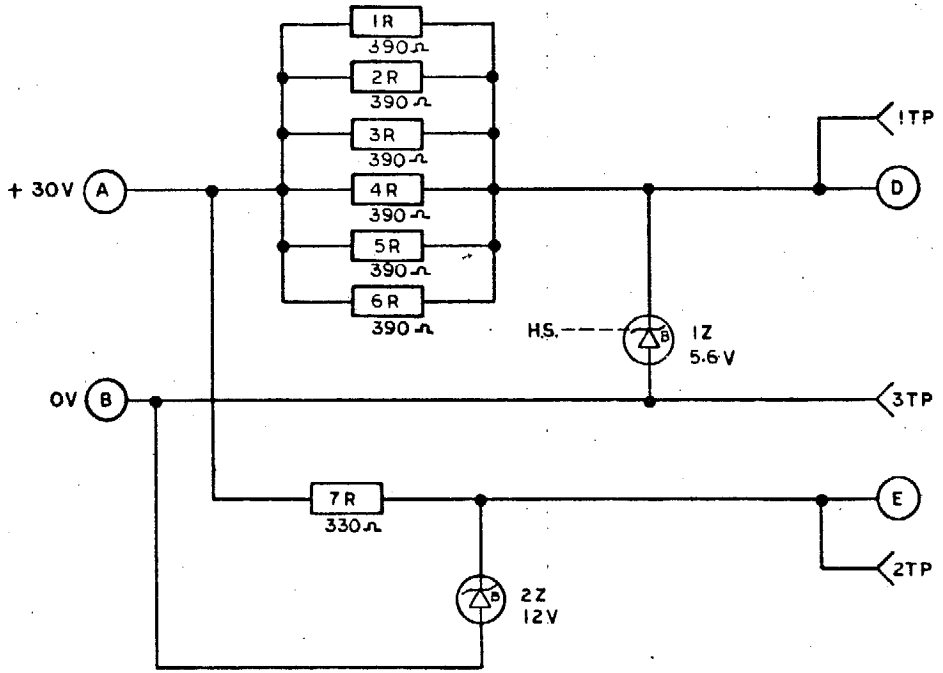
| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| | ✓ | ± | ± | ± |

TITLE

LSS PWR SUPPLY

FIRST MADE FOR

ELEM DIAGRAM




REVISIONS

PRINTS TO

11

LLP-2FB

FG-W

MADE BY Ray Bouzie 9-11-72

CHECKED D.K. Moyer Dept. 12, 1972

APPROVED R Lufsen 9/11/72

DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA.

LOCATION

44B306144

SH NO

3

FORM 908-F REV-71
PRINTED IN U.S.A.

GENERAL ELECTRIC

44B306148

| UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — | | | | | REV NO |
|---|---|---|---|---|---|
| APPLIED PRACTICES | SURFACES | TOLERANCES IN MULTIPLE DECIMAL ONLY | | | |
| | | FRACTIONS | DECIMALS | ANGLES | |
| | ✓ | ± | ± | ± | |

CONT ON SHEET — SH NO

TITLE
POWER CABLE WIRING, ELECTRONIC BOILER FEED PUMP TRUBINE CONTROL

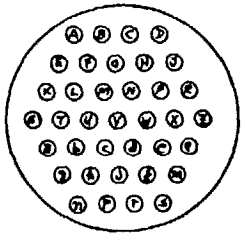
FIRST MADE FOR 3S7513TC300A10

200,300 POWER CABLE

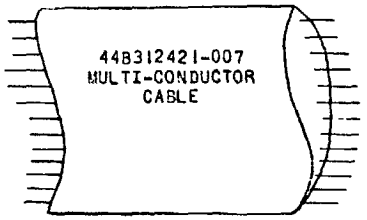
## ASSEMBLY NOTES

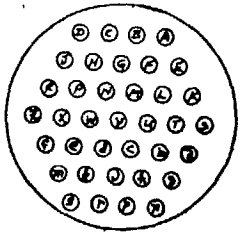
THE SPECIFIC COLOR OF WIRE CONNECTED TO A PIN IS NOT IMPORTANT, BUT THE SAME WIRE SHOULD BE CONNECTED TO THE SAME PIN AT THE OTHER CABLE END AND CONNECTOR.
EXAMPLE: IF A RED WIRE IS CONNECTED TO PIN "A" ON THE CONNECTOR PLUG, THE OTHER END OF THAT RED WIRE MUST BE CONNECTED TO PIN "A" OF THE SOCKET CONNECTOR AT THE OTHER END.

| CONNECT SAME COLOR WIRE TO THESE PINS AT BOTH CABLE ENDS | DO NOT CONNECT ANY WIRES TO THESE PINS |
|---|---|
| A | B |
| C | D |
| T | G |
| E | L |
| F | M |
| | N |
| H X | P |
| J Z | R |
| K | S |
| | U |
| | V |
| | W |

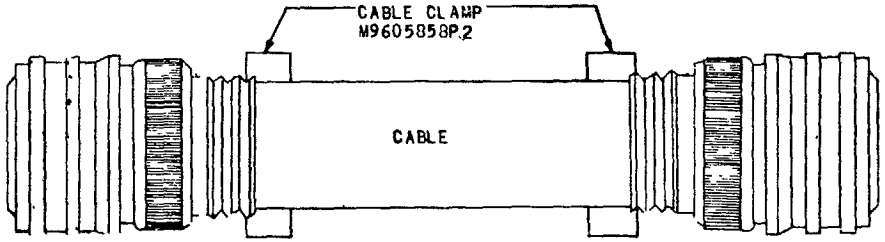


SOCKET
M9605859P18



PLUG
M9605859P42




SOCKET END OF
CABLE TO EHC REMOTE CONTROL PANEL.

PLUG END OF
CABLE TO EHC CABINET

| REVISIONS | PRINTS TO |
|---|---|
| | 11 |
| | APPVD |
| | FG-W |

MADE BY A. Templeton 10-18-72
APPROVED BY R. Morgan Oct. 20, 1972 P. Culver
COMM. & CONT. DEVICES DIV OR DEPT WAYNESBORO, VA. LOCATION

44B306148
CONT ON SHEET — SH NO

GENERAL ELECTRIC

44B306197 CONT ON SHEET SH NO.

DRAWING NO. 44B306197 CONT ON SHEET SH NO.

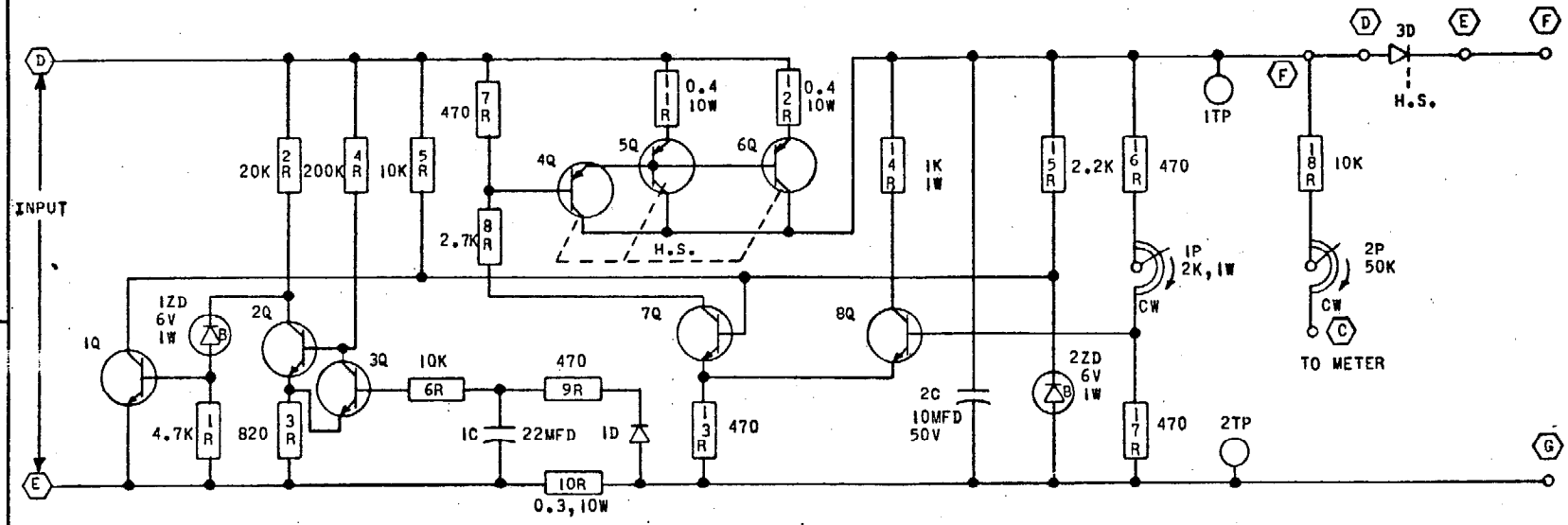
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| | ✓ | ± | ± | ± |

REV NO. 44B306197 CONT ON SHEET SH NO.

TITLE SERIES REGULATOR ISR

FIRST MADE FOR 44B236371-G01




| REVISIONS | PRINTS TO |
|---|---|
| | 11 |
| | FG-W |
| | 12P-DAB 2 |

MADE BY Lemon Ellison 9/29/73
ISSUED D. Moyer April 26, 1973
APPROVALS R G Shepard, R J Gilson
DRIVE SYSTEMS PROD. DEPT. DEPT. OR DEPT. WAYNESBORO, VA. LOCATION
44B306197 CONT ON SHEET SH NO.

FF-806-F (5-72) PRINTED IN U.S.A.

3

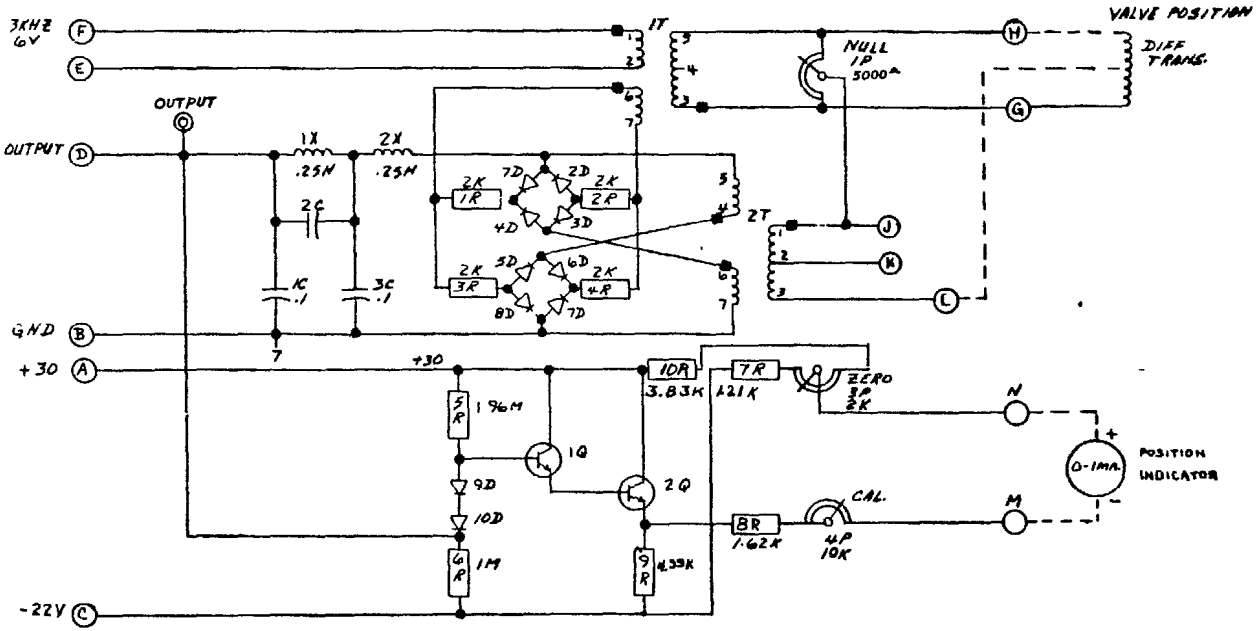
GENERAL ELECTRIC

44C300609

| UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — | | | | |
|---|---|---|---|---|
| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS | DECIMALS | ANGLES |
| | ✓ | ± | ± | ± |

TITLE: DETECTOR INDICATOR 1DI

FIRST MADE FOR 44D381585-G01




NOTES:

1. CONNECTIONS

| | | | |
|---|---|---|---|
| A | +30V | G, H, J, K, L | DIFFERENTIAL TRANSFORMER CONNECTIONS |
| B | GROUND | | |
| C | -22V | | |
| D | OUTPUT | | |
| E, F | 6V, 3 KHZ | M, N | OUTPUT METER 0-1MA |

44C300609 SHEET NO. 2 REV NO. O

| REVISIONS | | | PRINTS TO |
|---|---|---|---|
| 1 | R M C MAY 19 1969 | | 11 |
| 2 | R M C MAY 19 1969 | | FG-W |
| 3 | R Mallow OCT 19 1969 | | 12P DFB |
| 4 | [illegible] DEC 18 1969 | | |
| 5 | [illegible] MAY 18 1970 | | |
| 6 | [illegible] OCT 19 1970 | | |

R. W. Templeton JAN 26 1968 APPROVED APR 17 1968

SPECIALTY CONTROL DEPT OR SECT WAYNESBORO, VA. LOCATION

44C300609 DWG NO

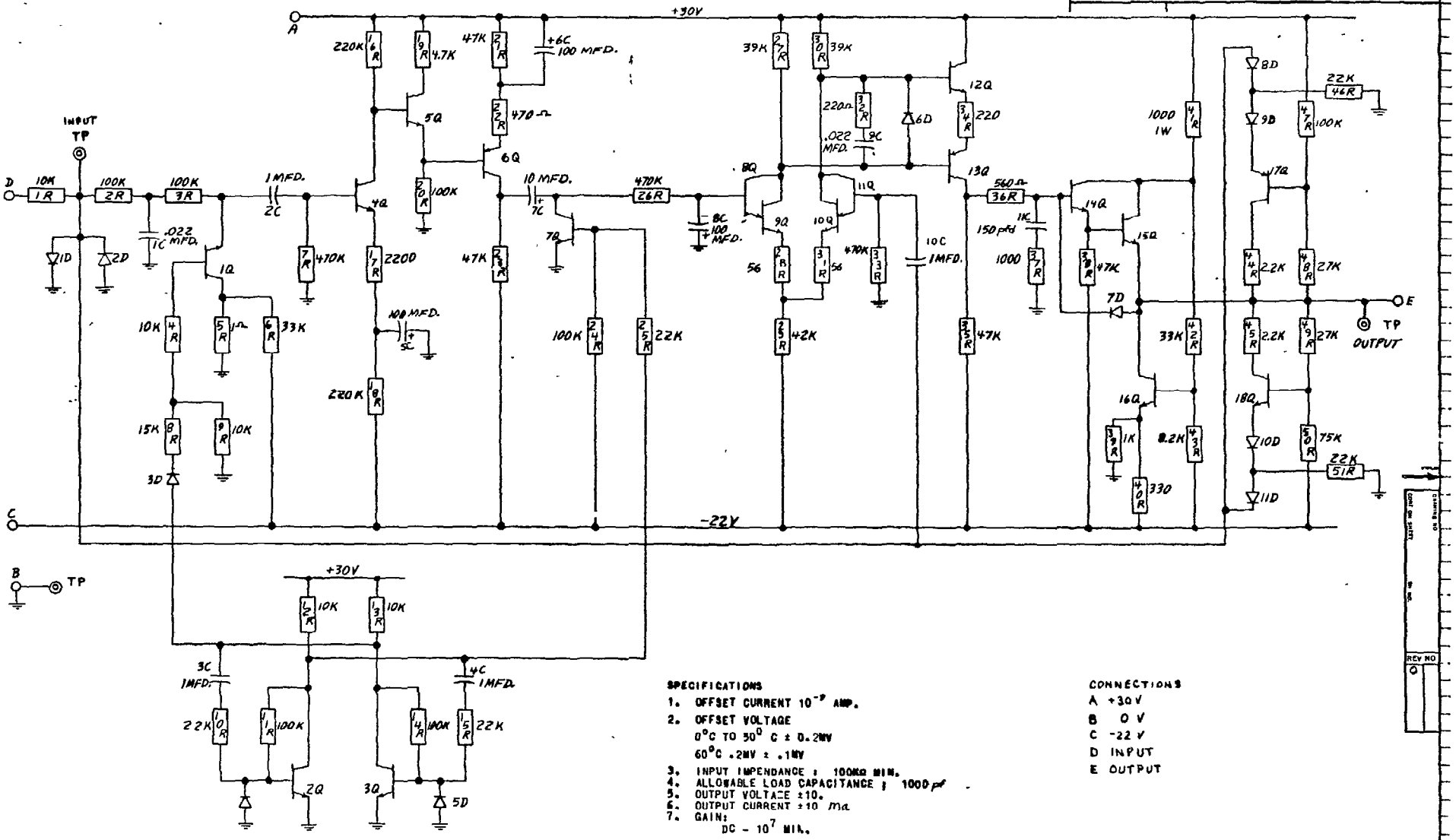
GENERAL ELECTRIC

44C300610

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
CHOPPER STABILIZED
OPERATIONAL AMPLIFIER
FIRST MADE FOR 44D38589-G01

ØA




SPECIFICATIONS

1. OFFSET CURRENT $10^{-9}$ AMP.
2. OFFSET VOLTAGE
   0°C TO 50° C ± 0.2MV
   60°C .2MV ± .1MV
3. INPUT IMPENDANCE : 100MΩ MIN.
4. ALLOWABLE LOAD CAPACITANCE : 1000 pf
5. OUTPUT VOLTAGE ±10.
6. OUTPUT CURRENT ±10 ma
7. GAIN:
   DC - $10^7$ MIN.
   AC - 5000 MIN. AT 100 CPS
   CHARACTERISTIC: BREAK FREQUENCY - 250 CPS
   20 db PER DECADE TO
   CROSSOVER.
8. CHOPPER STABILIZER RECOVERY TIME: 1 MINUTE
9. POWER REQUIREMENTS: 30 VOLTS - 25 ma
   -22 VOLTS - 20 ma
10. AMPLIFIER PROTECTION:
    A. OUTPUT CURRENT LIMITED TO APPROX 10 ma
    B. OUTPUT LIMITED TO 10 VOLTS
    C. ALLOWABLE INPUT VOLTAGE BEFORE DAMAGE - 70 VOLTS.

CONNECTIONS

A +30 V
B 0 V
C -22 V
D INPUT
E OUTPUT

REVISIONS

| | | |
|---|---|---|
| 1 | Nov 17, 1967 | |
| 2 | Release Aug 9, 1968 | |

PRINTS TO: 11, 12P-DFB

| MADE BY | APPROVED | | DRAWING NO. |
|---|---|---|---|
| J C Glenn JAN 22 1968 | MAR 12 1968 | SPECIALTY CONTROL WAYNESBORO, VA | 44C300610 |

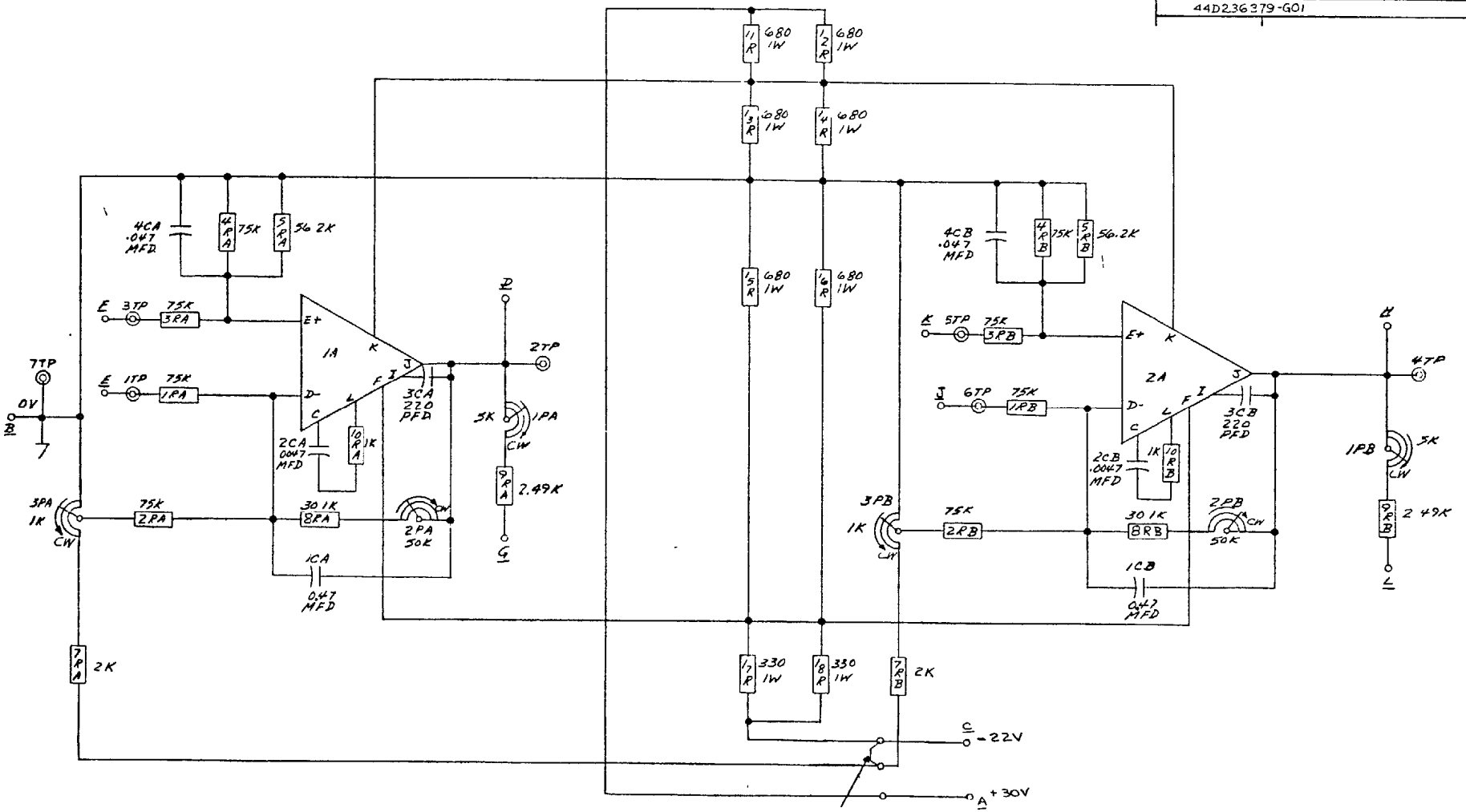
GENERAL ELECTRIC

44C30-

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ANALOG OUTPUT 1MØ2

FIRST MADE FOR
44D236379-G01




JUMPER POINTS PROVIDED ON THE PRINTED CIRCUIT BOARD SO RESISTORS 7RA & 7RB MAY BE CONNECTED TO EITHER +30V OR -22V.

CONNECTIONS

A = +30V
B = 0V
C = -22V
D = 1A VOLTAGE OUTPUT
E = 1A INPUT
G = 1A CURRENT OUTPUT
H = 2A VOLTAGE OUTPUT
J = 2A INPUT
L = 2A CURRENT OUTPUT

44C301216 REV N

| REVISIONS | | PRINTS TO |
|---|---|---|
| 1 | SEP 30 1968 | 11 |
| 2 | OCT 9 1969 | 11 DF 44 |
| 3 | FEB 3 1970 | |
| 4 | | |

N. M. Templeton SEP 16 1968 ISSUED
SEP 24 1968
APPROVALS MBC
SPECIALTY CONTROL WAYNE, N.C.
44C301216

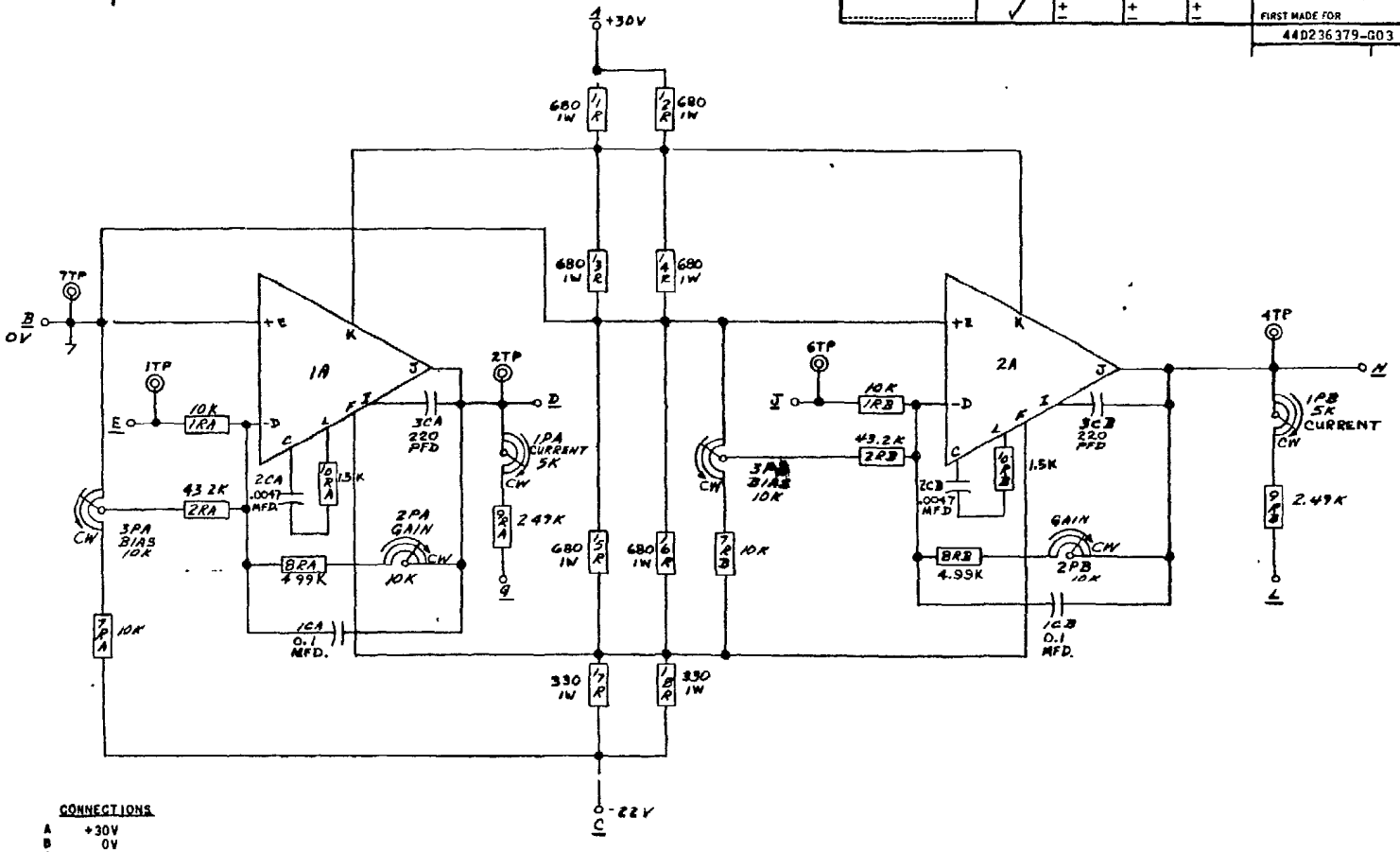
GENERAL ELECTRIC

44C301622

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON INCHOLD DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ANALOG OUTPUT 3A#2

FIRST MADE FOR
44D236379-G03




CONNECTIONS

| | |
|---|---|
| A | +30V |
| B | 0V |
| C | -22V |
| D | 1A VOLTAGE OUTPUT |
| E | 1A INPUT |
| G | 1A CURRENT OUTPUT |
| H | 2A VOLTAGE OUTPUT |
| J | 2A INPUT |
| L | 2A CURRENT OUTPUT |

44C301622 REV NO

| REVISIONS | | PRINTS TO |
|---|---|---|
| 1 | SEP 30 1968 | 11 |
| 2 | Nov 27, 1968 | |
| 3 | Aug 3 1970 | |

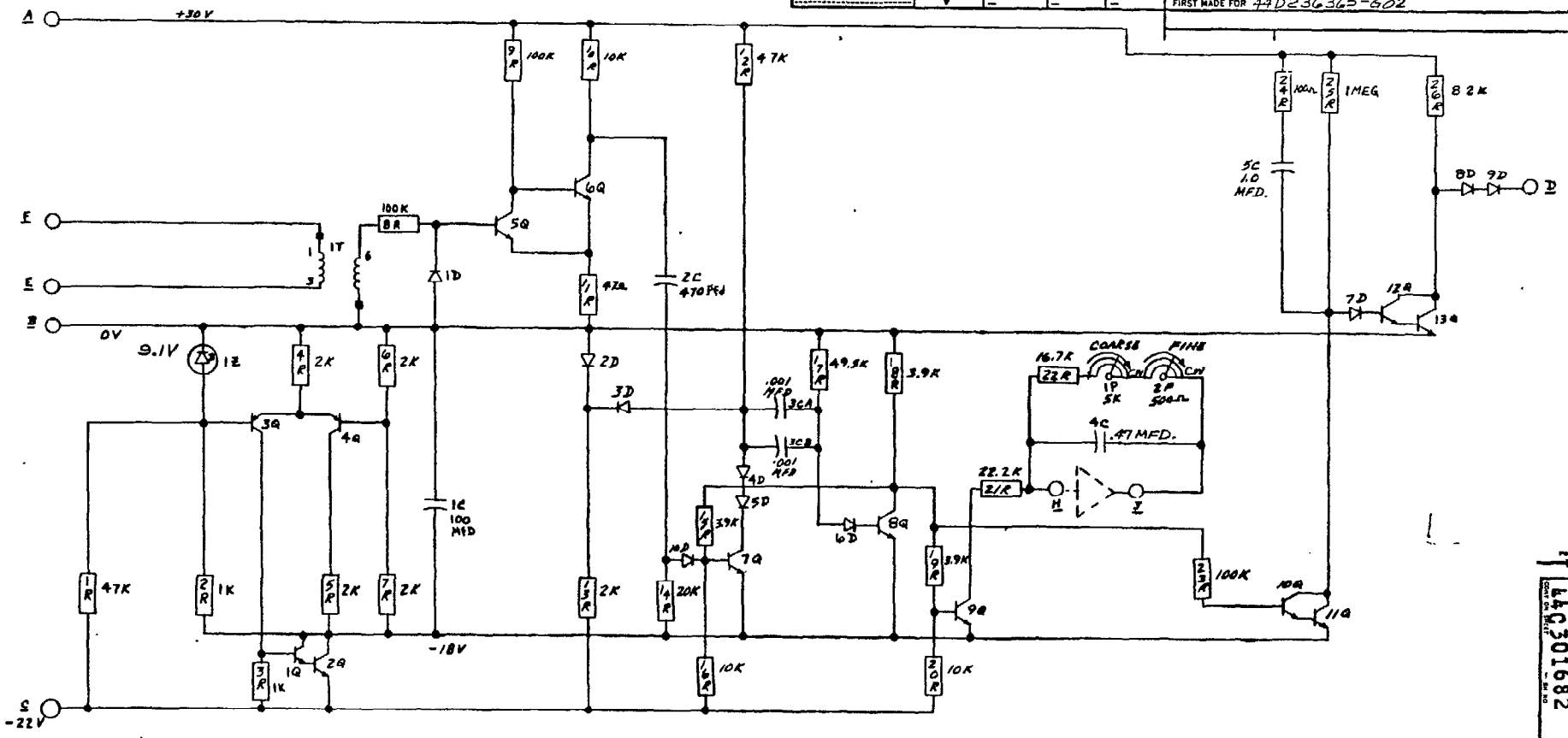
E Templeton SEP 17 1968
SEP 19 1968
M8C
SPECIALTY CONTROL WAYNESBORO
44C301622

GENERAL ELECTRIC

44C301682

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES OR MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
FREQUENCY-VOLTAGE CONVERTER (2FVC)
FIRST MADE FOR 44D236365-G02



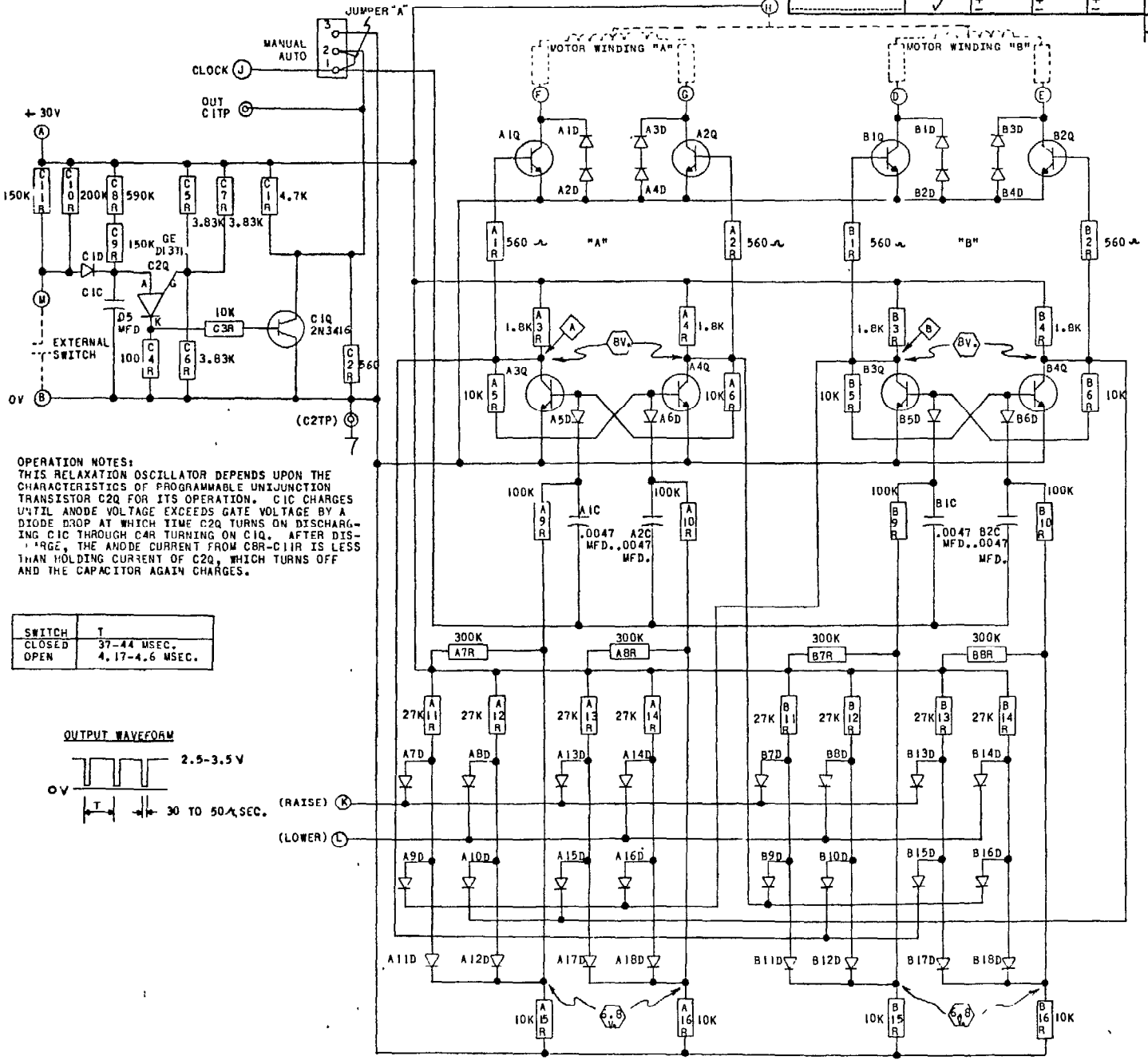

| REVISIONS | | PRINTS TO |
|---|---|---|
| 1 | QRL Dwg 11.A31 | 11. |
| 2 | C.L. Card Dwg 6; Aug 3, 1971 | 12P-DFB |
| 3 | C.O. Release Aug 3, 1971 | FG-W |

R. W. Templeton NOV 15 1968
ISSUED OCT 9 1968
R. L. Tabor
SPECIALTY CONTROL
WAYNESBORO

44C301682

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
CLOCK & STEPPING MOTOR DRIVE
FIRST MADE FOR 44D331918-G01



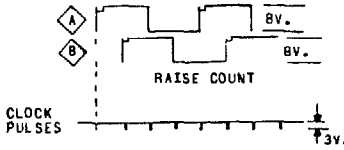

1. THIS BOARD CONSISTS OF A RELAXATION OSCILLATOR CLOCK IDENTIFIED BY "C" COMPONENTS AND ALSO TWO FLIP-FLOPS "A" AND "B" WITH DIODE LOGIC TO FORM A REVERSIBLE COUNTER. OUTPUT OF THE FLIP-FLOPS FEEDS POWER AMPLIFIER A1Q, A2Q, B1Q & B2Q. THE STEPPING MOTOR WINDINGS ARE CONNECTED DIRECTLY TO THE COLLECTORS OF THESE FOUR TRANSISTORS.
2. THE CLOCK PULSES ARE THREE VOLTS AMPLITUDE WITH FAST FALL TIME. NORMAL CLOCK FREQUENCY IS 240 PULSES PER SECOND.
3. RAISE COUNTING RESULTS FROM 0 TO 1 VOLT ON TERMINAL (K) & +6 VOLTS OR MORE ON TERMINAL (L). REVERSING THE VOLTAGE ON TERMINALS (K) & (L) RESULTS IN LOWER COUNTING. A POSITIVE VOLTAGE ON TERMINALS (K) & (L) WILL INHIBIT COUNTING. 0 TO 1 VOLT ON BOTH (K) & (L) IS NOT PERMISSIBLE.
4. CONNECT JUMPER "A" TO 1 & 2 FOR MANUAL SPEED REFERENCE; CONNECT JUMPER "A" TO 2 & 3 FOR AUTO SPEED REFERENCE.

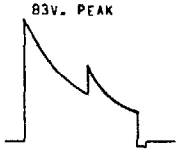
OPERATION NOTES:
THIS RELAXATION OSCILLATOR DEPENDS UPON THE CHARACTERISTICS OF PROGRAMMABLE UNIJUNCTION TRANSISTOR C2Q FOR ITS OPERATION. C1C CHARGES UNTIL ANODE VOLTAGE EXCEEDS GATE VOLTAGE BY A DIODE DROP AT WHICH TIME C2Q TURNS ON DISCHARGING C1C THROUGH C4R TURNING ON C1Q. AFTER DISCHARGE, THE ANODE CURRENT FROM C8R-C11R IS LESS THAN HOLDING CURRENT OF C2Q, WHICH TURNS OFF AND THE CAPACITOR AGAIN CHARGES.

| SWITCH | T |
|---|---|
| CLOSED | 37-44 MSEC. |
| OPEN | 4.17-4.6 MSEC. |

OUTPUT WAVEFORM

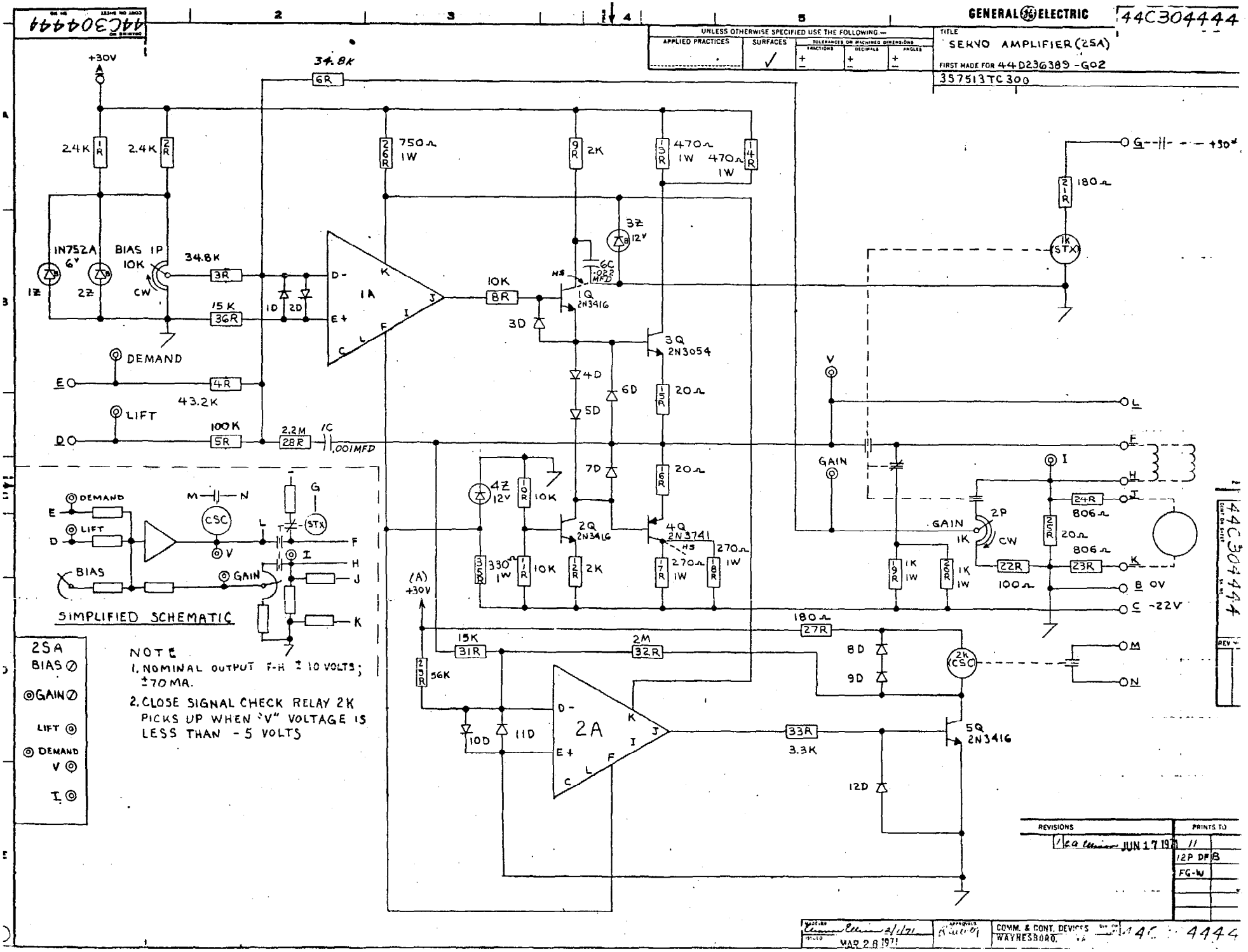



| REVISIONS | | PRINTS TO |
|---|---|---|
| 1 | E. MAR 9 1970 | 11 |
| 2 | [illegible] July 24, 1971 | 7P-D-B3 |
| 3 | [illegible] Nov 22, 1971 | F3 W |

DRAWN BY [illegible] 7-24-70 ISSUED MAR 3 1970
APPROVALS MSC
COMM. & CONT. DEVICES DIV. OR DEPT. WAYNESBORO, VA. LOCATION
44C303567


44C304444
GENERAL ELECTRIC
44C304444
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE
SERVO AMPLIFIER (2SA)
FIRST MADE FOR 44D236389-G02
357513TC300
+30V
34.8K
6R
2.4K 1R
2.4K 2R
750Ω 1W 26R
9R 2K
470Ω 1W 13R
470Ω 1W 14R
1N752A 6V
BIAS 1P 10K CW
34.8K 3R
15K 3GR
1Z 2Z
1D 2D
1A
D- K E+ J I F C L
10K 8R
1Q 2N3416
NS
6C .022 MFD
3Z 12V
3D
3Q 2N3054
4D 5D 6D
15R 20Ω
DEMAND
E
4R
43.2K
LIFT
D
100K 5R
2.2M 28R
1C .001MFD
7D
16R 20Ω
4Z 12V
10K 10R
2Q 2N3416
4Q 2N3741
NS
270Ω 1W 17R
8R
330Ω 1W 35R
10K 1R
2K 2R
G
L
M N
CSC
STX
V
I
GAIN
BIAS
F
H
J
K
SIMPLIFIED SCHEMATIC
2SA
BIAS
GAIN
LIFT
DEMAND
V
I
NOTE
1. NOMINAL OUTPUT F-H ± 10 VOLTS; ±70 MA.
2. CLOSE SIGNAL CHECK RELAY 2K PICKS UP WHEN "V" VOLTAGE IS LESS THAN -5 VOLTS
(A) +30V
15K 31R
2M 32R
56K 13R
10D 11D
2A
D- K E+ J I F C L
33R 3.3K
5Q 2N3416
12D
180Ω 27R
8D 9D
2K CSC
M N
G +30V
180Ω 21R
1K STX
V
L
GAIN
2P 1K CW
1K 1W 9R
1K 1W 10R
I
F H J K
24R 806Ω
20Ω 22R
806Ω 23R
100Ω
B 0V
C -22V
REVISIONS
PRINTS TO
JUN 17 1971
11
12P DF B
FG-W
MADE BY
APPROVALS
MAR 28 1971
COMM. & CONT. DEVICES
WAYNESBORO, VA
44C304444
REV

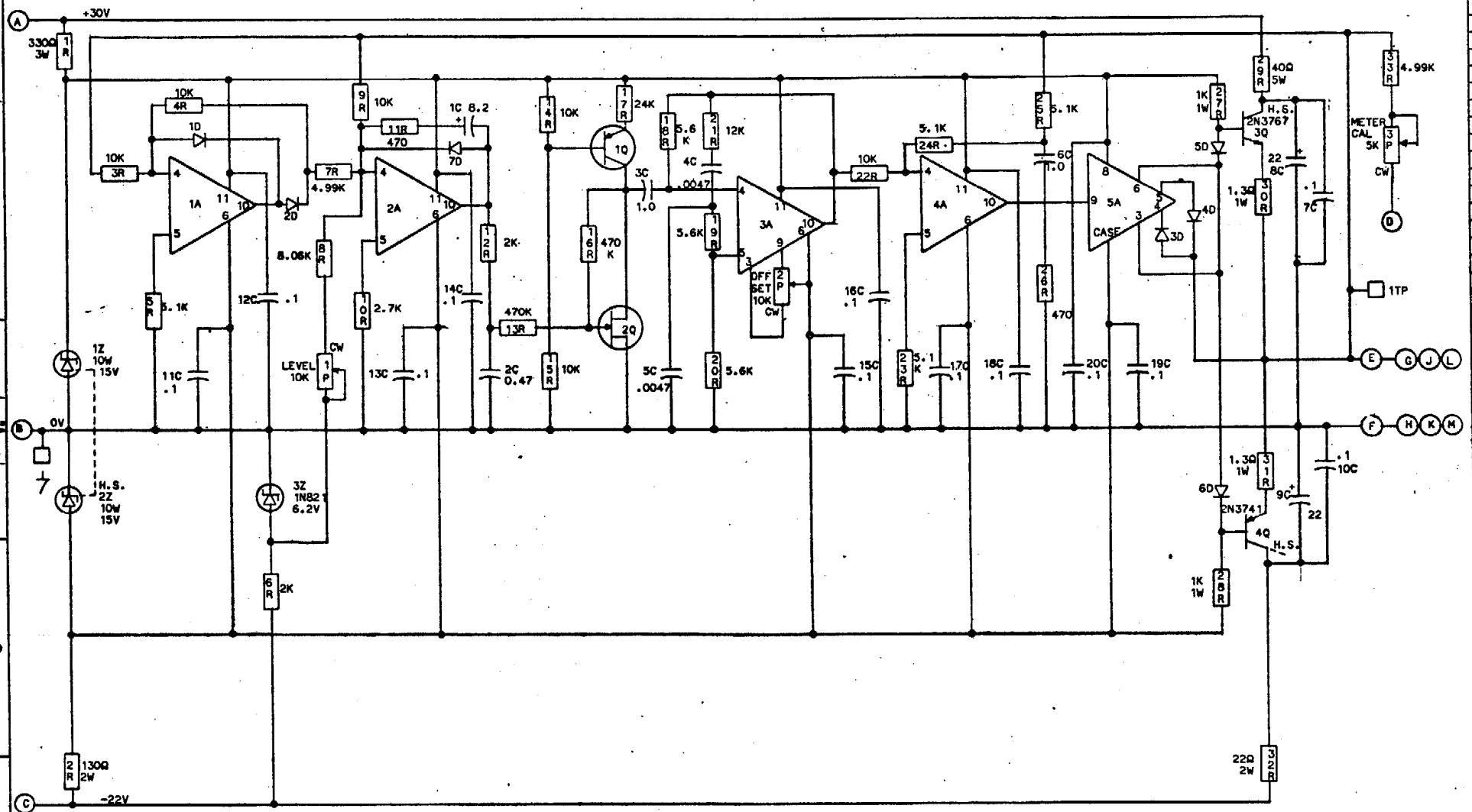
GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| | ✓ | ± | ± | ± |

TITLE: 3KHZ OSCILLATOR

FIRST MADE FOR EHC 7-OSC




ALL CAPACITORS IN MFD UNLESS OTHERWISE NOTED.

FREQUENCY: 3000HZ±300HZ
AMPLITUDE: 6.00VRMS (17V PEAK TO PEAK)
DC OFFSET: ADJUSTABLE TO ZERO
MINIMUM LOAD IMPEDANCE: 25Ω
AMPLITUDE DRIFT: 0.1% FIFTEEN HOURS

| REVISIONS | PRINTS TO |
|---|---|
| | 7/1 |
| | FG-W |

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44C321478

1 of 1 PL

GENERAL ELECTRIC

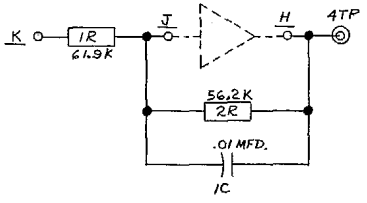
44C301648 CONT ON SHEET SH NO.

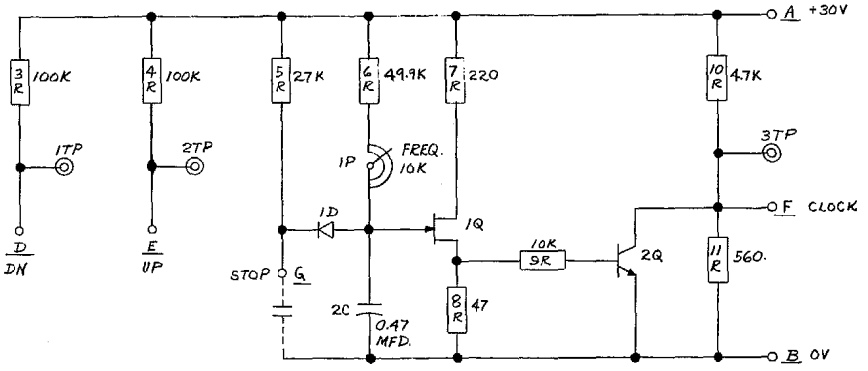
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| | √ | ± | ± | ± |

TITLE SIGNAL CONVERTER (2SC)

FIRST MADE FOR 44D2363G7 G02




NOTES:

1. CONNECTIONS
   - A +30 V
   - B 0V
   - C -22 V
   - D UP STEER
   - E DN STEER
   - F CLOCK
   - G STOP
   - H OUT } FEEDBACK
   - J IN } AMPLIFIER
   - K DEMOD. INPUT
2. SEE 44C300615 FOR STEPPING MOTOR DRIVE

DRAWING NO. 44C301648 SH NO. REV NO.

| REVISIONS | | PRINTS TO | |
|---|---|---|---|
| 1 | R.M. BUKOSKEY NOV 4 1968 | | 11 |
| 2 | H.T. SEP 30 1969 | | 12P-DFB |
| | | | FG-W |

MADE BY R.M. BUKOSKEY ISSUED NOV 18 1968

APPROVALS MBC

SPECIALTY CONTROL DIV OR DEPT WAYNESBORO, VA. LOCATION

44C301648 CONT ON SHEET SH NO.

GENERAL ELECTRIC

44C305259

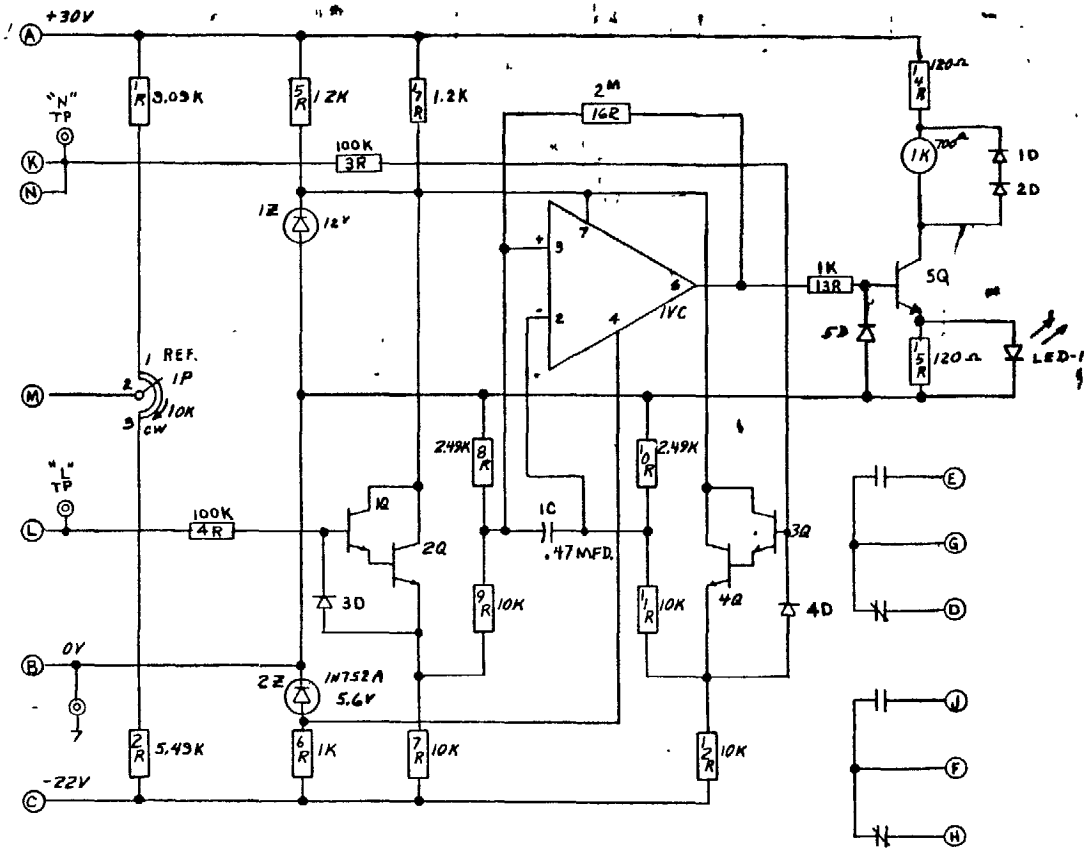
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—

| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| | ✓ | ± | ± | ± |

TITLE: VOLTAGE COMPARATOR

1VC

FIRST MADE FOR 44B331761-G01

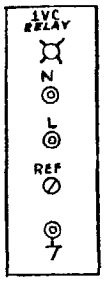



CONNECTIONS

| | | |
|---|---|---|
| A | +30V | |
| B | 0 V | |
| C | -22V | |
| D | | |
| E | | |
| F | | |
| G | | |
| H | | |
| J | | |
| N, K | | INPUT |
| L | | SEE NOTE 2 |
| M | | |

NOTES:

1. IN NORMAL OPERATION THE RELAY IS ENERGIZED. IT CAN BE ADJUSTED TO DROPOUT ON EITHER INCREASING OR DECREASING SIGNALS BETWEEN PLUS AND MINUS 10 VOLTS.
2. DROPOUT ON — INPUT — JUMPER — REF. TP — DROPOUT
   INCREASING SIGNAL — N — L-M — L — N>L
   DECREASING SIGNAL — L — N-M — N
3. RELAY CONTACTS RATED 2A, 28VDC.
4. RELAY ENERGIZED WHEN LIGHT IS ON.




FACEPLATE

44C305259 REV NO

| REVISIONS | PRINTS TO |
|---|---|
| 1 G.C. Aug. 7, 1973 | 11 |
| | 12P DPB |
| | FG-W |

| DRAWN BY | APPROVED | SPECIALTY CONTROL | 44C305259 |
|---|---|---|---|
| [illegible] 10/24/72 | G. Luber | WAYNESBORO VA | |
| E. Kotz Dec 8, 1972 | [illegible] | | |

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE: ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A17, 091-41761 TB 153090
ELEMENTARY DIAGRAM
LOWER COLORADO RIVER AUTHORITY
PRINTED CIRCUIT BOARD LOCATIONS FAYETTE #1, TB153090 FAYETTE #2, TB153091

CONT ON SHEET 1A SH NO 1

**RACK 1**

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LP STOP VALVE 1DI | | HP STOP VALVE 1DI | | CONTROL VALVE 1AØ2 | | SPEED SIGNAL 3AØ2 | | SPEED SIGNAL 3AØ2 | | OVERSPEED TEST 1ØSI | | 1ØA | REF. LVG 1ØA | | 2LVG | LOCAL 1VC | MANUAL REFERENCE CL & SMD | 1DI | | | ← AUTO REFERENCE → 1ØA | | 2SC | | CL & SMD | 1DI | |

**RACK 2**

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PRI SPEED 1ØA | 2FVC | | SEC. SPEED 1ØA | 2FVC | | SIGNAL LVG 1ØA | 1ØA | 1LVG | | REGULATION & COMPENSATION 1ØA | 1CA | SERVO VALVE LOOP 2ØA | | 1DI | | 3KHZ OSC 6OSC | SPEED SENSING 10% 105% 110% 1VC | 1VC | 1VC | 200 RPM 1VC | SPEED INDICATION AND SWITCHING 1ØA | 2FVC | | LOW SPEED SW. LSS | | LSS PWP SUP LPS | |

**RACK 3**

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1DB | 1DB | 1DB | | 30 VOLT SUPPLY LINE A 1REC | | 1SR | | LINE B 1SR | | 1REC | | −22 VOLT SUPPLY LINE A 1REC | | 1SR | | LINE B 1SR | | 1REC | | 24 VOLT SUPPLY LINE A 1REC | | 1SR | | LINE B 1SR | | 1REC | |

FRONT VIEW

REVISIONS

| | | | |
|---|---|---|---|
| 1 | R.P. Hicks | AUG 4 1975 | |
| 2 | C.A. Clay | MAY [illegible] | |
| 3 | W. Jones | OCT 1 1976 | FG-W |
| 4 | D. Wise | APR [illegible] 1977 | |

PRINTS TO: 711 | 12P-DRB2

MADE BY [signature] ISS'D APR 18 1975
APPROVED [signature]
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C309397
CONT ON SHEET 1A SH NO 1

REV NO.

PRINTED IN U.S.A.

1 2 3 4 5 6

3+1PL

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS | DECIMALS | ANGLES

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A17, 091-41761 TB 153090
ELEMENTARY DIAGRAM
LOWER COLORADO RIVER AUTHORITY
FAYETTE #1, TB153090
FAYETTE #2, TB153-01

SHEET 2 SH NO 1A

PRINTED CIRCUIT BOARD AND SUB. ASM. LIST

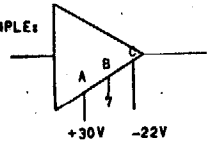
PRINTED CIRCUIT BOARD

| SYMBOL | TITLE | ELEM. DIAG. | ORDER NO. | QTY. |
|---|---|---|---|---|
| 1AØ2 | ANALOG OUTPUT BOARD | 44C301216 | 44D236379-G01 | 1 |
| CL& SMD | CLOCK & STEPPING-MOTOR DRIVE | 44C303567 | 44D331918-G01 | 2 |
| 1CR | COMPENSATION & REGULATION | 44A301400 | 44D236388-G01 | 1 |
| 1DB | DIODE BOARD | 44A300243 | 44D236370-G01 | 3 |
| 1DI | DETECTOR INDICATOR | 44C300609 | 44D381585-G01 | 5 |
| 3AØ2 | ANALOG OUTPUT | 44C301622 | 44D236379-G03 | 2 |
| 1LVG | SIGNAL LOW VALUE GATE | 44A304763 | 44D236372-G02 | 1 |
| 2FVC | FREQUENCY TO VOLTAGE CONVERTER | 44C301682 | 44D236365-G02 | 3 |
| 1REC | SERIES REG. BD. #1 | 44A306392 | 44B317827-G01 | 6 |
| 1SR | SERIES REG. BD. #2 | 44B306197 | 44D236371-G01 | 6 |
| 1ØA | OPERATIONAL AMPLIFIER | 44C300610 | 44D381589-G01 | 10 |
| 1ØSI | OVERSPEED INTEGRATOR | 44A302692 | 44D236373-G02 | 1 |
| 2SA | SERVO AMPLIFIER | 44C304445 | 44D236389-G02 | 1 |
| LPS | LOW SPEED SWITCH POWER SUPPLY | 44B306144 | 44B337328-G01 | 1 |
| LSS | LOW SPEED SWITCH | 44C306444 | 44B337327-G01 | 1 |
| 1VC | VOLTAGE COMPARATOR | 44C305259 | 44B331761-G01 | 5 |
| | | | | 1 |
| 2LVG | REF. LOW VALUE GATE | 44A300281 | 44D236378-G01 | 1 |
| 60SC | 3KHZ. OSC. BOARD | 44C304462 | 44B331717-G01 | 1 |

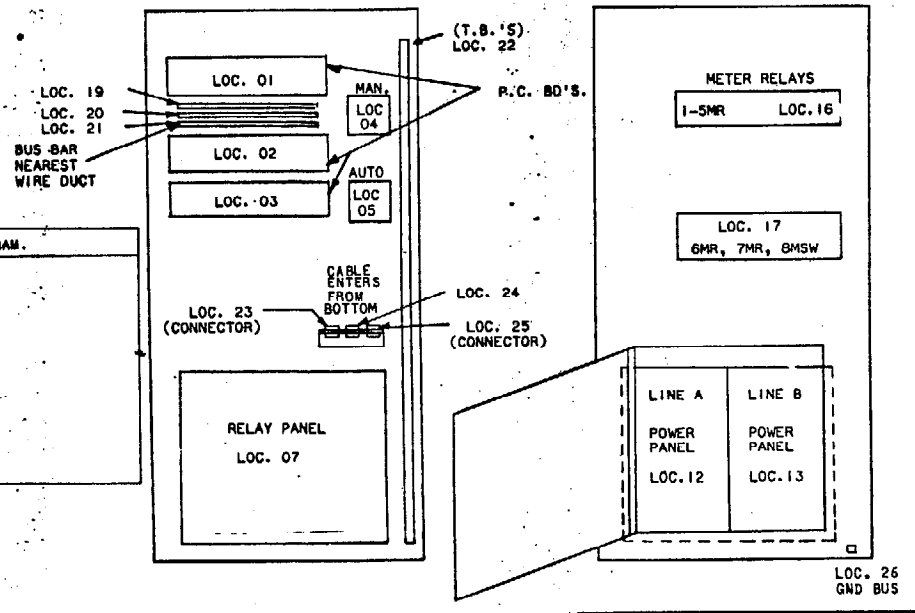
CUSTOMER NOTES:

1. INDICATES ZERO REFERENCE POTENTIAL, NOT CASE GROUND.
2. INDICATES STATION GROUND.
3. ALL OPERATIONAL AMPLIFIERS HAVE ADDITIONAL CONNECTIONS NOT SHOWN ELSEWHERE

EXAMPLE:




4. THE LOCATION OF THE PRINTED CIRCUIT BOARDS IS SHOWN BY THE NUMBERS APPEARING IN OR NEAR THE SQUARE WHICH REPRESENTS THE P.C. BD. EXAMPLE 0215 WOULD BE RACK #2 AND SLOT #15.
5. 1MR TO 7MR (SHEET 10) ARE METER RELAYS.
6. * INDICATES ITEMS NOT FURNISHED BY WAYNESBORO.
7. INDICATES SHIELDED CABLE.
8. INDICATES TWISTED WIRES.
9. INDICATES TERMINALS AT EHC CASE FOR CUSTOMER CONNECTION
10. INDICATES TERMINALS AT TURBINE
11. INDICATES SUB-PANEL & PC BOARD TERMINALS IN EHC CASE
12. CONTACTS FROM VOLTAGE COMPARATOR RELAY (1VC)




SUB. ASSEMBLIES - PANELS -

| TITLE | ORDER NO. | CONNECTION DIAGRAM. |
|---|---|---|
| CONTROL PANEL | 3S7513KL300A15 | |
| POWER PANEL | 3S7513KF300A5 | 44C306526 |
| METER PANEL | 3S7513KF306A9 | 44C305834 |
| METER PANEL | 3S7513KF306A11 | 44C306262 |
| RELAY PANEL | 3S7513K307A | 44C |
| DIFF. XFMR. DRIVE | 44D315178-G01 | 44C300882 |

REVISIONS
1 J.G. Clemons MAY 27 1975
2 D.W.D. Tech.

PRINTS TO
1 1
12P-DRB 2
FG-W

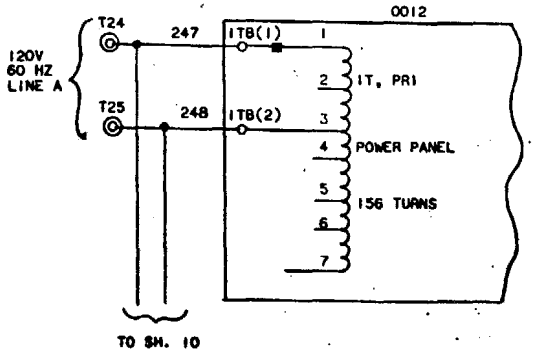
ISSUED JUN 18 1975
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C309397
SH NO 1A

3+1PL

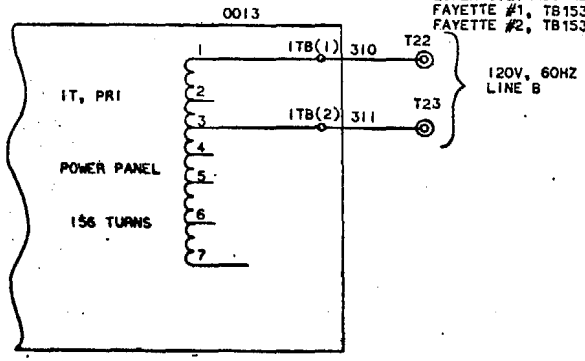
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 357513TC300A17, 091-41761 TB 153090
ELEMENTARY DIAGRAM
LOWER COLORADO RIVER AUTHORITY
FAYETTE #1, TB153090
FAYETTE #2, TB153091

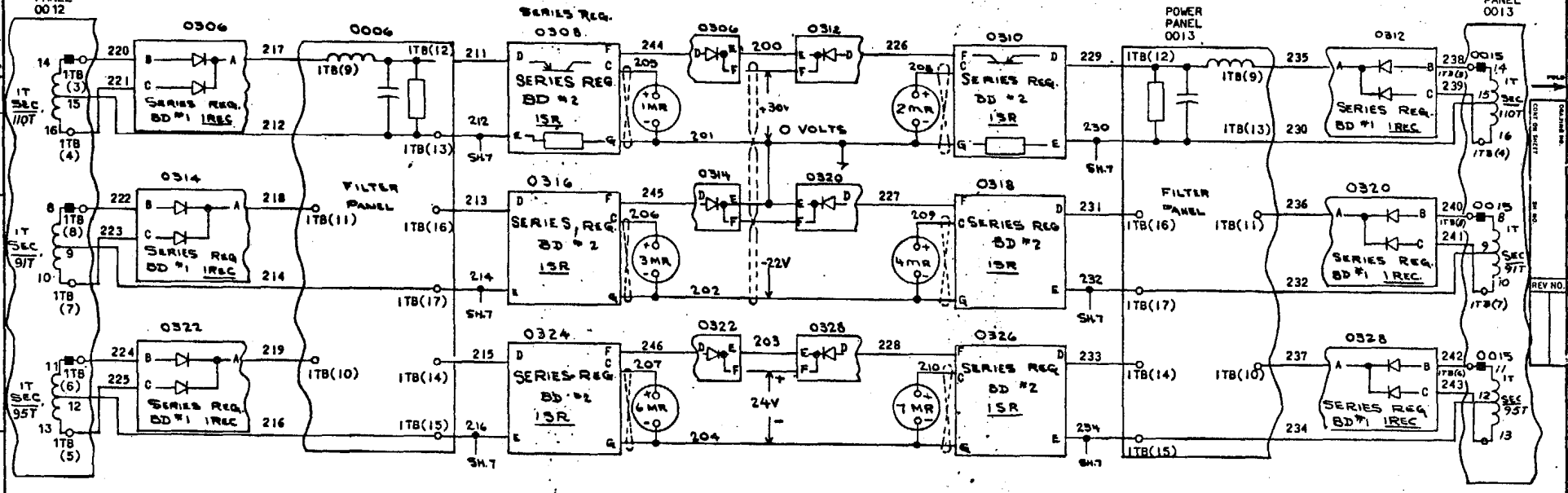
CONT ON SHEET 3 SH NO 2




LINE A POWER SUPPLY




LINE B POWER SUPPLY




REVISIONS

PRINTS TO
711
12P-DFB²
FG-W

Designed by Crumpler 4-11-75
Issued APR 18 1978
Approved R.H. Jr. R. Culver
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C309397
CONT ON SHEET 3 SH NO 2
3+1PL

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES | TOLERANCES OR MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A17, 091-41761 TB 153090
ELEMENTARY DIAGRAM
LOWER COLORADO RIVER AUTHORITY
FAYETTE #1, TB153090
FAYETTE #2, TB153091

SIGNAL CONVERTER POSITION FROM SH. 4 — 421

0108 — 3AØ2 — IN, GAIN, OUT, D, E, G, I OUT, BIAS, H, J, L, A, B, C — 200 +30V, 201 0V, 202 −22V

418 TO FEEDWATER CONTROLLER SH. 4

0110 — 3AØ2 — IN, GAIN, OUT, D, E, G, I OUT, BIAS, H, J, L, A, B, C — +30V 0V −22V

335 — T91 N10 — TI-1 0-1MA
T92 N9 — N11
339 — T94 — 0-1MA — ANALOG SPEED SIGNAL FOR CUSTOMER USE
201 — T93

1CO-M ▲ 337 — 0-8000 RPM — 201 — 1CO-N
1CO-V

30V 200 | 0V 201 | −22V 202 | 0224 — COARSE A | B | C FINE — 0222 — 1ØA — 317 H | D | 12 | E | D | J

329 SPEED SIGNAL FROM SH. 6 330 — F, E — 2FVC — SH. 7

316

+30 0V −22 — 200 201 202 — 0218 — L, M, N — A, B, C — D, G, E, H, F, J — 10% SPEED — IVC — 505 SH. 5 — 321 SH. 5

VOLTAGE COMPARATORS ARE NORMALLY PICKED UP AND WILL DROP OUT AT OPERATING POINT OR SPEED

0220 — L, M, N — D, G, E, H, F, J — A, B, C — 110% SPEED — IVC — 200 201 202 +30V 0V −22V
203 — SH. 12
1263 — SH. 12 (110SX)

SEE SH. 10 — 5MR — 322 — 201 — 300

0217 — D — F,H,K,M — E,G,J,L — 70SC AMPLIFIER — A B C — 200 201 202 +30V 0V −22V
6V, RMS 3KHZ SH. 4,5,8,9 LVDT EXCITATION

0219 — L, M, N — D, G, E, H, F, J — A, B, C — 105% SPEED — IVC — 200 201 202 +30V 0V −22V
320 — SH. 11(105X)
203 — +24V

0221 — L, M, N — D, G, E, H, F, J — A, B, C — 200 RPM SPEED — IVC — 200 201 202 +30V 0V −22V
316
1250 — TO LSS SH. 5
1249 — TO SH. 12
514 — SH. 5
510 — SH. 5

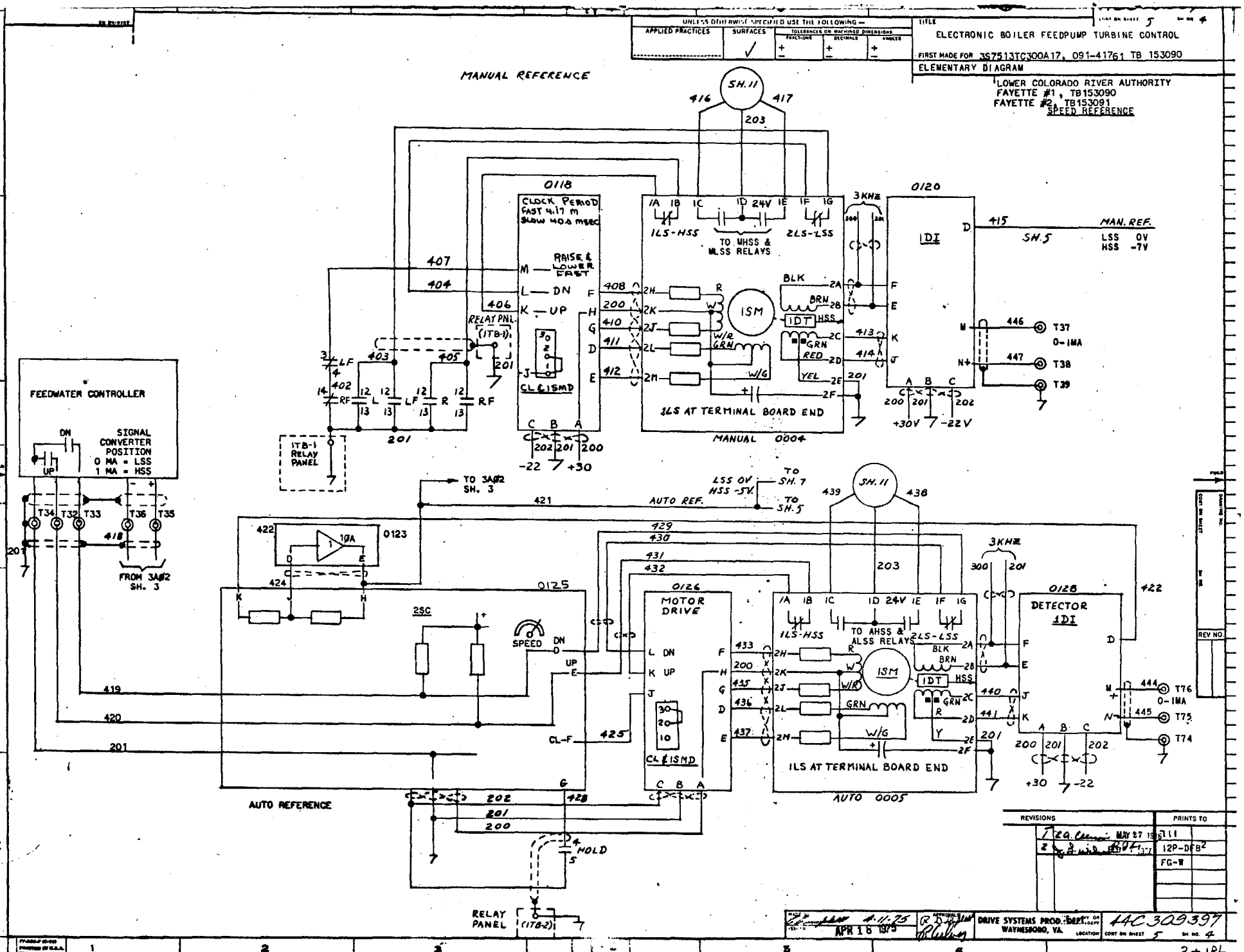
▲ INDICATES ITEM MOUNTED ON REMOTE CONTROL PANEL

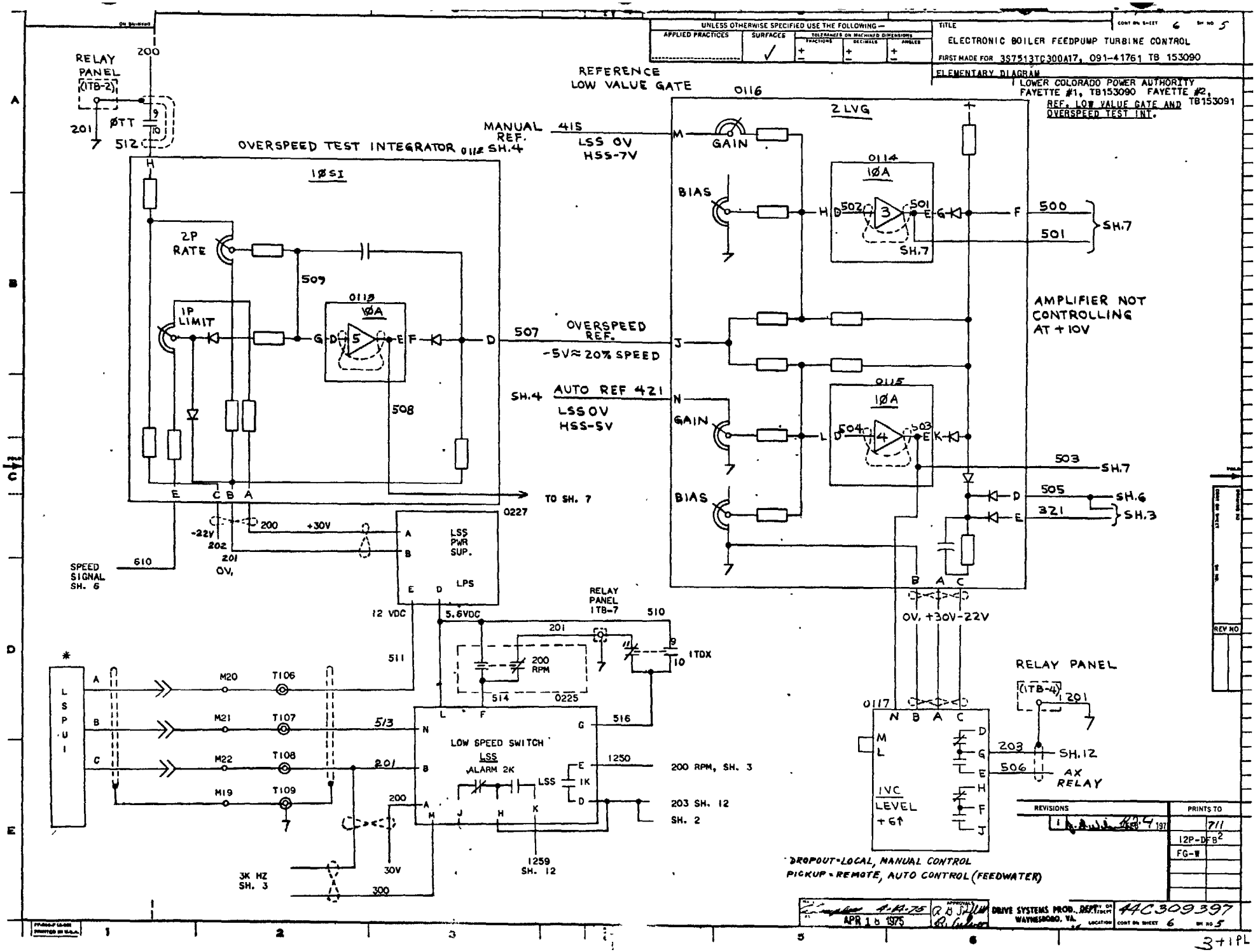
CONT ON SHEET — REV NO

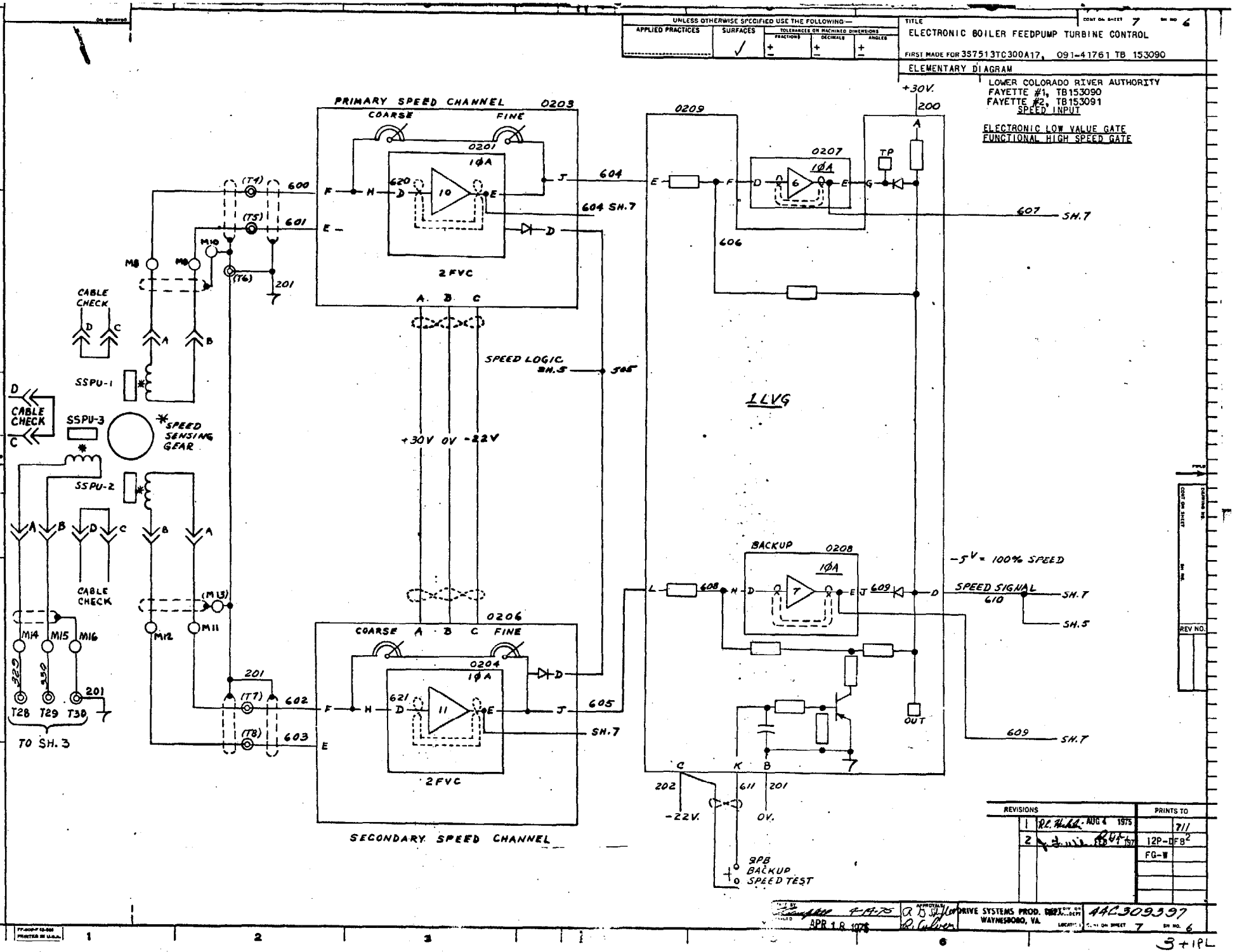
REVISIONS | PRINTS TO
1 — MAY 27 1976 — 711
2 — FEB 1 1977 — 12P-DFB
3 — JUN 8 1977 — FG-W

APR 18 1975 — DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA. — 44C309397 — SH. NO 4 — SH. NO 3

3+1PL


UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A17, 091-41761 TB 153090
ELEMENTARY DIAGRAM
LOWER COLORADO RIVER AUTHORITY
FAYETTE #1, TB153090
FAYETTE #2, TB153091
SPEED REFERENCE
MANUAL REFERENCE
FEEDWATER CONTROLLER
SIGNAL CONVERTER POSITION
0 MA = LSS
1 MA = HSS
FROM 3A@2 SH. 3
TO 3A@2 SH. 3
ITB-1 RELAY PANEL
RELAY PNL (ITB-1)
CLOCK PERIOD
FAST 4.17 m
SLOW 40.8 msec
RAISE & LOWER FAST
DN
UP
CL & ISMD
ILS-HSS
TO WHSS & MLSS RELAYS
2LS-LSS
ILS AT TERMINAL BOARD END
MANUAL 0004
ISM
IDT
HSS
3KHz
IDI
MAN. REF.
LSS 0V
HSS -7V
T37 0-1MA
T38
T39
+30V
-22V
AUTO REF.
LSS 0V HSS -5V
MOTOR DRIVE
TO AHSS & ALSS RELAYS
DETECTOR
T76 0-1MA
T75
T74
AUTO 0005
AUTO REFERENCE
HOLD
RELAY PANEL (ITB-2)
REVISIONS
PRINTS TO
12P-DFB2
FG-W
DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA.
44C309397
APR 16 1975


UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A17, 091-41761 TB 153090
ELEMENTARY DIAGRAM
LOWER COLORADO POWER AUTHORITY
FAYETTE #1, TB153090 FAYETTE #2, TB153091
REF. LOW VALUE GATE AND OVERSPEED TEST INT.
REFERENCE
LOW VALUE GATE
0116
2 LVG
MANUAL REF.
415
LSS 0V
HSS-7V
GAIN
BIAS
0114
1ØA
SH.7
500
501
AMPLIFIER NOT CONTROLLING AT +10V
OVERSPEED TEST INTEGRATOR 0112 SH.4
1ØSI
2P RATE
1P LIMIT
509
0113
ØA
508
507
OVERSPEED REF.
-5V≈20% SPEED
TO SH. 7
0227
SH.4
AUTO REF 421
LSS 0V
HSS-5V
0115
503
505
SH.6
321
SH.3
200
-22V
202
+30V
201
0V,
LSS PWR SUP.
LPS
SPEED SIGNAL SH. 6
610
12 VDC
5.6VDC
RELAY PANEL 1TB-7
510
511
200 RPM
514
0225
1TDX
0V, +30V-22V
RELAY PANEL
(1TB-4)
201
0117
IVC LEVEL +6↑
203
506
SH.12
AX RELAY
LSPUI
M20
M21
M22
M19
T106
T107
T108
T109
513
201
LOW SPEED SWITCH
LSS
ALARM 2K
LSS
1K
1250
200 RPM, SH. 3
203 SH. 12
SH. 2
1259
SH. 12
3K HZ
SH. 3
30V
300
DROPOUT=LOCAL, MANUAL CONTROL
PICKUP=REMOTE, AUTO CONTROL (FEEDWATER)
REVISIONS
PRINTS TO
711
12P-DFB2
FG-W
APR 16 1975
DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA.
44C309397
CONT ON SHEET 6
SH NO 5
3+1PL


UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A17, 091-41761 TB 153090
ELEMENTARY DIAGRAM
LOWER COLORADO RIVER AUTHORITY
FAYETTE #1, TB153090
FAYETTE #2, TB153091
SPEED INPUT
ELECTRONIC LOW VALUE GATE
FUNCTIONAL HIGH SPEED GATE
PRIMARY SPEED CHANNEL 0203
COARSE
FINE
0201
1ØA
2FVC
SECONDARY SPEED CHANNEL
0206
0204
0209
0207
0208
BACKUP
+30V.
200
604
604 SH.7
605
606
607 SH.7
608
609
610
609 SH.7
SH.7
SH.5
SPEED LOGIC SH.5
565
+30V 0V -22V
1LVG
-5V = 100% SPEED
SPEED SIGNAL
OUT
600
601
602
603
201
202
611
-22V.
0V.
(T4)
(T5)
(T6)
(T7)
(T8)
(M13)
M8
M9
M10
M11
M12
M14
M15
M16
329
330
T28
T29
T30
TO SH. 3
CABLE CHECK
SSPU-1
SSPU-2
SSPU-3
SPEED SENSING GEAR
SPB BACKUP SPEED TEST
REVISIONS
PRINTS TO
12P-DFB2
FG-W
DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA
44C309397
APR 18 1975
CONT ON SHEET 7
SH NO 6
REV NO.
3+1PL

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| | ✓ | ± | ± | ± |

TITLE

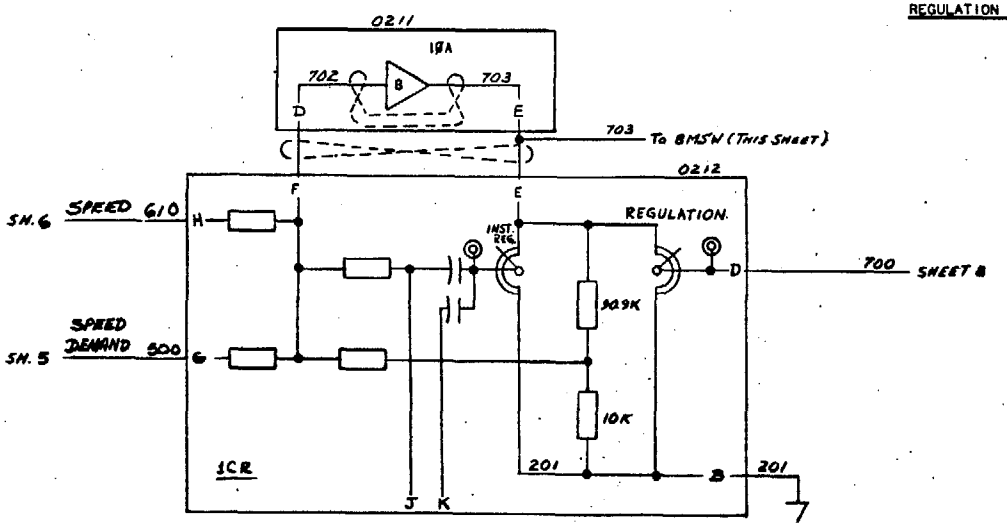
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

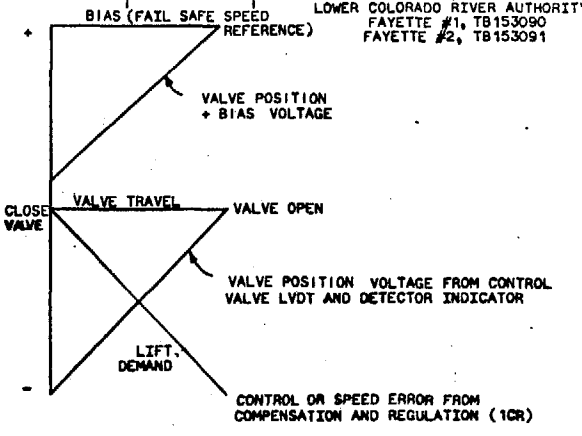
FIRST MADE FOR 3S7513TC300A17 091-41761 T8 153090

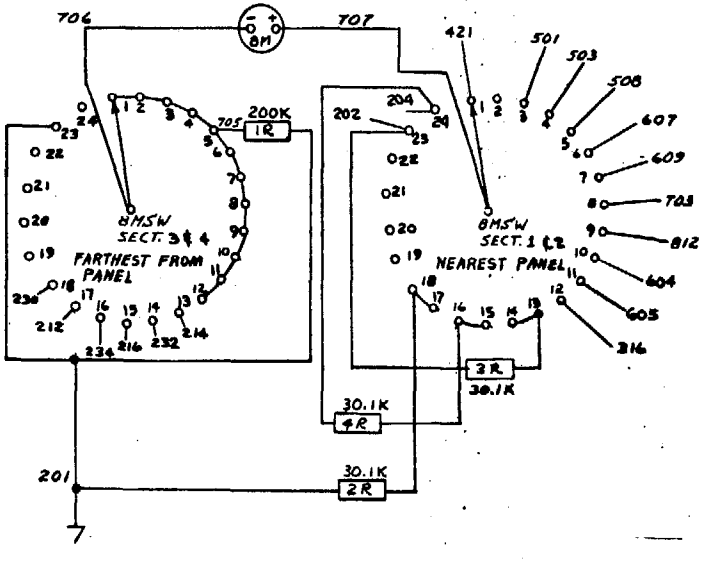
ELEMENTARY DIAGRAM

LOWER COLORADO RIVER AUTHORITY
FAYETTE #1, TB153090
FAYETTE #2, TB153091

## REGULATION AND STABILIZATION




AMPLIFIER

| SWITCH POSITION | NUMBER | SYMBOL | FUNCTION | RACK LOCATION | APPROXIMATE OUTPUT VOLT RANGE | SHEET NO. |
|---|---|---|---|---|---|---|
| 1 | 1 | 2SC | SIG. CONV. | 0123 | 0 TO -5VDC | 4 |
| | | | | | | 4 |
| 3 | 3 | 2LVG | MAN. REF. | 0114 | 10VDC TO -5VDC | 5 |
| 4 | 4 | 2LVG | AUTO REF. | 0115 | 10VDC TO -5VDC | 5 |
| 5 | 5 | O'SPD | OVERSPEED | 0113 | 10VDC TO -5VDC | 5 |
| 6 | 6 | 1LVG | PRI. SPD. | 0207 | 10VDC TO -7VDC | 6 |
| 7 | 7 | 1LVG | SEC. SPD. | 0208 | 10VDC TO -7VDC | 6 |
| 8 | 8 | 1CR | COMP. & REG. | 0211 | 10VDC TO -10VDC | 7 |
| 9 | 9 | 2SA | SERVO AMP. | 0214 | 7VDC TO -7VDC | 8 |
| 10 | 10 | 2FVC | PRI. SPD. | 0201 | 10VDC TO -7VDC | 6 |
| 11 | 11 | 2FVC | SEC. SPD. | 0204 | 0 TO -7VDC | 6 |
| 12 | 12 | 2FVC | SPD. IND. | 0222 | 0 TO -7VDC | 3 |

| POWER SUPPLY | | | | | APPROXIMATE OUTPUT CURRENT RANGE | |
|---|---|---|---|---|---|---|
| 13 | LINE A | 1SR | -22VDC | 0316 | 5 AMPS | 2 |
| 14 | LINE B | 1SR | -22VDC | 0318 | | 2 |
| 15 | LINE A | 1SR | +24VDC | 0322 | | 2 |
| 16 | LINE B | 1SR | +24VDC | 0328 | | 2 |
| 17 | LINE A | 1SR | +30VDC | 0308 | | 2 |
| 18 | LINE B | 1SR | +30VDC | 0310 | 5 AMPS | 2 |

REVISIONS

| | | | |
|---|---|---|---|
| 1 | R.E. Hall | AUG 4 1975 | |
| 2 | | FEB 1977 | |

PRINTS TO

| | |
|---|---|
| | 711 |
| | 12P-DFB2 |
| FG-8 | |

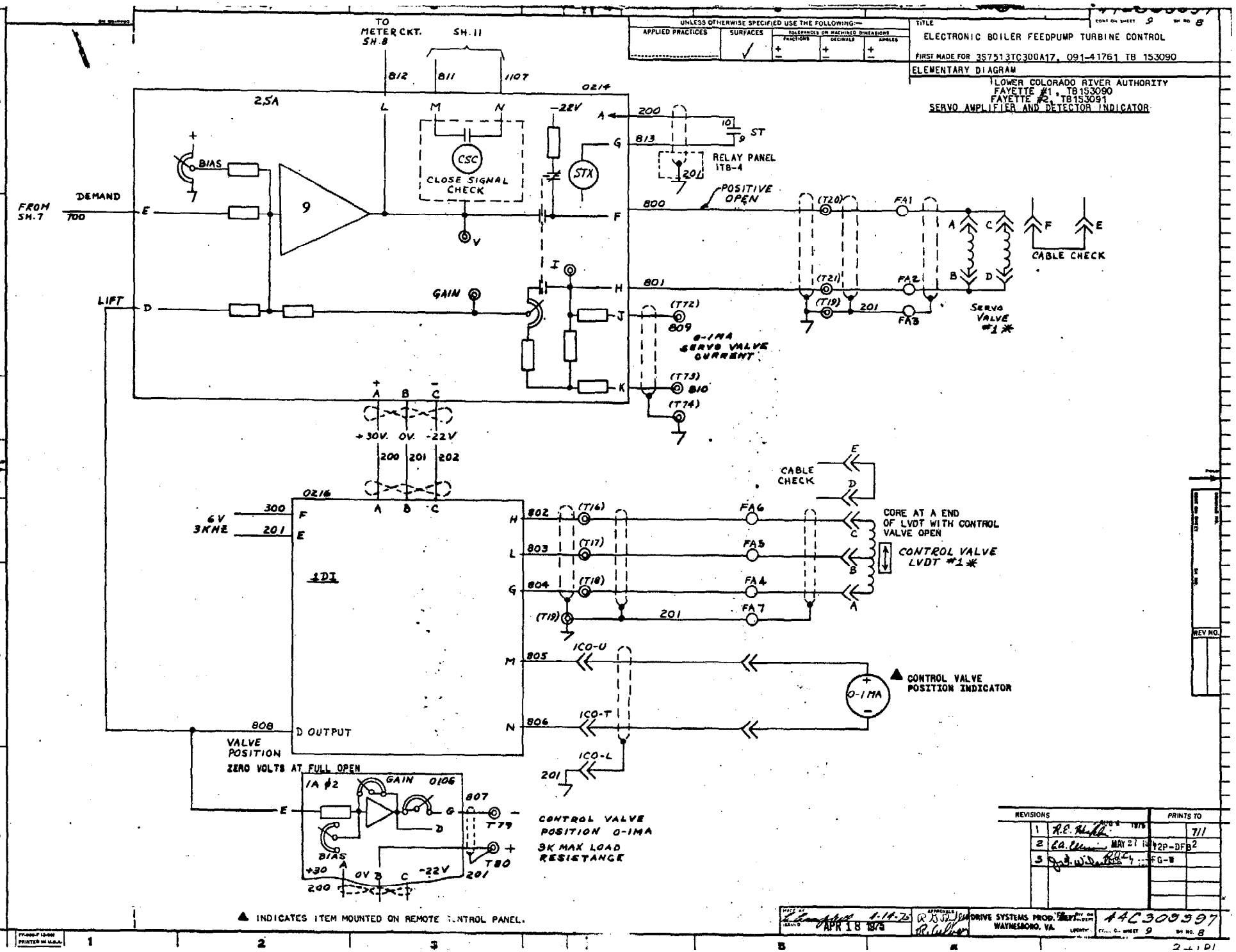
APR 18 1975

DRIVE SYSTEMS PROD. WAYNESBORO, VA.

44C309597

SH. NO. 7

3+IPL

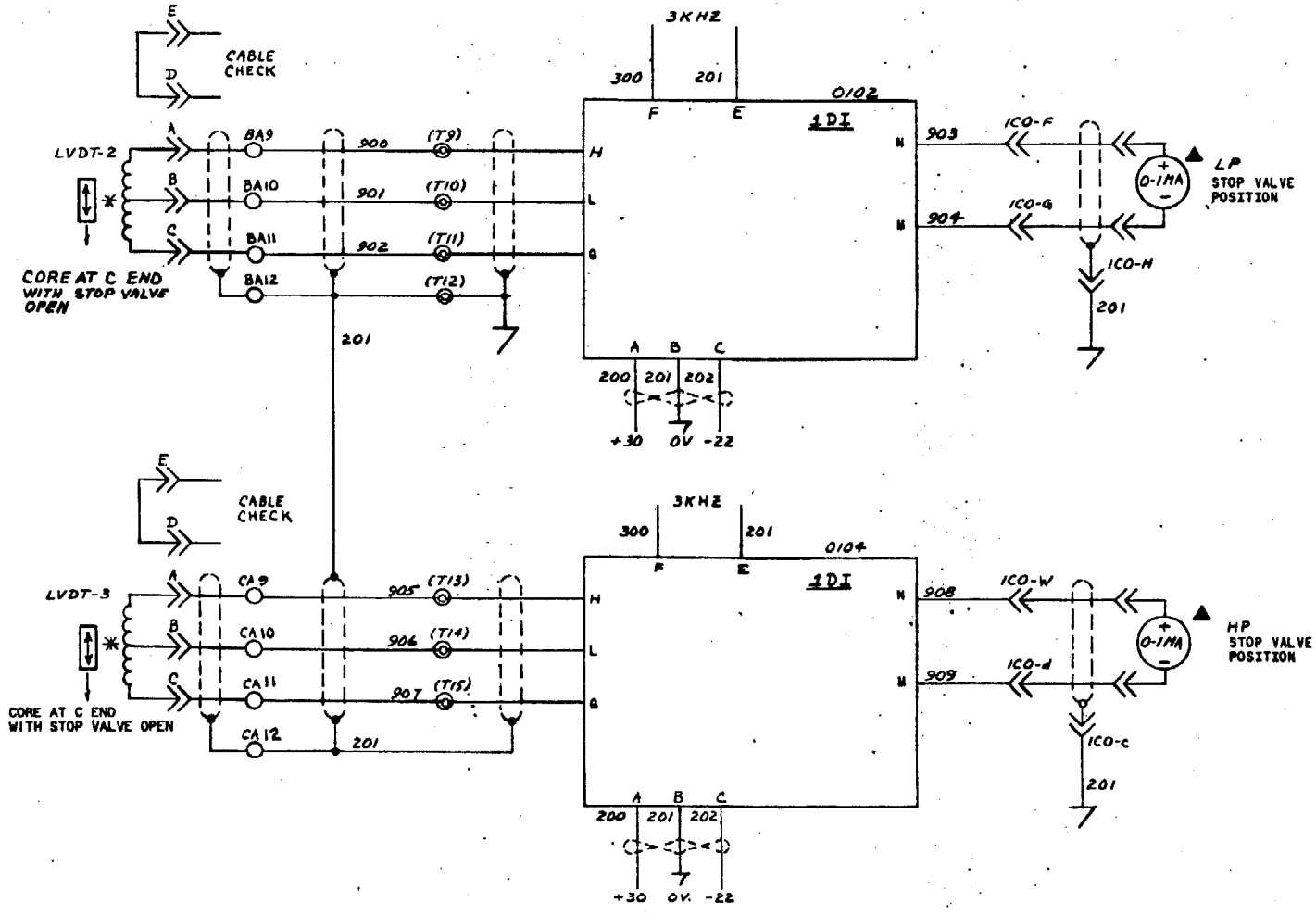

TO METER CKT. SH.8
SH.11
812
811
1107
0214
2SA
L
M
N
-22V
A
200
10
9
ST
G
813
RELAY PANEL 1TB-4
20
CSC
CLOSE SIGNAL CHECK
STX
BIAS
9
FROM SH.7
DEMAND
700
E
V
F
800
POSITIVE OPEN
(T20)
FA1
A
C
F
E
CABLE CHECK
I
H
801
GAIN
LIFT
D
(T21)
FA2
B
D
(T19)
201
FA3
SERVO VALVE #1 *
J
(T72)
809
0-1MA SERVO VALVE CURRENT
K
(T73)
810
(T74)
+
A
B
C
-
+30V.
0V.
-22V
200
201
202
0216
300
6V 3KHZ
201
F
E
1DI
H
802
(T16)
FA6
CABLE CHECK
E
D
CORE AT A END OF LVDT WITH CONTROL VALVE OPEN
L
803
(T17)
FA5
C
B
CONTROL VALVE LVDT #1 *
G
804
(T18)
FA4
A
(T19)
201
FA7
M
805
ICO-U
0-1MA
+
-
▲ CONTROL VALVE POSITION INDICATOR
N
806
ICO-T
ICO-L
201
808
D OUTPUT
VALVE POSITION
ZERO VOLTS AT FULL OPEN
1A #2
GAIN
0106
807
E
G
D
T79
T80
+
BIAS
A
+30
0V
B
C
-22V
201
200
▲ INDICATES ITEM MOUNTED ON REMOTE CONTROL PANEL.
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A17, 091-41761 TB 153090
ELEMENTARY DIAGRAM
LOWER COLORADO RIVER AUTHORITY
FAYETTE #1, TB153090
FAYETTE #2, TB153091
SERVO AMPLIFIER AND DETECTOR INDICATOR
REVISIONS
1
2
3
PRINTS TO
12P-DFB2
FG-B
APR 18 1975
DRIVE SYSTEMS PROD.
WAYNESBORO, VA.
44C300397
2-1DI

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| | ✓ | ± | ± | ± |

CONT ON SHEET 10 SH NO 9

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 357513TC300A17, 091-41761 TB 153090
ELEMENTARY DIAGRAM
LOWER COLORADO RIVER AUTHORITY
FAYETTE #1, TB153090 FAYETTE #2, TB153091
STOP VALVE POSITION




▲ INDICATES ITEMS MOUNTED ON REMOTE CONTROL PANEL

| REVISIONS | | | PRINTS TO |
|---|---|---|---|
| 1 | R.R. Hall AUG 4 1975 | | 7/11 |
| 2 | G.A. Cline MAY 27 1976 | | 12P-DFB2 |
| 3 | [illegible] | | FG-W |

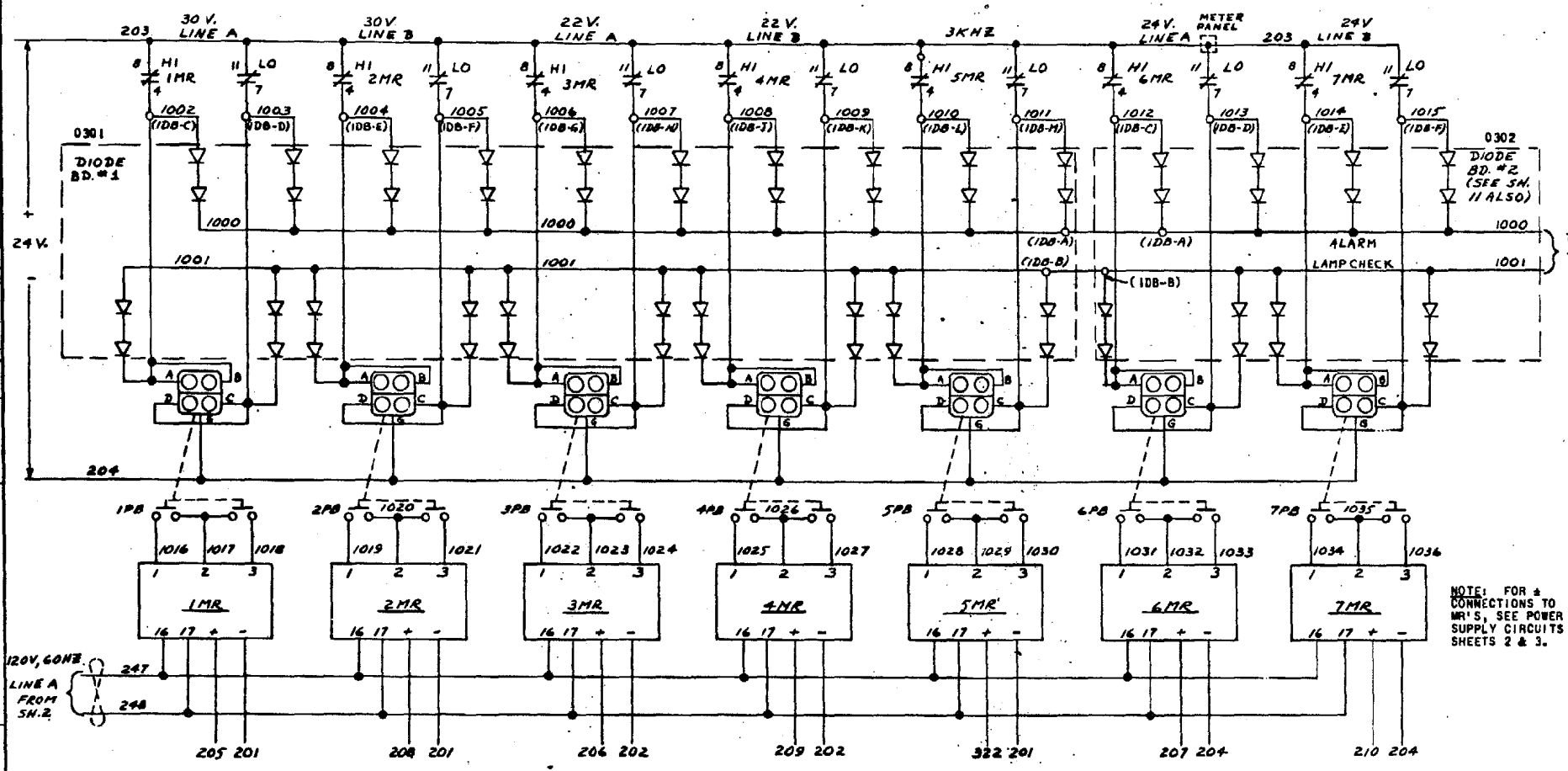
| | | | | |
|---|---|---|---|---|
| ISSUED APR 18 1975 | APPROVALS G.A. Riley R. Calvin | DRIVE SYSTEMS PROD. WAYNESBORO, VA. | DEPT. OR DEPT. LOCATION | 44C309397 CONT ON SHEET 10 SH NO. 9 |

3+1PL

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

CONT ON SHEET 11 SH NO 10

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A17, 091-41761 TB 153090
ELEMENTARY DIAGRAM
LOWER COLORADO RIVER AUTHORITY
FAYETTE #1, TB153090
MONITOR FAYETTE #2, TB153091



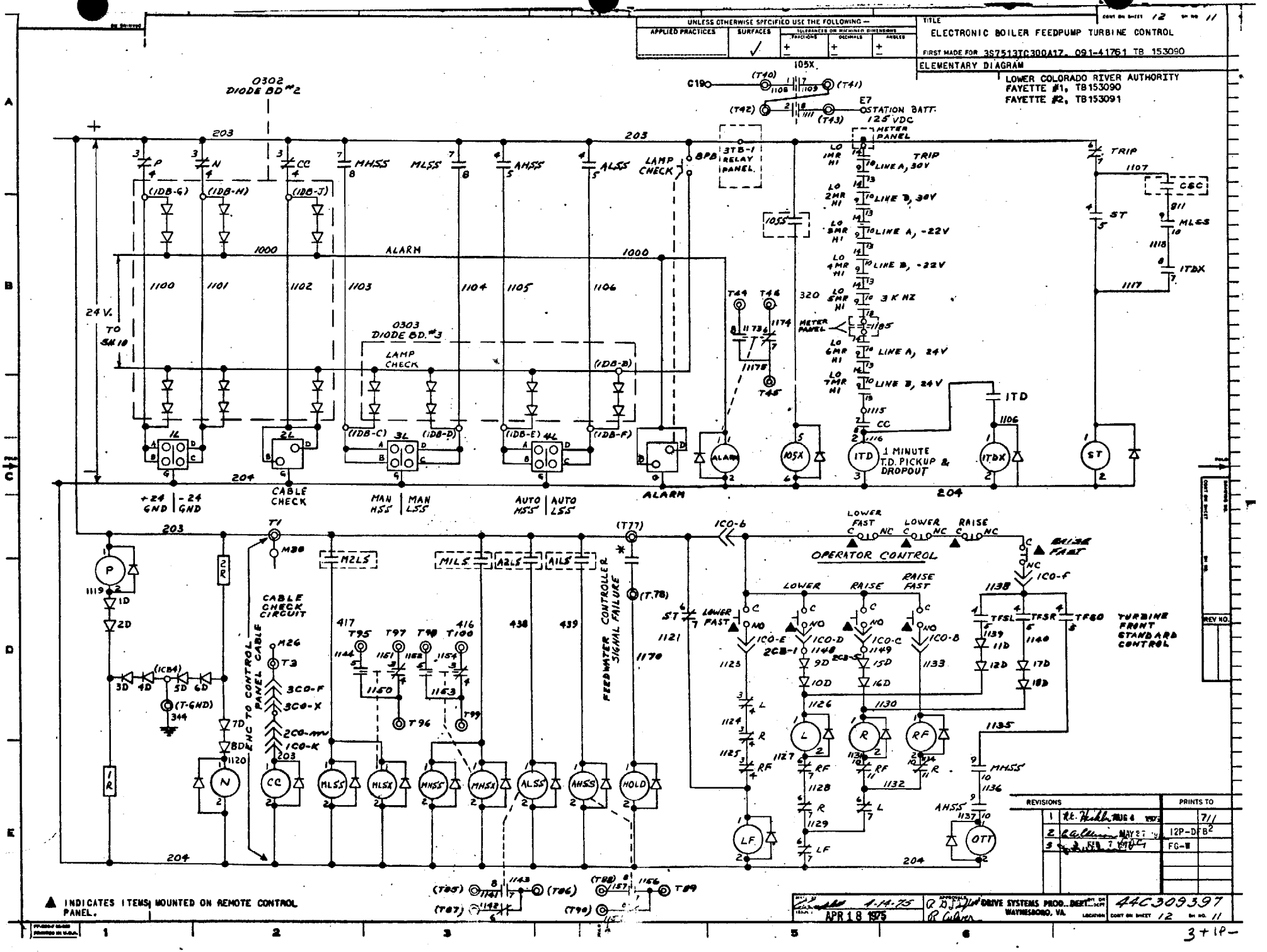

NOTE: FOR ± CONNECTIONS TO MR'S, SEE POWER SUPPLY CIRCUITS SHEETS 2 & 3.

REVISIONS
1 MAY 27 1977
2 JUN 1 1977

PRINTS TO
7/11
12P-DFB 2
FG-W

MADE BY APR 18 1975
APPROVED
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
4C309397
SHEET 11 SH NO. 10

3+1PL


UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A17, 091-41761 TB 153090
ELEMENTARY DIAGRAM
LOWER COLORADO RIVER AUTHORITY
FAYETTE #1, TB153090
FAYETTE #2, TB153091
105X
0302 DIODE BD. #2
0303 DIODE BD. #3
203
204
1000
ALARM
1100
1101
1102
1103
1104
1105
1106
24 V. TO SH 10
LAMP CHECK
3TB-1 RELAY PANEL
METER PANEL
TRIP
LINE A, 30V
LINE B, 30V
LINE A, -22V
LINE B, -22V
3 K HZ
LINE A, 24V
LINE B, 24V
1 MINUTE T.D. PICKUP & DROPOUT
ITD
ITDX
ST
C&C
MLSS
CABLE CHECK
MAN HSS | MAN LSS
AUTO HSS | AUTO LSS
+24 GND | -24 GND
OPERATOR CONTROL
LOWER FAST
LOWER
RAISE
RAISE FAST
TURBINE FRONT STANDARD CONTROL
CABLE CHECK CIRCUIT
FEEDWATER CONTROLLER SIGNAL FAILURE
ENC TO CONTROL PANEL CABLE
▲ INDICATES ITEMS MOUNTED ON REMOTE CONTROL PANEL.
REVISIONS
PRINTS TO
DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA.
44C309397
APR 18 1975

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3875133C300A17, 091-41761 TB 153090

ELEMENTARY DIAGRAM

LOWER COLORADO RIVER AUTHORITY
FAYETTE #1, TB153090
FAYETTE #2, TB153091

CONT ON SHEET 13 DWG NO 12

OPERATOR'S CONTROL PANEL INDICATING LIGHTS

3TB RELAY PANEL

203

OPEN WHEN ON MANUAL CONTROL

REM

1R LSS OPEN ABOVE 2 RPM

1250

200 RPM OPEN ABOVE 200 RPM

1249

1TDX

1216

T110 T112

1256 1258

1267

T111

1248

24 V.

506

2R LSS ALARM

T113 T115

1260 1262

1261

T114

1269

204

110S OPEN ABOVE 110% SPEED

1263

T47 T49

1264 1266

1265

T48

AX 1219

MLSS 1222

MHSS 1223

ALARM 1224

ALARM 1225

2FL 1226

21D

22D

2CR-20

ICO-S 1267

REMOTE CONTROL PANEL

ICO-R ICO-P ICO-mu ICO-X

LAMP TEST

ICO-L

MANUAL/AUTO CONTROL

OVERSPEED TRIP TEST

MANUAL LSF

MANUAL HSS

NORMAL RED

YELLOW FLASH ABNORM POWER

3R

2TB RELAY PANEL

AX

LSSX ENERGIZED BELOW TURNING GEAR SPEED

LSS ALARM X

110 SX

(TB2)

(TB8) (TB4)

ENERGIZED WHEN ON AUTO CONTROL

(T59) 1227 (2785) 1X

1C 40MFD

(ICB-20) (T-GND)

125 V. BATTERY

2C 40MFD

1228 (T60) (2786-6) 2X 1230

(T54) 1229 (T27)

F1 F3 F5 M17

PB14 INCREASE SPEED P

PB15 DECREASE SPEED

PB19 OVERSPEED TRIP TEST

PS10 CLOSED ON TRIP

M18

G1 G3 G5

1240 1241 1243 1244

(T55) (T56) (T58) (T26)

TPSA TPSL TPSA TRIP

1230

▲ INDICATES ITEMS MOUNTED ON CONTROL PANEL

REVISIONS

| | | | |
|---|---|---|---|
| 1 | T.R. Kuster | MAY 6 1975 | |
| 2 | H.E. Hinkle | AUG 4 1975 | |
| 3 | F.A. Knight | DEC 15 1975 | |
| | | MAY 27 1976 | |
| 4 | C.G. Clemens | JUNE 7 1976 | |
| 5 | | | |

PRINTS TO

| | |
|---|---|
| 711 | |
| 12P-DRB | 2 |
| FG-W | |

DRAWN R.B.D. APR 18 1975
APPROVED R. Gilmore
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C309397
CONT ON SHEET 13 DWG NO 12

3+1PL

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

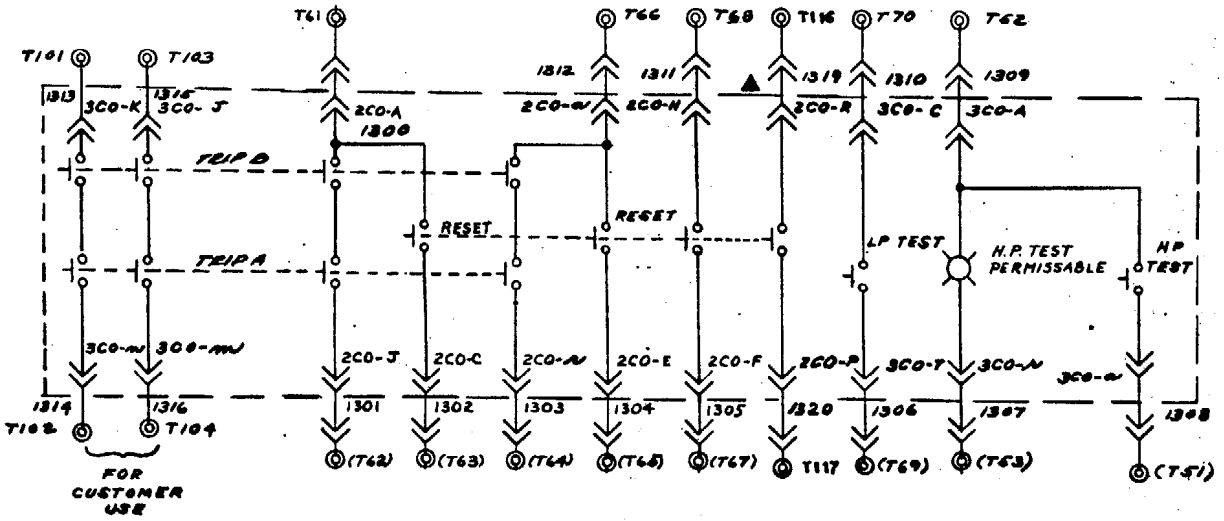
DRWG SH SHEET FNL SH NO 13

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A17, 091-41761 TB 153090

ELEMENTARY DIAGRAM

LOWER COLORADO RIVER AUTHORITY
FAYETTE #1, TB153090
FAYETTE #2, TB153091




▲ INDICATES ITEMS MOUNTED ON CONTROL PANEL

| REVISIONS | | | PRINTS TO |
|---|---|---|---|
| 1 | R.C. Hill AUG 4 1975 | | 711 |
| 2 | [illegible] DEC 15 1975 | | 12P-DFB2 |
| 3 | [illegible] FEB 1 1976 | | FG-W |
| 4 | [illegible] MAR 1 1977 | | |

APR 18 1978

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44C309397

FNL SH NO 13

341PL

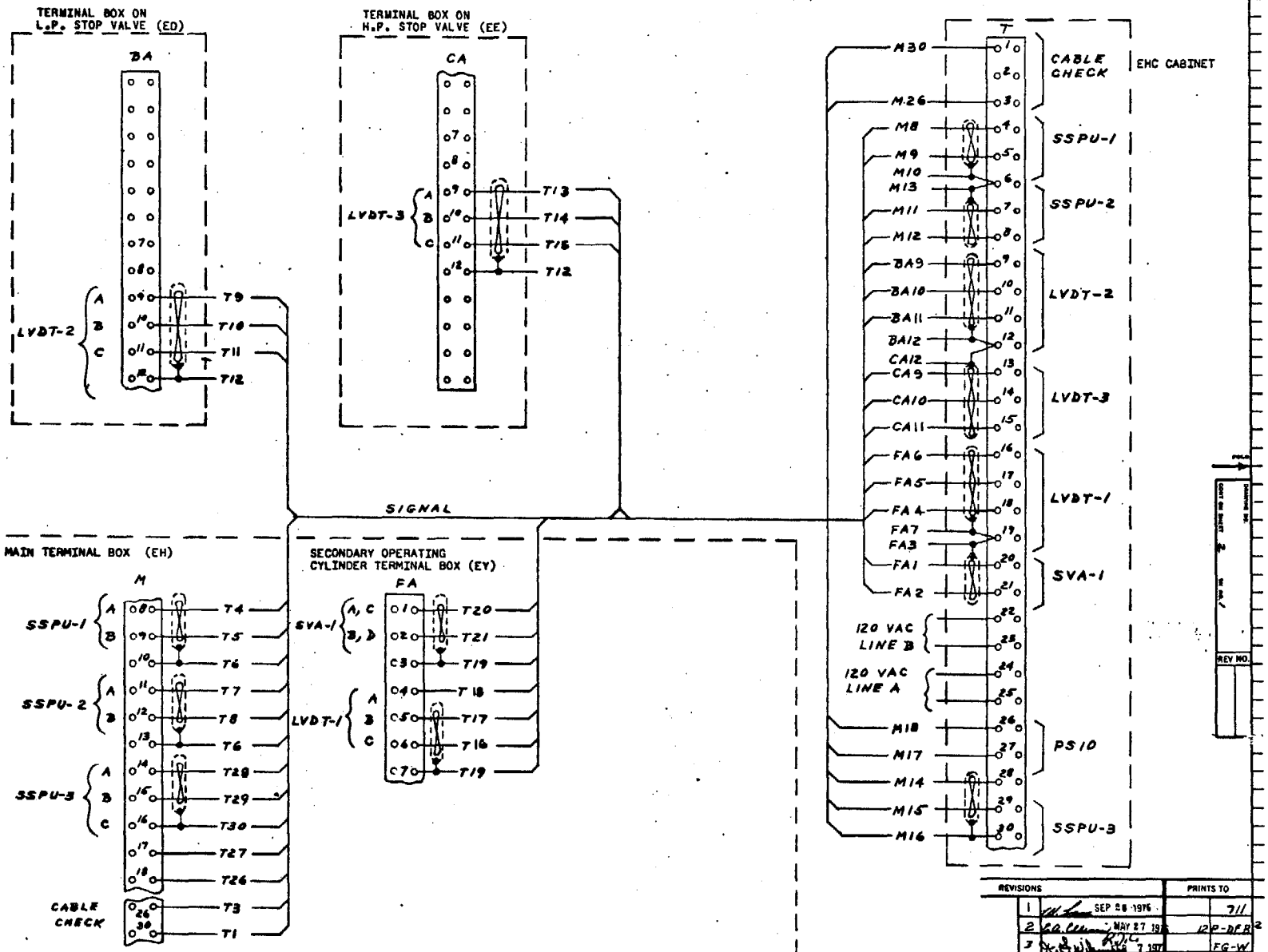
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A17 TURBINE: 153090 091-41761
INTERCONNECTION DIAG.
FAYETTE #1 LOWER COLORADO RIVER AUTHORITY, 153090
FAYETTE #2 LOWER COLORADO RIVER AUTHORITY, 153091

CONT ON SHEET 2 SH NO. 1

NOTE:

1. REFER TO 44C309397 (EHC ELEMENTARY) AND 134B550XF (TURBINE WIRING DWG.) FOR REFERENCE.




REVISIONS

| | | |
|---|---|---|
| 1 | SEP 26 1976 | |
| 2 | MAY 27 1977 | |
| 3 | JUL 7 1977 | |

PRINTS TO
711
12P-DFB 2
FG-W
12P-F 5 2

K.E. Hall 8/1/75 AUG 5 1975 | R.A. Sutton | DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA. | 44C309398 | CONT ON SHEET 2 SH NO. 1

1 2 3 4 5 6

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A17 TURBINE 153090 091-41761
INTERCONN. DIAG.
FAYETTE #1 LOWER COLORADO RIVER AUTHORITY TB153090
FAYETTE #2 LOWER COLORADO RIVER AUTHORITY TB153091

EHC CABINET
EHC REMOTE PANEL
TRIP
T
0103 CUSTOMER USE
0104
0105
0106 M20
0107 M21
0108 M22
0109 M19
LOW SPEED SWITCH
0110
0111
0112
0113
0114
0115
LSSX
LSS ALARMX
FOR CUSTOMER USE
EHC REMOTE PANEL
RE-SET FB
0116 B7
0117 B3
0118

MAIN TERMINAL BOX ON OIL TANK (EH)
D
3 T117
7 T116
28 T53
C
19 T40
G
1 T55
3 T56
5 T58
M
19 T109
LSPU A 20 T106
B 21 T107
C 22 T108
F
1 T54
3
5
E
7 T43
30 T52

TERMINAL BOX ON HP STOP VALVE (EE)
CB
7 T61
8

EHC CABINET
T
310
320 UP
330 DN
340
350 AUTO SPEED REFERENCE POSITION 0-1 MA
360
370 MANUAL SPEED REFERENCE POSITION 0-1MA
380
390
INPUT FROM FEEDWATER CONTROLLER
FOR CUSTOMER USE
C19 400 105X
410
420 105X
E7 430
440 EHC ABNORMAL CONDITION ALARM
450
460
FOR CUST. USE
470 110% SPEED
480
490 EHC REMOTE PANEL
500
C37 510
E30 520 HP TEST PERMISSIBLE
D28 530
F1 540
G1 550
G3 560
570
G6 580
590
600
STATION BATTERY 125 VDC + −

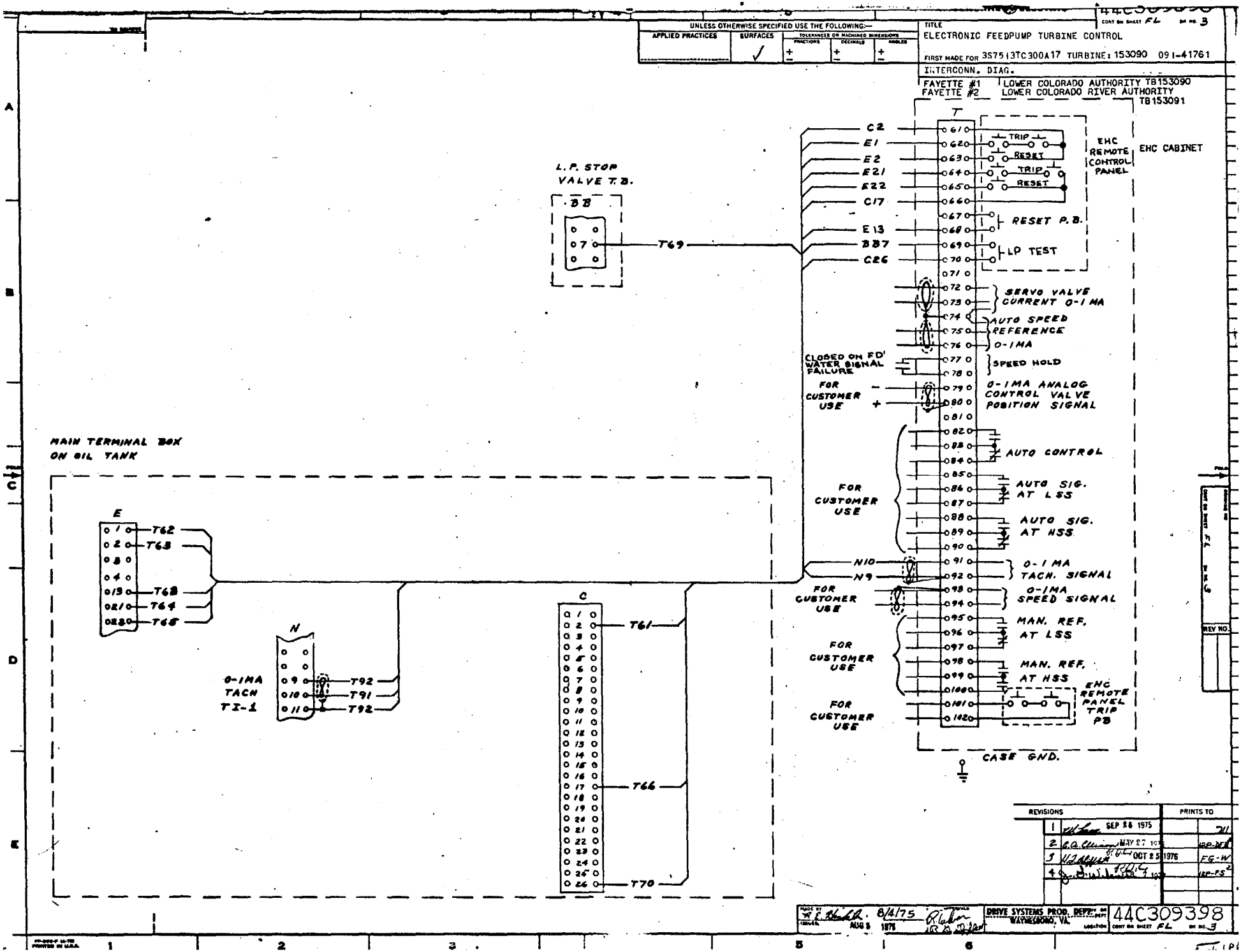
CONTROL PANEL
1CO
2CO
3CO
3 MULTICONDUCTOR 110 FT CABLES WITH PLUGS FURNISHED WITH EHC.

REVISIONS
1 SEP 26 1975
2 MAY 27 1977
3 JUN 1 1977

PRINTS TO
711
UEP-DFB 2
PG-W
UEP-TS 2

DRAWN R.P. Hall 8/1/75
TRACED AUG 5 1975
APPROVED R.A. Stafford
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C309398
CONT ON SHEET 3 SH NO 2

5 7 4 1 PL


UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE
ELECTRONIC FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S75 | 3TC300A17 TURBINE: 153090 091-41761
INTERCONN. DIAG.
FAYETTE #1 LOWER COLORADO AUTHORITY TB153090
FAYETTE #2 LOWER COLORADO RIVER AUTHORITY TB153091
L.P. STOP VALVE T.B.
TB
T69
MAIN TERMINAL BOX ON OIL TANK
E
T62
T63
T68
T64
T65
N
0-1MA TACH TI-1
T92
T91
T92
C
T61
T66
T70
T
C2
E1
E2
E21
E22
C17
E13
BB7
C26
EHC REMOTE CONTROL PANEL
TRIP
RESET
TRIP
RESET
RESET P.B.
LP TEST
EHC CABINET
SERVO VALVE CURRENT 0-1 MA
AUTO SPEED REFERENCE
0-1MA
SPEED HOLD
CLOSED ON FD' WATER SIGNAL FAILURE
FOR CUSTOMER USE
0-1MA ANALOG CONTROL VALVE POSITION SIGNAL
AUTO CONTROL
FOR CUSTOMER USE
AUTO SIG. AT LSS
AUTO SIG. AT HSS
N10
N9
0-1 MA TACH. SIGNAL
FOR CUSTOMER USE
0-1MA SPEED SIGNAL
MAN. REF. AT LSS
FOR CUSTOMER USE
MAN. REF. AT HSS
EHC REMOTE PANEL TRIP PB
FOR CUSTOMER USE
CASE GND.
REVISIONS
PRINTS TO
SEP 26 1975
OCT 28 1976
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C309398

GENERAL ELECTRIC 44C3C6444

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE: LOW SPEED SWITCH

FIRST MADE FOR 355T/8TC300A1

## CLOCK

6VRMS 3KHZ (M) — ½ W 6R 1K — 21 — 02 — 4TL — 21 — 1 CL

5.6 — 5.6V — 7C .0022MF — 0V

3D — 0V

21 — 2 CL 1500 HZ

## COUNTER

15SPU

2 CL 1500 HZ → 02 → 03 → 03 → 04 → 04 → 05 → 1 2 4 8 16 32

05 → 06 → 06 → 07 → 07 → 08 → 08 → 4096 → 255PU

64 128 256 512 3TP 1024 2048 4096

## ALARM

0V — 4C .015 MF — 200 (F) — 09 — 09 — 09 — 09 — ITDX

64 — 11

10 — 10 — Z — 10 — 10 — SET — 10 — 11 — ITDX — 3TL

355PU — SET — 20

0V ICL — 19 — 2TL — ALARM

5.6V — 10R 1K ½ W — ALARM RELAY

30V — 7R 180Ω ½ W — 2K — 5D — 4D — 27K ½ W 8R — 2Q — 5R 10Ω ½ W — 0V

K H J

## SPEED DETECTOR

1024 256 — 16 — 16 — 16 — 15 — 8

1024 512 — 16 — 16

4096 2048 — 17 — 17

## SPEED INHIBIT

SET — 15 — 15 — 15 — 47P

## STANDSTILL

0V ICL — 14 — 13 — 13 — SET

## SPEED

1TL — 0V ICL 55PU — 14 — 13 — 13

0V ICL — 12

0V ICL — 12

## DIGITAL OUTPUT

ALARM — 01 — 01 — 11 — 170

## LSS RELAY

+30V — A — 180Ω 2R ½ W — 1D — 2D — 1K — 1Q — 4R 10Ω ½ W — 0V

3R 27K ½ W — D — E

.015 MF 5C — ITDX — G — 01 — 01 — 01

1K, ½ W 1P — 5.6V — 6C .22MF — 0V

REVISIONS | PRINT NO | DRIVE SYSTEMS PROD WAYNES ORO

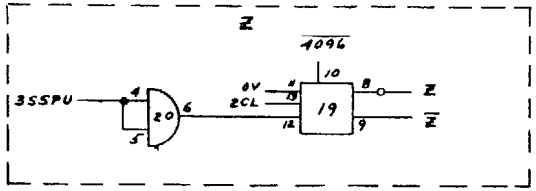
GENERAL ELECTRIC

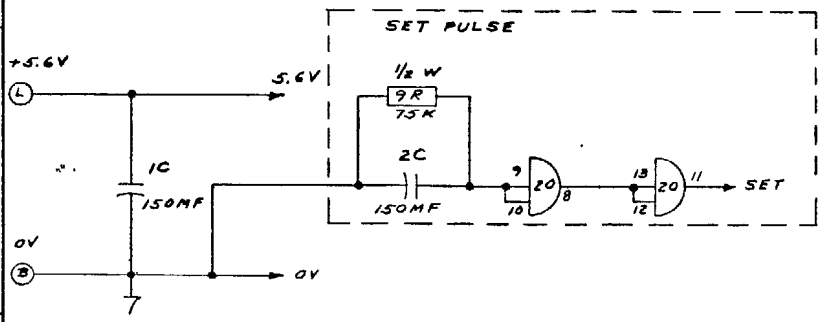
UNLESS OTHERWISE SPECIFIED IS TO BE THE FOLLOWING —

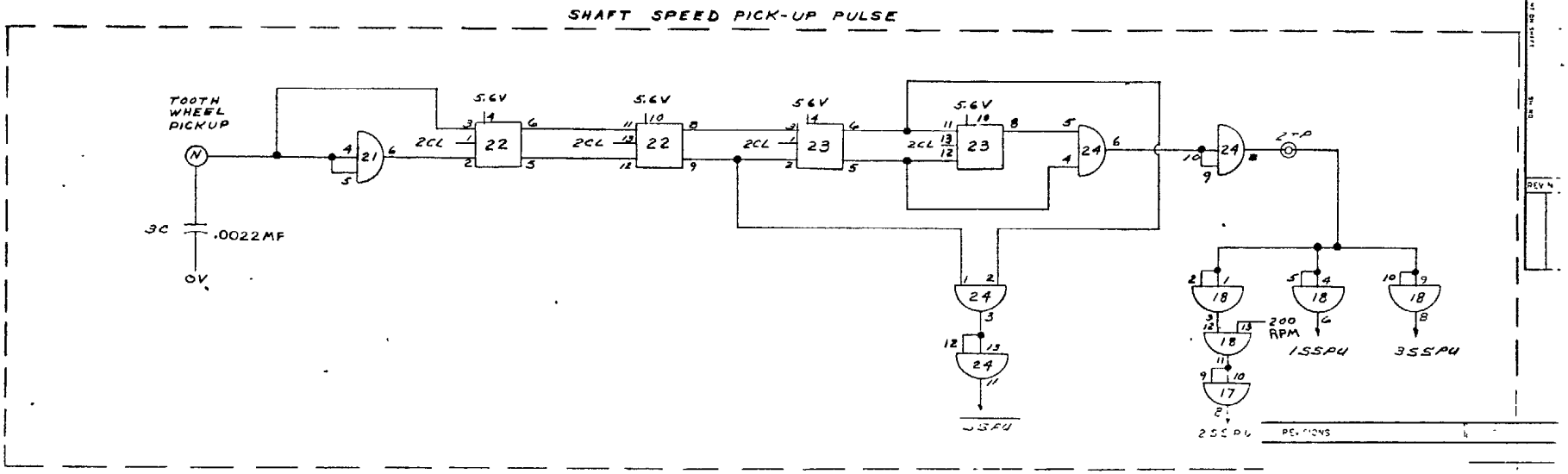
| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ± |
|---|---|---|---|---|
| | ✓ | | | |

TITLE: LOW SPEED SWITCH

FIRST MADE FOR 357513TC300A11



GENERAL ELECTRIC

4AC306526

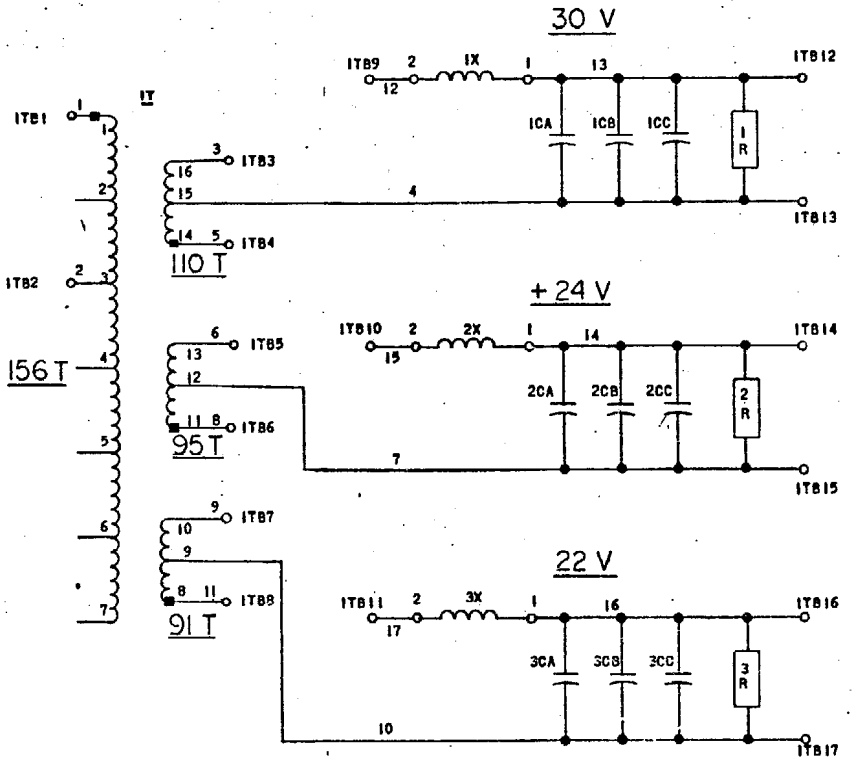
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

| APPLICO PRACTICES | SURFACES | TOLERANCES OF MACHINED DIMENSIONS FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
| | ✓ | ± | ± | ± |

TITLE

LINE POWER SUPPLY

FIRST MADE FOR 357513KF300A5




| CKT. NO. | WIRE SEQUENCE, WIRE M9729494P1918 #18 WH. |
|---|---|
| 1 | 1TB(1),1T(1) |
| 2 | 1TB(2),1T(3) |
| 3 | 1TB(3),1T(16) |
| 4 | 1CC(1),1CB(1),1CA(1),1T(15),1R(1),1TB(13) |
| 5 | 1T(14),1TB(4) |
| 6 | 1T(13),1TB(5) |
| 7 | 2CC(1),2CB(1),2CA(1),2R(1),1T(12),1TB(15) |
| 8 | 1T(11),1TB(6) |
| 9 | 1T(10),1TB(7) |
| 10 | 3CC(1),3CB(1),3CA(1),3R(1),1T(9),1TB(17) |
| 11 | 1T(8),1TB(8) |
| 13 | 1CC(2),1CB(2),1CA(2),1R(2),1TB(12) |
| 14 | 2CC(2),2CB(2),2CA(2),2R(2),1TB(14) |
| 16 | 3CC(2),3CB(2),3CA(2),3R(2),1TB(16) |

1X

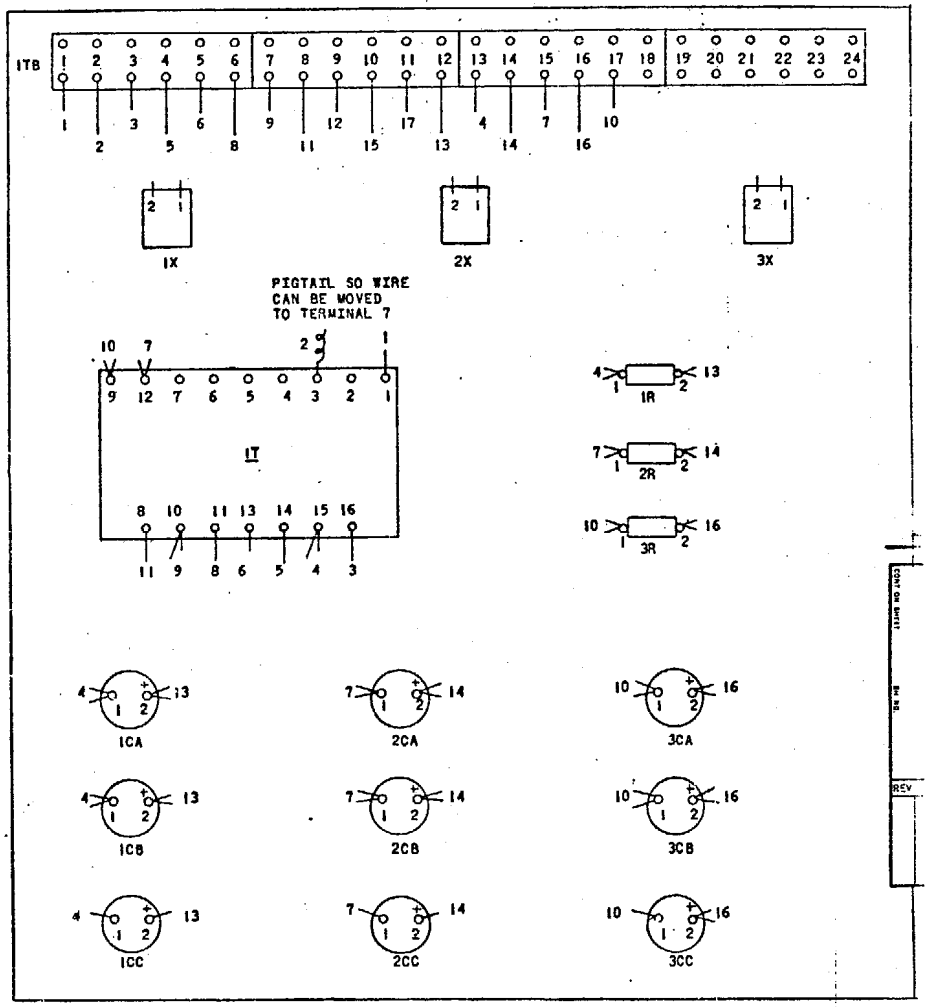
| CKT. | LEAD NO. | WIRE TO |
|---|---|---|
| 13 | 1 | 1TB(12) |
| 12 | 2 | 1TB(9) |

2X

| CKT. | LEAD NO. | WIRE TO |
|---|---|---|
| 14 | 1 | 1TB(14) |
| 15 | 2 | 1TB(10) |

3X

| CKT. | LEAD NO. | WIRE TO |
|---|---|---|
| 16 | 1 | 1TB(16) |
| 17 | 2 | 1TB(11) |



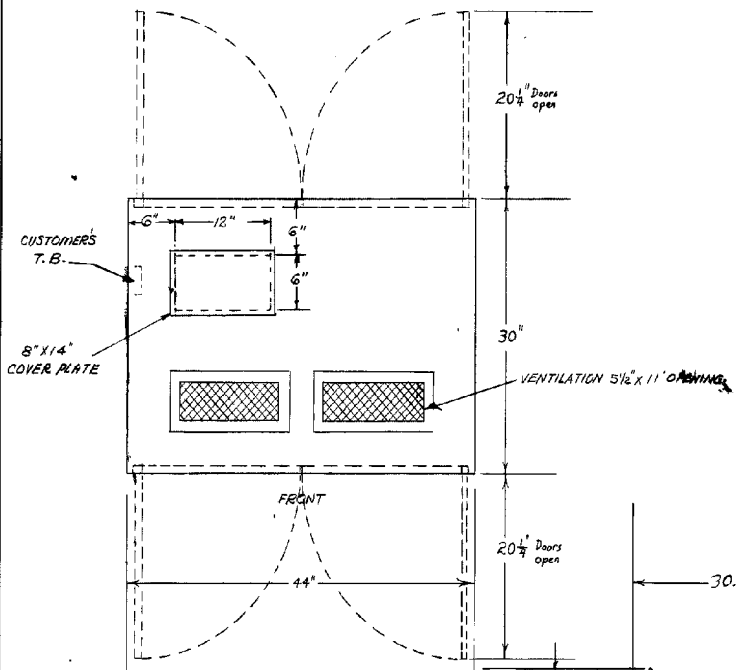

| REVISIONS | | | PRINTS TO |
|---|---|---|---|
| 1 | C.R.P. | Nov. 3, 1972 | |
| 2 | EC Correction | May 10, 1973 | |
| 3 | W4 Drawing | May 30, 1973 | |

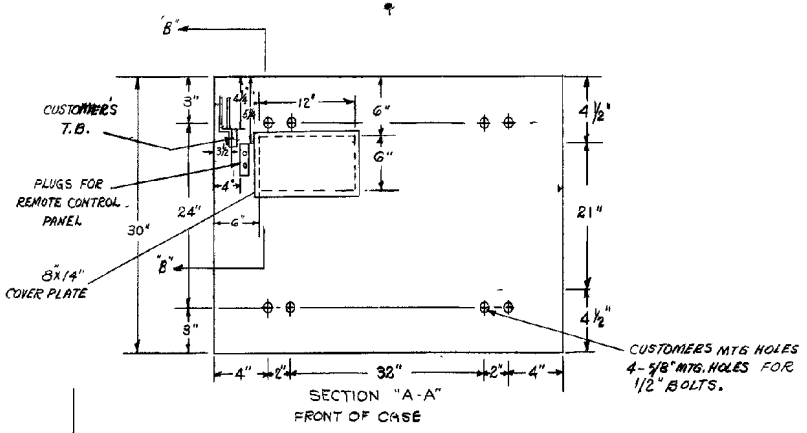
| | | | |
|---|---|---|---|
| MADE BY W. Templeton 10-12-72 | APPROVALS C. Cubbon | DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA. | 4AC306526 |
| ISSUED Oct 20, 1972 D. Meyer | | | |

GENERAL ELECTRIC

| UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING— | | | | | | REV NO | TITLE |
|---|---|---|---|---|---|---|---|
| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS | | | | 44D300043 | ELECTRONIC FEEDPUMP TURBINE CONTROL |
| | ✓ | FRACTIONS ± | DECIMALS ± | ANGLES ± | | | FIRST MADE FOR 38TS13TC300A1 |

CABINET OUTLINE

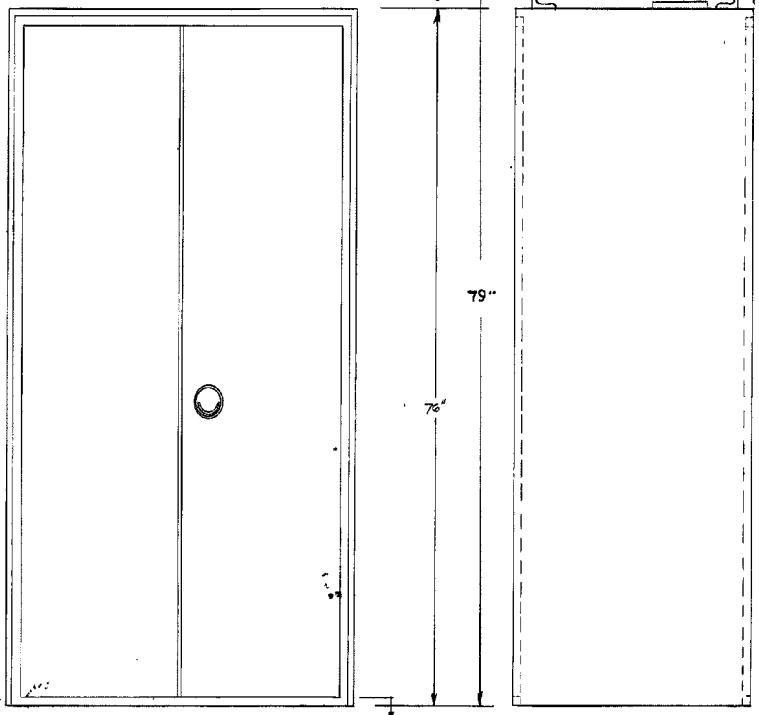


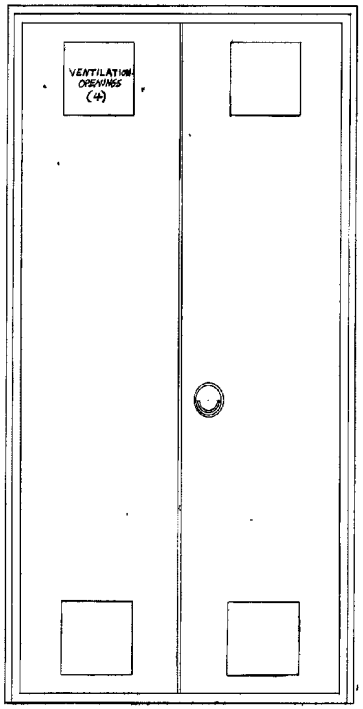

SECTION "A-A"
FRONT OF CASE

NOTES:

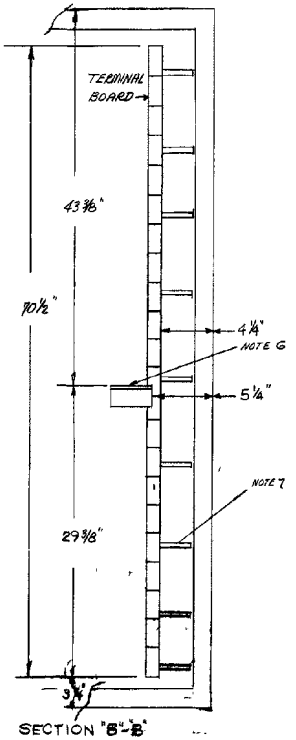
1. ONE CABINET REQUIRED FOR EACH BOILER FEED PUMP TURBINE.
2. ESTIMATED WT. 1500 LBS.
3. MAXIMUM AMBIENT TEMP. 40°C
4. FINISH ASA 61 GRAY
5. CABINET HAS LOOSE PIN REMOVABLE HINGED DOORS.
6. DISCONNECTION OF CABINET/REMOTE CONTROL CABLE ENTRANCE EITHER FROM TOP ☐ OR BOTTOM ☐ AS CUSTOMER DESIRES. (CUSTOMER TO CHECK ONE BLOCK.)
7. CUSTOMER'S PROVISIONS FOR STRAPPING CABLE RUN NEXT TO CUSTOMER'S TERMINAL BOARDS.
8. IF NON-STANDARD FINISH REQUIRED SPECIFY ON ORDERING DRAWING
9. POWER CONSUMPTION/DISSIPATION 750W.




FRONT VIEW — SIDE VIEW




BACK VIEW




SECTION "B-B"

| REVISIONS | | | PRINTS TO |
|---|---|---|---|
| 1 | | | 711 |
| 2 | | | 720-FB |
| 3 | | | FS-W |
| 4 | | | |
| 5 | | | |
| 6 | | | |
| 7 | | | |
| 8 | | | |
| 9 | REMOVED INSIDE COVER | | |

SPECIALTY CONTROL DEPT. WAYNESBORO, VA. 44D300043

3+1PL

GENERAL ELECTRIC 44D308148

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE ENG BOILER FEEDPUMP ALIGNMENT DIAGRAM FAYETTE #1, TB153090, FAYETTE #2, TB153091
FIRST MADE FOR 357513TC300AM, 031-417PG1
LOWER COLORADO RIVER AUTHORITY, FAYETTE #2

PRIMARY CONTROL-5.00V
BACK-UP CONTROL-5.05V
AT RATED SPEED SSPU4218 · HZ

ALIGNMENT INSTR.
TURBINE RATED SPEED: 5385 RPM @ 4218
VOLTAGE COMPARATOR SETTINGS

| LOC. | FUNCTION | OPERATING POINT |
|---|---|---|
| 0218 | 10% SPEED | D0 422 HZ SSPU-3 |
| 0220 | 110% SPEED | D04640 HZ SSPU-3 |
| 0219 | 105% SPEED | D04429 HZ SSPU-3 |
| 0114 | REMOTE/LOCAL | D0+6.5V N-TP |
| 0221 | 200 RPM | D0 ISTHZ SSPU-3 |

| VALVE LIFT | STOP | | "OUT" VOLTS |
|---|---|---|---|
| | CONTROL | L P H P | 1D1, 0216 |
| C.O.T. | 0" | 0" 0" | (1) |
| EFF.O | 7/8" | | |
| FULL OPEN | 13" | 5.5". 3.58" | (2) |
| FULL LOAD LIFT | 4.438 | | |

GAIN = $\frac{(2)-(1)}{13}$ = VOLT/INCH

REF. STOPS 2LVG, 0116 OUTPUT TP

| | MAN | AUTO |
|---|---|---|
| LSS | 0V | 1.762 |
| HSS | 5.440 | 5.440 |

COMPENSATION & REG. - 0212
TERMINALS J-K OPEN
INST. REG. EFF. ROT.
REG. 6% - SEE INSTRUCTIONS

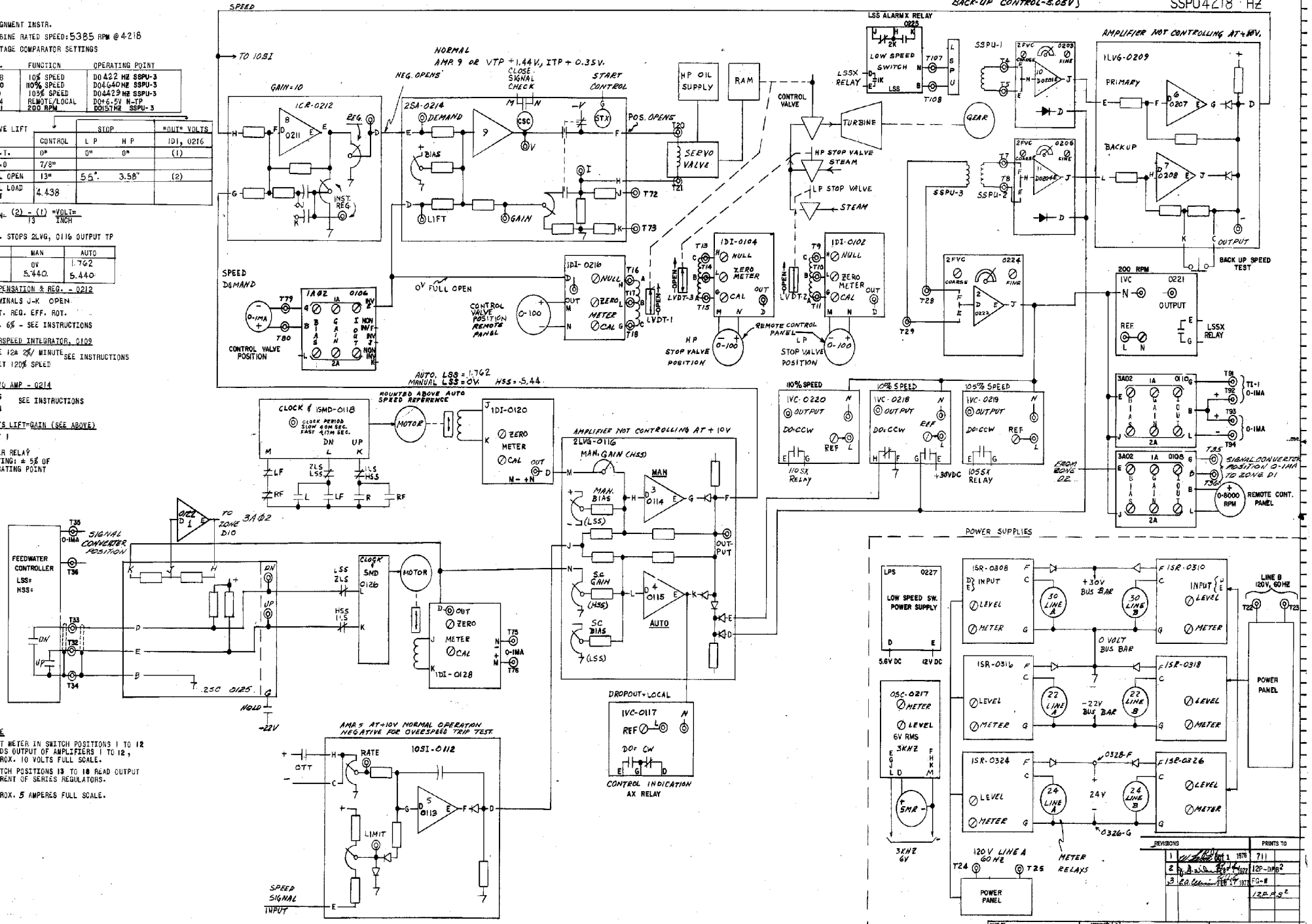
OVERSPEED INTEGRATOR, 0109
RATE 12± 2% MINUTE SEE INSTRUCTIONS
LIMIT 120% SPEED

SERVO AMP - 0214
BIAS SEE INSTRUCTIONS
GAIN

VOLTS LIFT=GAIN (SEE ABOVE)
VOLT 1

METER RELAY SETTING: ± 5% OF OPERATING POINT

NOTE
TEST METER IN SWITCH POSITIONS 1 TO 12 READS OUTPUT OF AMPLIFIERS 1 TO 12, APPROX. 10 VOLTS FULL SCALE.
SWITCH POSITIONS 13 TO 18 READ OUTPUT CURRENT OF SERIES REGULATORS.
APPROX. 5 AMPERES FULL SCALE.



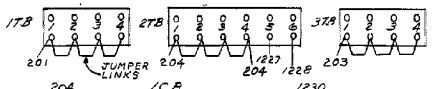

| REVISIONS | | PRINTS TO |
|---|---|---|
| 1 | W. Oct 1 1976 | 711 |
| 2 | [illegible] | 12P-DMB2 |
| 3 | [illegible] FEB 17 1977 | FG-M |
| | | 12P-A-S2 |

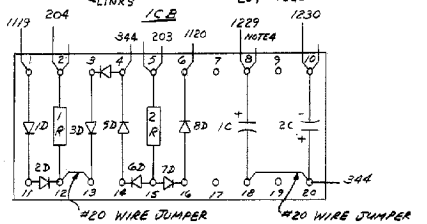
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA. 44D308148

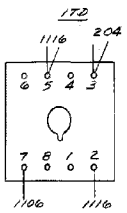
54HPL

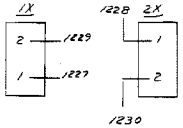
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS, DECIMALS, ANGLES | 44D308137

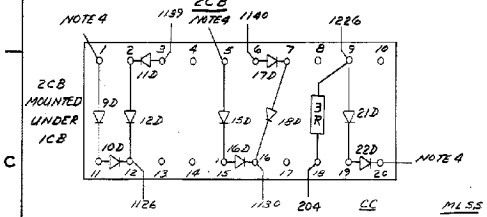
TITLE: RELAY PANEL
FIRST MADE FOR 35751A, 4A307A16

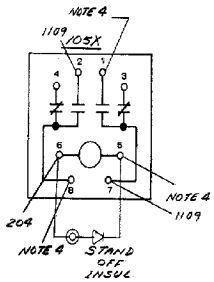



WIRE TABLE

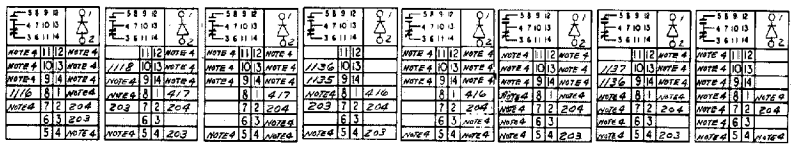
| CIRCUIT | WIRE SEQUENCE |
|---|---|
| 201 | RF,R,L,LF,1TDX,1TB |
| 203 | CC,MLSS,MLSS,MHSS,UHSS,ALSS,AHSS,ST, ALARM,P,P,N,AX,TRIP, ICB,3TB |
| 204 | 105X,CC,MLSS,MLSX,MHSS,MHSX,ALSS,AHSS, HOLD,OTT,LF,LF,ST,110SX,ALARM,N,1TDX, AX,LSS ALARMX,LSSX,ICB,1TD,2TB-1 |
| 204 | 2CB-1B, 2TB |
| 344 | ICB-4,ICB-20 |
| 402 | LF,RF |
| 403 | L,LF |
| 405 | R,RF |
| 416 | MHSS,MHSX |
| 417 | MLSS,MLSX |
| 1106 | 1TD,1TDX |
| 1107 | TRIP,ST |
| 1116 | 1TD,1TD,CC |
| 1117 | ST,ST,1TDX |
| 1118 | 1TDX,MLSS |
| 1119 | ICB,P |
| 1120 | N,ICB |
| 1121 | RF,LF,ST |
| 1124 | L,R |
| 1125 | R,RF |
| 1126 | L,2CB |
| 1127 | L,RF |
| 1128 | RF,R |
| 1129 | R,L,LF |
| 1130 | R,2CB |
| 1131 | R,RF |
| 1132 | RF,R,L |
| 1134 | RF,R |
| 1135 | TFSO,MHSS |
| 1136 | MHSS,AHSS |
| 1137 | AHSS,OTT |
| 1138 | TFSO,TFSR,TFSL |
| 1139 | TFSL,2CB |
| 1140 | TFSR,2CB |
| 1225 | ALARM,2FL |
| 1226 | 2FL,2CB |
| 1230 | TRIP,TFSO,TFSR,TFSL,ICB |
| 1248 | LSSX,1TDX |
| 1109 | 105X,105X |

FACTORY NOTES:
1. USE #20 WIRE, M5729494P161B
2. DO NOT CORD-TIE WIRES. INTERCONNECTING WIRES TO BE ADDED AFTER PANEL IS MOUNTED IN CASE ASM.
3. CONNECT DIODES TO EACH RELAY SOCKET AS SHOWN IN RELAY SYMBOL.
4. WIRE CONNECTION SHOWN ON MASTER WIRE TABLE.

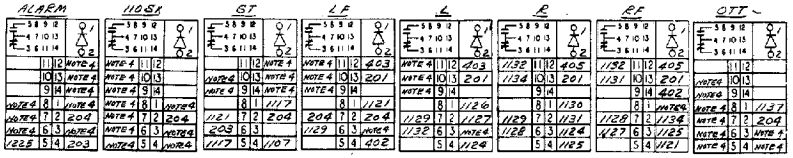
REACTOR WIRE TABLE

| CIRCUIT | CONNECTIONS |
|---|---|
| 1227 | 1X, 2TB |
| 1228 | 2X, 2TB |
| 1229 | 1X, ICB |
| 1230 | 2X, ICB |

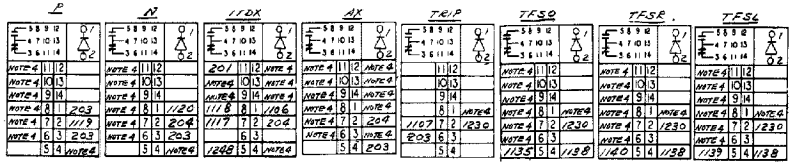
CC | MLSS | MLSX | MHSS | MHSX | ALSS | AHSS | HOLD



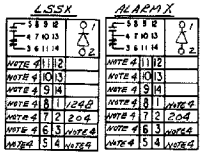
ALARM | 110SX | ST | LF | L | R | RF | OTT

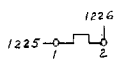


P | N | 1TDX | AX | TRIP | TFSO | TFSR | TFSL



LSSX | LSS ALARMX





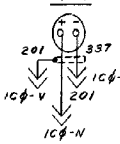

REVISIONS
1 | B. Hughes | SEP 4 1974

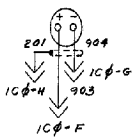
PRINTS TO
11
182-268
FG-14

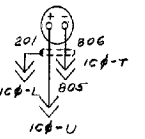
APPROVED: Thomas Clemens 4/10/75 | APR 15 1975 | R. Rubes
DRIVE SYSTEMS PROD. DEPT. | WAYNESBORO, VA.
44D308137

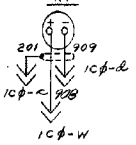
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES: SURFACES: √ TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± DECIMALS ± ANGLES ± REV NO. 44D309933 CONT ON SHEET DR NO.

TITLE: CONTROL PANEL
FIRST MADE FOR 3S7513KL300A15

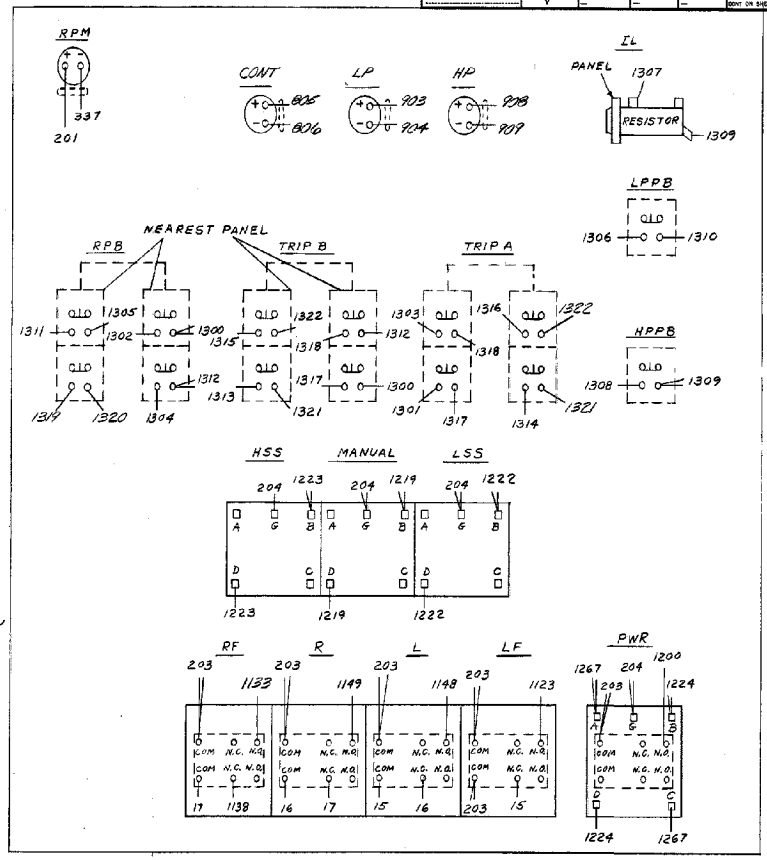
RPM


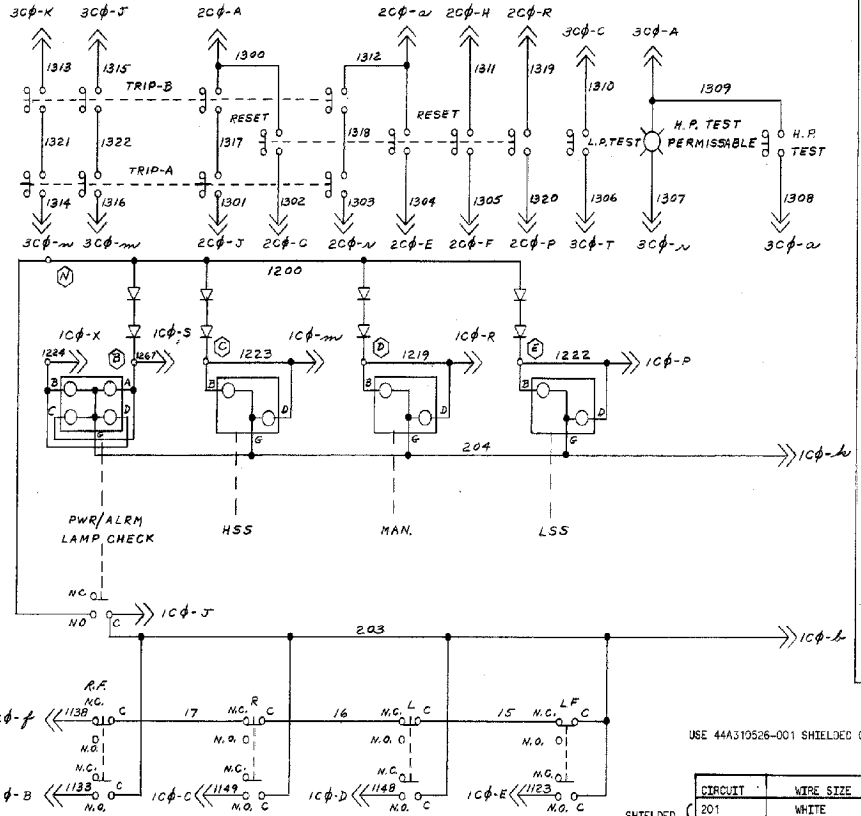

LP



CONT.


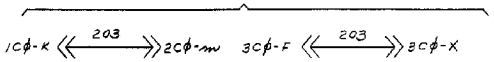

HP



BACK VIEW

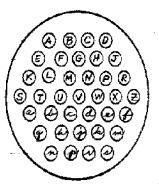



CABLE CHECK




1Cϕ, 2Cϕ, 3Cϕ




⬡ INDICATES DIODE BOARD TERMINALS

USE 44A310526-001 SHIELDED CABLE & #20 WHITE M9729494P161B

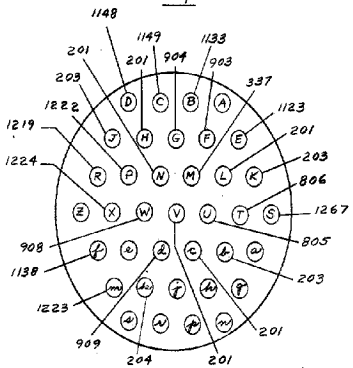
| | CIRCUIT | WIRE SIZE | CONNECTIONS |
|---|---|---|---|
| SHIELDED CABLE | 201 | WHITE | 1Cϕ,RPM |
| | 337 | BLACK | 1Cϕ,RPM |
| | 201 | SHIELD | 1Cϕ-V |
| SHIELDED CABLE | 805 | WHITE | 1Cϕ,CONT |
| | 806 | BLACK | 1Cϕ,CONT |
| | 201 | SHIELD | 1Cϕ-L |
| SHIELDED CABLE | 903 | WHITE | 1Cϕ-LP |
| | 904 | BLACK | 1Cϕ-LP |
| | 201 | SHIELD | 1Cϕ-H |
| SHIELDED CABLE | 908 | WHITE | 1Cϕ-HP |
| | 909 | BLACK | 1Cϕ-HP |
| | 201 | SHIELD | 1Cϕ-c |
| | 15 | #20 | LF,L |
| | 16 | #20 | L,R |
| | 17 | #20 | R,RF |
| | 203 | #20 | 1Cϕ,PWR,LF,LF,L,R,R,RF,1Cϕ |
| | 203 | #20 | 1Cϕ-K,2Cϕ-m |
| | 203 | #20 | 3Cϕ-F,3Cϕ-X |
| | 204 | #20 | HSS,MAN,LSS,PWR,1Cϕ |
| | 1123 | #20 | LF,1Cϕ |
| | 1133 | #20 | RF,1Cϕ |
| | 1148 | #20 | L,1Cϕ |
| | 1149 | #20 | R,1Cϕ |
| | 1200 | #20 | PWR,DB |
| | 1219 | #20 | MAN,MAN,DB,1Cϕ |
| | 1222 | #20 | LSS,LSS,DB,1Cϕ |
| | 1223 | #20 | HSS,HSS,DB,1Cϕ |
| | 1224 | #20 | PWR,PWR, ,1Cϕ |
| | 1267 | #20 | PWR,PWR,DB,1Cϕ |
| | 1138 | #20 | 1Cϕ,RF |

#16 WIRE, M9729494P211B

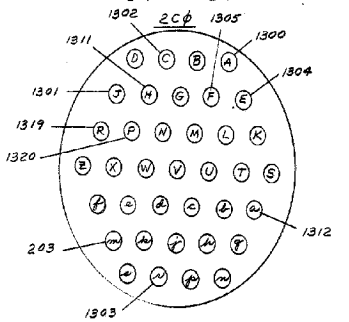
| CIRCUIT | WIRE SIZE | CONNECTIONS |
|---|---|---|
| 1300 | #16 | TRIP-B,RPB,2Cϕ |
| 1301 | #16 | TRIP-A,2Cϕ |
| 1302 | #16 | RPB,2Cϕ |
| 1303 | #16 | TRIP-A,2Cϕ |
| 1304 | #16 | RPB,2Cϕ |
| 1305 | #16 | RPB,2Cϕ |
| 1306 | #16 | LPPB,3Cϕ |
| 1307 | #16 | IL,3Cϕ |
| 1308 | #16 | HPPB,3Cϕ |
| 1309 | #16 | IL,HPPB,3Cϕ |
| 1310 | #16 | LPPB,3Cϕ |
| 1311 | #16 | RPB,2Cϕ |
| 1312 | #16 | TRIP-B,RPB,2Cϕ |
| 1314 | #16 | TRIP-A,3Cϕ |
| 1313 | #16 | TRIP-B,3Cϕ |
| 1316 | #16 | TRIP-A,3Cϕ |
| 1315 | #16 | TRIP-B,3Cϕ |
| 1317 | #16 | TRIP-A,TRIP-B |
| 1318 | #16 | TRIP-A,TRIP-B |
| 1319 | #16 | RPB,2Cϕ |
| 1320 | #16 | RPB,2Cϕ |
| 1321 | #16 | TRIP-A,TRIP-B |
| 1322 | #16 | TRIP-A,TRIP-B |

CONNECTORS - DIODE BOARD (BACK VIEW)

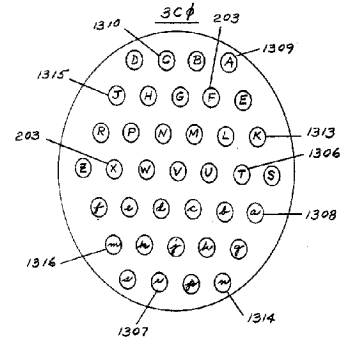
1Cϕ



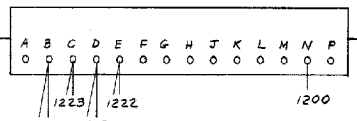

2Cϕ




3Cϕ




DB




REVISIONS: 1 OCT 1 1976
PRINTS TO: 711, 42-DFB, FG-W

DRAWN BY: [illegible] 4/25/76 ISSUED MAY 24 1976
APPROVALS
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA
REV LTR 44D309933 CONT ON SHEET DR NO.

SCS-204 (4-55)

| | |
|---|---|
| **GENERAL ELECTRIC CO.**<br>**WAYNESBORO, VA.** | **REGULATOR SUMMARY** 44X308922<br>APR 18 1975 |
| **COMPILED BY:** R.J. Culver | **DATE:** April 7, 1975 1 **SHEETS; SHEET** 1 |
| **CUSTOMER:** MDTD LOWER COLORADO RIVER AUTHORITY FAYETTE #1 | |
| TURBINES: 153090 | REQN: 091-41704 |

| ITEM | MATERIAL | QUAN. |
|---|---|---|
| 1 | 3S7513TC300A17 Electronic Feedpump Turbine Control | 1 |
| | Outline Dwg: 44D300043 | |
| | Wire Table: 44A308806 | |
| | Elementary Dwg: 44C309397 | |
| | Connection Dwg: 44C309398 | |
| | Alignment Dwg: 44D308148 | |
| | Alignment Instructions: 44A308807 | |
| | Renewal Parts: 44A335636 | |
| | | |
| 2 | 3S7513KL300A15 Control Panel | 1 |
| | Outline Dwg: 44B309405 | |
| | | |
| 3 | Cabinet-Control Panel Power Cable | 2 |
| | 44B306148 | |
| | | |
| 4 | Cabinet-Control Panel Control Cable | 1 |
| | 44B300423 | |

**REVISIONS**

| **DISTRIBUTION** | |
|---|---|
| | 11 |
| | 12P DFB$^2$ |
| | FG-W |
| | 12P-FS$^2$ |
| | |
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GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS DEPARTMENT
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